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Muhammad Zain

CFA LEVEL 1

STUDY GUIDE

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MUHAMMAD ZAIN

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Special Credit for Contribution

I am grateful to Ms. Maha Zahid for being the Co-Author in this book.

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Let's Connect With Each Other

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PREFACE

Every thread of knowledge woven into the tapestry of my understanding is a divine gift from the Supreme Architect, the Almighty Allah. It is His infinite mercy and blessing that empowered me to conquer the daunting peaks of the Certified Public Accountant (CPA), Certified Management Accountant (CMA), Certified Internal Auditor (CIA), and Masters of Business Administration (MBA) exams in my maiden attempt.

My heart thrums with gratitude as I recall the unceasing support of my family. Their enduring sacrifices – the surrendering of resources and time – have fueled my growth in all dimensions: moral, physical, and spiritual. I extend a profound token of thanks to my mentors, whose wisdom, experience, and teachings have sculpted me into the person I am today.

This book reflects the symphony of wisdom bestowed upon me by Allah, in conjunction with the tapestry of experiences and learnings acquired over a lifetime. My thirst for knowledge has led me on countless quests, diving into the endless seas of information found on the Internet, Blogs, Social Media, and Wikipedia. To all the scribes and curators of Google, Blogs, Social Media, and Wikipedia, I owe a debt of gratitude for feeding my insatiable curiosity and illuminating my path with their wisdom.

Yet, as I delved deeper, a profound realization dawned upon me: our human understanding is but a mere droplet in the boundless ocean of knowledge yet to be explored and discovered. This very human curiosity sparks a cascade of innovations, discoveries, and ideas, nudging us ever so slightly closer to the vast unknown.

In the grand scheme of this infinite wisdom, if my words happen to echo any copyrighted material, I assure you it is nothing but a coincidence. Any perceived resemblance is unintentional, a serendipitous concurrence of thoughts and ideas.

I warmly welcome you, dear readers, to freely explore this book for your personal growth and enlightenment, devoid of any time or device constraints. To make this treasure trove of knowledge accessible to all, I have consciously kept the price minimal, thereby encouraging genuine engagement with the material.

I strive for accuracy and integrity in every word that this book carries, yet I am aware of the fallibility of human knowledge. If you stumble upon any discrepancies or inaccuracies, I graciously invite your critique and correction for future updates.

In the spirit of learning and wisdom, I implore our Lord, the Supreme Master and Judge, to bless us with greater understanding and wisdom in this world, and eternal grace in the Life Hereafter. Ameen.

14 October 2024

LETTER FROM MUHAMMAD ZAIN

Dear *CFAs,*

May Peace, Blessings, and Mercy of Allah be upon you and, in particular, on the Noble Messenger Prophet Muhammad (Peace Be Upon Him), his Family, and his Companions.

Be a symbol of excellence in your life. Always dream big and think beyond the dimensions of the Universe. Man is made to conquer the seven Heavens. Explore the purpose of your existence and discover the enormous potential that is within oneself. Having faith and trust in Creator will give you the light in the darkness and unchartered territories. There is always a silver lining beneath the dark skies. A creative mindset makes life simple. Work on your passion by synchronizing your soul, heart, and mind. We all will die one day, but only a few dare to live the life they wish for.

The Creator has created the entire Universe in six days. There is great potential to discover the magnificent beauty that remains unexplored to date. This is only possible by seeking knowledge and applying them in our daily lives.

We are witnessing a moment in time that humanity has not ever experienced before. This is the digital transformation age. Business norms are artificial intelligence, Blockchain Technology, Cryptocurrency, Business Intelligence, and Big Data. AI is on the path of achieving consciousness.

All the information is available in the blink of an eye. Whatever we think in mind comes in front of our screens. These advancements will change the dynamics of the whole world we live in today. Cloud computing will replace all the traditional and so-called “modern” work methods. The work of accountants, doctors, engineers, and pilots will no longer exist. The irredeemable paper money will be replaced by electronic money. Central Governments will only exist in name only. Universal Government and a unified taxation system will emerge. Virtual reality will be ordinary. Blind will be able to see, deaf will be able to hear, without limbs persons will be able to run, and mentally disabled people will utilize the maximum brain capacity through mental chip implants. Teleportation of humans will be done in a blink of an eye.

I advise all readers worldwide to focus on entrepreneurship after the certification. This is the only way of survival. Only those businesses that are operational who have inelastic demand for their products or services and who are on cloud computing / virtual workplaces. Furthermore, invest surplus funds in real assets such as Gold, Silver, and property. They are effective hedges against inflation and devaluation. They generate positive returns even in times of economic distress.

I highly recommend that my potential readers pay their interest-bearing debt at the earliest to avoid the debt trap and never go for this easy money for the foreseeable future, even credit cards. These are all the means to enslave the human race. Always spend out of your realized income. Save some funds for your family as a contingency measure.

Allow me the opportunity to present to you the **2024** edition of ***Chartered Financial Analyst (CFA) Level 1 Study Guide***. It covers all the essential and relevant 2,116 study points that will be tested in the CFA Exam. It also includes the 2,092 True / False questions to reinforce the core concepts. It have 999 fill in the blanks and 763 word search puzzles. After reading this guide, you will feel the difference. The practice of exam questions is essential from CFA Level 1 Question Bank 2024, which have 2,594 questions, available from the Zain Academy website.

Study with complete dedication and commitment. Make the goal of learning something new and different each day. Replace your fear with curiosity.

Let's work together towards the common goal of earning a Chartered Financial Analyst (CFA) credential. My support and guidance will be with you TILL YOU PASS THE EXAMS. Furthermore, you can ask as many questions as you wish, either through WhatsApp or email, and I will answer to the best of my ability.

Your work is going to fill a large part of your life and the only way to be truly satisfied is to do what you believe is great work. The only way to do great work is to love what you do. If you haven't found it yet, keep looking. Don't settle. As with all matters of the heart, you will know when you find it.

Have the courage to follow your heart and intuition. They somehow already know what you truly want to become. Everything else is secondary.

Your imagination is everything. It is the preview of life's coming attractions. Only those who believe anything is possible can achieve things most would consider impossible.

Don't let the noise of others' opinions drown out your own inner voice.

Remembering that you are going to die is the best way I know to avoid the trap of thinking you have something to lose. You are already naked. There is no reason not to follow your heart.

I dedicate this work to the Prophet Muhammad (Peace Be Upon Him), Mercy to all the Creation, who has been humanity's source of inspiration and guidance.

May the Knowledge delivered by me shall be a continuing blessing for me in the Life Hereafter (Ameen).

With Love and Care,

Muhammad Zain

SECTION A – QUANTITATIVE METHODS

STUDY POINTS

S.No	Description
1.	Explain the **Time Value of Money**? The **time value of money** is a concept that states that cash received today is more valuable than cash received in the future. If a person agrees to receive payment in the future, he foregoes the option of earning interest had he invested that amount of money today. On the contrary, cash received today is more valuable due to inflation. It is imperative to remember that inflation tends to reduce the purchasing power of money over time.
2.	Discuss the **Key Components of the Time Value of Money?** **Discount rate or interest rate**: The rate of discounting or compounding that you apply to an amount of money to calculate its present or future value. **Time periods**: The number of periods over which a sum's present or future value is calculated. These time periods can be annually, semi-annually, quarterly, monthly, weekly, etc. **Present value (PV)**: The amount of money you have today (or at time T = 0) is the present value. **Future value (FV)**: The accumulated amount of money you get after investing the original sum at a certain interest rate and for a given time period, say, two years.
3.	How many **Ways Are There to Interpret Interest Rates**? There are **three ways** to interpret interest rates: a. **Required rate of return**: The minimum return an investor expects to earn to postpone their current consumption. b. **Discount rate**: The rate discounting future cash flows to determine the present value. c. **Opportunity cost**: The interest foregone when investors spend money on current consumption rather than saving or investing.

4.	What is **Interest**? **Interest** is a reward paid by a borrower for using an asset, usually capital, belonging to a lender. It is the compensation paid for the loss or value depreciation occasioned by using the asset. We could also describe it as the opportunity cost of alternative investments. At the time of lending, the lender most likely has a portfolio of investment vehicles to choose from. As such, they must charge a premium for the 'loss' of the alternative investment opportunities. We express interest as an annual percentage, from which we can calculate monthly, quarterly, or semiannual equivalents. The level of interest rate is a function of several risks.
5.	Formulate **Interest Rate**? An interest rate is composed of a real risk-free interest rate plus a set of four premiums that represent compensation for bearing distinct types of risk: r = Risk free rate + Inflation premium + Liquidity premium + Maturity premium + Default premium. Where r is the interest rate.
6.	Describe the **Risk-Free Rate?** The **risk-free rate** is the rate of return offered by assets largely considered risk-free. Such assets usually include government securities or local authority bonds. The real risk-free interest rate is the single-period return on a risk-free security assuming zero inflation.
7.	List the **Types of Risk Premiums?** a. Inflation risk premium. b. Liquidity risk premium. c. Default risk premium. d. Maturity risk premium.
8.	Explain **Inflation, Liquidity, Default, and Maturity Risk Premium**?

Inflation Risk Premium

Inflation risk is the loss of purchasing power of money as a result of the increase in prices of consumer goods. The inflation risk premium is the additional return that investors demand aside from the real risk-free rate. The risk of a decrease in purchasing power validates the inflation risk premium.

Liquidity Risk Premium

Liquidity refers to the ease with which an investment can be converted into cash without significantly sacrificing market value. Liquidity risk premium refers to the additional return an investor demands from any investment that cannot be instantly liquidated for cash in the market.

Default Risk Premium

Default risk describes a situation where a borrower may cease to repay borrowed funds due to bankruptcy. This might result in significant losses on the side of the lender. A default premium is the additional return the lender or investor requires from a borrower for their (lender's) default risk assumption.

Maturity Risk Premium

The maturity risk premium is the additional return an investor requires for assuming an investment's interest rate and reinvestment risk resulting from a longer maturity timeline. Maturity risk premium increases with an increase in the maturity timeline. In other words, the longer the maturity timeline of an investment, the higher the maturity risk premium.

9. What is meant by an **Effective Annual Rate of Interest**?

The **effective annual rate of interest** (EAR) refers to the rate of return earned by an investor in a year, taking the effects of compounding into account.

10. Generate the **Formula for an effective annual rate of interest**?

EAR = (1 + Periodic rate) m – 1

	Where, $$\text{Periodic rate} = \frac{\text{Stated annual rate}}{\text{m}}$$ And m is the number of compounding periods per year.
11.	Why is the **Effective Annual Rate of Interest Important?** The EAR is an essential concept in financial management because it compares two or more projects that calculate compound interest differently. For example, assume that you have two projects, X and Y. Project X pays 5% interest compounded monthly, while Project Y pays 5% interest compounded quarterly. Calculating the EAR represented by each rate, you could pick the more profitable project. Furthermore, the higher the EAR, the higher the return an investment offers.
12.	How is **Future Value Calculated?** The future value (FV) of an investment is given by: FV = PV (1 + r) N FV= future value PV= present value r= interest rate N= number of Years
13.	How **Present Value is Calculated**? PV = FV $(1 + r)^{-N}$ FV= future value PV= present value r= interest rate N= number of Years
14.	Elaborate on the **Time Value of Money with Different Frequencies of Compounding?**

	In calculating an investment's present or future value with multiple compounding periods per year, ensuring that the interest rate corresponds to the number of compounding periods present per year is the most important. The future value (FV) of an investment is given by: $FV_N = PV\ (1 + rs\ /m)^{\ mN}$ Where; rs = quoted annual rate N= number of years m= compounding periods (per year) To find the present value of an investment, make PV the subject of the above formula. $PV = FV\ \{(1 + rs/m)^{\ -m\times n}\}$
15.	Clarify the **Future Value (FV) of a Single Sum of Cash Flow?** The Future Value (FV) of a single sum of money is the amount that money invested today at a given interest rate (r) for a specified period will translate into the future. Denoted by FVN, the future value of a single sum of money is given by: $FV_N = PV\ (1 + r)^N$ Where, PV = present value of the investment FVN = future value of the investment N periods from today r= rate of interest per period N= number of periods (Years) Note that the formula above is based on the time value of money.
16.	What is a **Future Value Factor**? The factor $(1 + r)^N$ is a future value factor.

	$$R2 = (\frac{Cov\ (X, Y\ )^2}{\sigma X\ \sigma Y})$$ Where: Cov (X, Y) = covariance between two variables, X and Y. σX = standard deviation of X. σy= standard deviation of Y.
255.	Elaborate on the **F-statistic in Simple Regression Model?** To check the statistical significance of a regression model, we use the F-test. The F-test requires us to calculate the F-statistic. The F-test confirms whether a regression model's slope (denoted by bi) equals zero. In a typical simple linear regression hypothesis, the null hypothesis is formulated as H0: b1 = 0: against the alternative hypothesis, H1: b1 ≠ 0. The null hypothesis is rejected if the confidence interval at the desired significance level excludes zero. The Sum of Squares Regression (SSR) and Sum of Squares Error (SSE) are employed to calculate the F-statistic. In the calculation, the Sum of Squares Regression (SSR) and Sum of Squares Error (SSE) are adjusted for the degrees of freedom. The Sum of Squares Regression is divided by the number of independent variables, k, to get the Mean Square Regression (MSR). That is: $$MSR\ = \frac{SSR}{k} = \frac{\Sigma ni =\ 1\ (Y\hat{}\ i\ - Y\bar{})^2}{k}$$ Since we only have k=1, in a simple linear regression model, the above formula changes to: $$MSR\ = \frac{SSR}{k} = \frac{\Sigma ni= 1\ (Y\hat{}\ i - Y\bar{})^2}{k} = \Sigma\ ^{n}i= 1\ (Y\hat{}\ i- Y^{-})^2$$ Therefore, in the Simple Linear Regression Model, MSR = SSR. Also, the Sum of Squares Error (SSE) is divided by degrees of freedom given by n – k – 1 (this translates to n – 2 for simple linear regression) to arrive at Mean Square Error (MSE). That is,

$$\text{MSE} = \frac{\text{The sum of Squares Error (SSE)}}{n - k - 1} = \frac{\Sigma ni = 1\,(\hat{Y} i - \bar{Y})^2}{n - k - 1}$$

For a simple linear regression model,

$$\text{MSE} = \frac{\text{The sum of Squares Error (SSE)}}{n - 2} = \frac{\Sigma ni = 1\,(\hat{Y} i - \bar{Y})^2}{n - 2}$$

Finally, we find the ratio of MSR to MSE to calculate the F-statistic for the linear regression. That is,

$$F - \text{statistic} = \frac{\text{MSR}}{\text{MSE}} = \frac{\frac{\text{SSR}}{k}}{\frac{\text{SSE}}{n-k-1}} = \frac{\frac{\Sigma ni = 1\,(\hat{Y} i - \bar{Y})^2}{k}}{\frac{\Sigma ni = 1\,(\hat{Y} i - \bar{Y})^2}{n-k-1}}$$

The F-statistic in simple linear regression is F-distributed with 1 and n – 2 degrees of freedom. That is,

$$\frac{\text{MSR}}{\text{MSE}} \sim F1, n - 2$$

256. Interpret the **F-test Statistic?**

A large F-statistic value proves that the regression model effectively explains the variation in the dependent variable and vice versa. On the contrary, an F-statistic of 0 indicates that the independent variable does not explain the variation in the dependent variable.

We reject the null hypothesis if the calculated value of the F-statistic is greater than the critical F-value. It is worth mentioning that F-statistics are not commonly used in regressions with one independent variable. This is because the F-statistic is equal to the square of the t-statistic for the slope coefficient, which implies the same thing as the t-test.

257. Clarify the **Standard Error of Estimate?**

Standard Error of Estimate, Se or SSE, is alternatively referred to as the root mean square error or standard error of the regression.

	It measures the distance between the observed and dependent variables predicted by the regression model. The Standard Error of Estimate is easily calculated from the ANOVA table using the following formula: $$Se = \sqrt{MSE}\sqrt{\frac{\Sigma ni = 1\,(Y\hat{}\,i - Y\bar{})^2}{n-2}}$$ The standard error of estimate, coefficient of determination, and F-statistic measures can be used to gauge the goodness of fit of a regression model. In other words, these measures tell how much a regression model syncs with data. The smaller the Standard Error of Estimate is, the better the regression line fit. However, the Standard Error of Estimate does not tell us how well the independent variable explains the variation in the dependent variable.
258.	Formulate a **Null and an Alternative Hypothesis** about a population value of a **regression coefficient?** Hypothesis testing tests whether the estimated regression coefficients bear any statistical significance. The confidence interval or t-test approach can be used in hypothesis testing. In this section, we will explore the t-test approach. **The t-test Approach** The following are the steps followed in the performance of the t-test: • Set the significance level for the test. • Formulate the null and the alternative hypothesis. • Calculate the t-statistic using the formula below: $$t = \frac{b\hat{}\,1 - b1}{s\hat{}b1}$$ Where: b1 = T rue slope coefficient.

	$\hat{b}_1$ = Point estimate for $s_{\hat{b}1}$ = Standard error of the regression coefficient. • Compare the absolute value of the t-statistic to the critical t value (t_c). Reject the null hypothesis if the absolute value of the t-statistic is greater than the critical t-value, i.e., $t > +$ tcritical or $t < -$ tcritical.
259.	Describe **Different Functional Forms of simple linear regressions?** For non-linear relationships, we can potentially use several functional forms to transform the data for linear regression. The following are the three often-used log transformation functional forms: • **Log-lin model**: In this log transformation, the dependent variable is logarithmic, but the independent variable is linear. It is represented as $\ln Y_i = b_0 + b_1X_i$. The slope coefficient in the log-lin model is the relative change in the dependent variable for an absolute change in the independent variable. A one-unit increment in X triggers a $100b_1$ percent change in Y. • **Lin-log model**: In this case, the dependent variable is linear while the independent variable is logarithmic. It is represented as $Y_i = b_0 + b_1\ln X_i$. The slope coefficient in the lin-log model is responsible for the absolute change in the dependent variable for a relative change in the independent variable. Put another way, if X increases by 1%, it will change by b_1 per 100 units. • **Log-log model**: This log transformation's dependent and independent variables are logarithmic. It is represented as $\ln Y_i = b_0 + b_1\ln X_i$. The slope coefficient in the log-log model is the relative change in the dependent variable for a relative change in the independent variable. In other words, if it increases by 1%, Y will change by b_1.
260.	How to **Select the Correct Functional Form?** To settle on the correct functional form, consider the following goodness of fit measures: • Coefficient of determination (R^2). A high value is better.

- F-statistic. A high value of F-statistic is better.
- Standard error of the estimate (Se). A low value of Se is better.

Aside from the factors cited above, the patterns in residuals can also be analyzed when evaluating a model. Residuals are random and uncorrelated in a good model.

SECTION A – QUANTITATIVE METHODS

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	The time value of money states that cash received today is less valuable than cash received in the future.	**FALSE.** The time value of money is a concept that states that cash received today is **more valuable** than cash received in the future.
2.	Inflation tends to enhance the purchasing power of money over time.	**FALSE.** Inflation tends to **reduce** the purchasing power of money over time.
3.	Data is a collection of facts that can be either raw or formatted.	**TRUE. Data** is a collection of facts such as numbers, characters, measurements, observations, audio, and videos and can be either raw or formatted.
4.	*Quantiles* are values that split sorted data or a probability distribution into equal parts.	**TRUE. Quartiles, quintiles, deciles, and percentiles** are values or cut points that partition a finite number of observations into nearly equal-sized subsets.
5.	Correlation is a measure of how two variables move together.	**FALSE. Covariance** is a measure of how two variables move together.
6.	The mean, the median, and the mode are the measures of central tendency.	**TRUE. Measures of central tendency** are values that tend to occur at the center of a well-ordered data set.
7.	Semi-variance can be used to estimate downside risk.	**TRUE. Semi-variance** is the average squared deviation below the mean.
8.	Probabilities are usually expressed	**TRUE. Probability** is a measure of the likelihood that something will happen and is usually expressed as percentages, from 0%

	as percentages, from 0% to 100%.	(impossible to happen) to 100% (guaranteed).
9.	The time value of money concept has a wide range of applications in finance.	**TRUE.** The **time value of money** concept has many applications in finance, including bond valuation, shares valuation, loan facilities pricing, and capital budgeting.
10.	A portfolio consists of assets such as stocks, bonds, or cash equivalents.	**TRUE.** A **portfolio** is a collection of investments held by a company, mutual fund, or individual investor.
11.	Data visualization is also called visual discovery.	**TRUE. Data visualization** refers to the presentation of data in a pictorial or graphical format using different graphs such as histograms, polygons, line charts, bar charts, etc.
12.	The first quartile is referred to as the upper quartile.	**FALSE.** The **first quartile**, Q1, is referred to as the **lower quartile**, and the **last**, Q4, is the **upper quartile**.
13.	The rate used to calculate an amount's present or future value is called a discount rate.	**TRUE. Discount rate** also called the **interest rate,** is the rate of discounting or compounding applied to an amount of money to calculate its present or future value.
14.	Opportunity cost is the interest foregone when investors spend on current consumption rather than saving or investing their money.	**TRUE. Opportunity cost** is a key factor in interpreting interest rates. It refers to the interest foregone when investors opt for an alternate option, such as spending on current consumption instead of saving or investing.
15.	Semi-deviation is also known as semi-standard deviation.	**TRUE. Semi-deviation** is the positive square root of semi-variance and is used to estimate downside risk, i.e., returns below the mean.
16.	Correlation is a measure of the	**TRUE. Correlation** divides the covariance by the product of the standard deviations of

	linear relationship between two variables.	both variables. As a result, its value ranges between -1 and +1 and is easier to interpret.
17.	An outcome is any quantity whose expected future value is not known beforehand.	**FALSE**. A **random variable** is any quantity whose expected future value is unknown, whereas an outcome is any possible value a random variable can take.
18.	The upper quartile subdivides the data into the lower 25% of the values.	**FALSE**. The first quartile splits the data into the lower 25% of the values and the upper 75%. Likewise, **the upper quartile subdivides the data into the lower 75% of the values** and the **upper 25%.**
19.	Investors usually tend to reduce the portfolio variance.	**TRUE. Portfolio variance** is a measure of risk; more variance translates to more risk. Therefore, investors usually reduce the portfolio variance by choosing assets with low or negative covariance, e.g., stocks and bonds.
20.	Present value is the accumulated amount of money you get after investing the original sum at a certain interest rate and for a given time period.	**FALSE. Future value** is the accumulated amount of money you get after investing the original sum at a certain interest rate and for a given time period.
21.	Numerical data represent values that can be measured or counted.	**TRUE. Numerical data** represent values that can be measured or counted, for example, age, height, weight, or the number of shares in a portfolio.
22.	The inflation risk premium is the additional return that investors demand aside from the real risk-free rate.	**TRUE.** The **inflation risk premium** is the additional return that investors demand aside from the real risk-free rate and is validated by the risk of a decrease in purchasing power.
23.	Liquidity refers to the ease with	**FALSE.** Liquidity refers to the ease with which an investment can be converted into

	which an investment can be converted into cash and making a significant market value sacrifice.	cash **without** significantly sacrificing market value.
24.	A set of data may have a mode or none.	**TRUE**. The **mode** is the value that occurs most frequently in a data set.
25.	Exhaustive events are such that one event precludes the occurrence of all the other events.	**FALSE. Mutually exclusive events** are such that one event precludes the occurrence of all the other events, whereas events are said to be exhaustive if they include all possible outcomes.
26.	The interquartile range is the difference between the upper and lower quartile.	**TRUE.** The difference between the upper and lower quartiles is known as the **interquartile range**, which indicates the spread of the middle 50% of the data.
27.	The required rate of return is the maximum return an investor expects to earn to postpone their current consumption.	**FALSE.** The required rate of return is the **minimum return** an investor expects to earn to postpone their current consumption.
28.	The risk-free rate is the rate of return offered by assets largely considered risk-free.	**TRUE.** The **risk-free rate** is the rate of return offered by assets, usually government securities or local authority bonds, which are largely considered risk-free.
29.	Quintiles split a set of data into four equal parts.	**FALSE**. Quintiles split a data set into **five** equal parts, i.e., fifths. Therefore, the second quintile splits data into the lower 40% of the values and the upper 60%.
30.	A correlation of 0 indicates that a linear relationship exists between the variables.	**FALSE.** A correlation of 0 (uncorrelated variables) indicates **no linear relationship between the variables.**

178.	A low R^2 explains variability better than a high R^2.	**FALSE.** The coefficient of determination, R^2, measures the proportion of the total variability of the dependent variable explained by the independent variable. A **high R^2 explains variability better than a low R^2**. If R^2 = 0.01, only 1% of the total variability can be explained. Alternatively, if R^2 = 0.90, over 90% of the total variability can be explained. Concisely, the higher the R^2, the higher the model's explanatory power.
179.	The standard error measures the variability obtained within one sample.	**FALSE.** The **standard deviation** measures the variability obtained within one sample, whereas the standard error estimates the variability among many samples.
180.	Data snooping bias occurs when analysts excessively analyze data, giving rise to statistically irrelevant and, sometimes, non-existent trends.	**TRUE. Data snooping** analyzes historical data to unearth trends and other inherent relationships between variables. Analysts may then use such trends to predict future behavior.
181.	The standard deviation gives an estimate of the variability among many samples.	**FALSE.** The **standard error** gives an estimate of the variability among many samples.
182.	The central limit theorem provides useful normal approximations to binomial and poisson distributions.	**TRUE.** The **central limit theorem** asserts that when we have simple random samples, each of size n from a population with mean μ and variance σ2, the sample mean X approximately has a normal distribution with mean μ and variance σ2/n as n (sample size) becomes large.
183.	The slope coefficient in the log-lin model is responsible for the absolute change in	**FALSE.** The slope coefficient in the **lin-log model** is responsible for the absolute change in the dependent variable for a relative change in the independent variable.

	the dependent variable for a relative change in the independent variable.	
184.	A scatter plot is a graph showing the relationship between two numerical variables.	**TRUE.** A **scatter plot** helps display and understand potential relationships between two variables, as one variable is plotted on the x-axis and another on the y-axis.
185.	False positive values are also known as Type II errors.	**FALSE.** False positive values, also known as **Type I errors,** are those that were actually negative but falsely predicted as positive.
186.	The number of tally sheets for each value or group is likely a class limit.	**FALSE.** The number of tally sheet counts for each value or a group is most likely known as **frequency,** as the frequency of a class is the number of data entries in the class.
187.	Heat maps can display frequency distributions and visualize the degree of correlation among different variables.	**TRUE.** A **heat map** is a graphical representation of data that uses a system of color-coding to represent different values. More intense color is displayed as the marks "heat up" due to their higher values or density of records.
188.	The arithmetic mean is used to measure growth rates.	**FALSE.** The **geometric mean** measures central tendency, mainly used to measure growth rates.
189.	The estimate's standard error is referred to as the root mean square error or standard error of the regression.	**TRUE.** The **standard error of estimate** measures the distance between the observed dependent variables and the dependent variables predicted by the regression model.
190.	A bubble line chart is the most appropriate for plotting stock price change over five years.	**FALSE.** A **line chart** displays the change of data series over time, so it is the most appropriate for plotting stock price change over five years.

191.	Backfill bias excludes financial vehicle information that no longer exists during the sampling period.	**FALSE. Survivorship bias** excludes financial vehicle information that no longer exists during the sampling period. Consequently, conclusions informed by survivorship bias may underestimate or overestimate the population parameters.
192.	Population variance is symbolized by σ2.	**TRUE.** The **population variance**, denoted by σ2, is the average of the squared deviations from the mean.
193.	A lower standard error value indicates a less precise estimation of the population mean.	**FALSE.** A lower standard error value indicates a **more precise** estimation of the population mean. In contrast, a larger standard error value indicates a less precise estimate of the population mean.
194.	The harmonic mean determines the average growth rates of economies or assets.	**TRUE.** Analysts use the **harmonic mean** to determine the average growth rates of economies or assets.
195.	False positive values are those that were actually positive but falsely predicted as negative.	**FALSE. False negative** values are those that were actually positive but falsely predicted as negative.
196.	The standard error becomes larger as the sample size increases.	**FALSE.** The standard error becomes **smaller as the sample size increases** as increasing the sample size ultimately brings the sample mean closer to the true value of the population mean.
197.	Lookahead bias arises when successful funds submit their past performance to a database, which then compiles all previous data.	**FALSE. Backfill bias** arises when successful funds submit their past performance to a database, which then compiles all previous data. Meanwhile, unsuccessful funds don't report to the database. Their poor performance is, therefore, excluded from the database.

198.	Both the dependent and independent variables are logarithmic in the log-log model.	**TRUE.** Both the dependent and independent variables are logarithmic in the **log-log model.** The slope coefficient in the log-log model is the relative change in the dependent variable for a relative change in the independent variable.
199.	The geometric mean is equal to the arithmetic mean when there is a variability in the observations.	**FALSE.** The geometric mean is equal to the arithmetic mean when there is **no variability** in the observations, which is when all the series' observations are the same.
200.	Time period bias involves an inappropriate generalization of time-specific results that only apply to certain seasons or periods.	**TRUE. Time period bias** involves an inappropriate generalization of time-specific results that only apply to certain seasons or periods. Most entities experience seasonal variation in performance, resulting in some months being more productive than others.
201.	The arithmetic mean should be used for estimating the average returns over more than one period.	**FALSE.** The arithmetic mean should be used for estimating the average return over a **one-period horizon**. In contrast, the geometric mean should be used for estimating the average returns over more than one period.

SECTION A – QUANTITATIVE METHODS

FILL IN THE BLANKS

1. The present value of an annuity is calculated using the formula PV = C × [1 - (1 + r)^(-n)] / r, where "C" stands for _________.

2. The _________ measures the dispersion of a data set around its mean and is calculated as the square root of the variance.

3. In hypothesis testing, a _________ error occurs when the null hypothesis is incorrectly rejected when it is actually true.

4. The _________ ratio measures the excess return per unit of risk and is calculated as (Expected Return - Risk-Free Rate) / Standard Deviation.

5. The _________ coefficient measures the degree and direction of linear association between two variables.

6. A probability distribution that is symmetric and bell-shaped is known as a _________ distribution.

7. The _________ value in a confidence interval is used to indicate the range within which the population parameter is expected to lie, with a certain level of confidence.

8. The formula for the future value of an investment compounded annually is FV = PV × (1 + r)^n, where "n" represents the _________.

9. _________ risk is the risk that can be diversified away by holding a well-diversified portfolio.

10. In a normal distribution, approximately _________% of the observations lie within one standard deviation of the mean.

11. The _________ is the point estimate in hypothesis testing used to make decisions about the null hypothesis.

12. The geometric mean return is generally _________ than the arithmetic mean return for the same set of data.

13. The _________ of an event is the likelihood that the event will occur, expressed as a number between 0 and 1.

14. A _________ distribution describes the probability of a given number of successes in a fixed number of independent trials, with only two possible outcomes.

15. The _________ of a sample is calculated as the sum of all sample values divided by the number of observations in the sample.

16. The _________ effect refers to the idea that the returns of a series of periodic investments compound on both the principal and the accumulated interest.

17. The _________ hypothesis is a statement that is tested in statistical hypothesis testing and is assumed to be true until evidence indicates otherwise.

18. The _________ coefficient of determination (R^2) explains the proportion of the variance in the dependent variable that is predictable from the independent variable(s).

19. In a _________ distribution, the probability of success remains constant across trials, and each trial is independent of the others.

20. The formula for calculating the coefficient of variation is _________ divided by the mean.

21. When calculating the internal rate of return (IRR), it is the rate that makes the net present value (NPV) equal to _________.

22. The _________ law of large numbers states that as the number of trials increases, the sample mean will get closer to the population mean.

23. The _________ value is a measure of central tendency that identifies the most frequently occurring value in a data set.

24. A _________ is a graphical representation of the relationship between two variables, typically used to assess the potential linear association between them.

25. In time series analysis, _________ refers to the statistical relationship between values separated by a given time interval.

26. The _________ ratio compares the likelihood of two competing hypotheses, given the observed data.

27. _________ data are numerical measurements that can be divided into meaningful subparts, such as age or income.

28. A _________ test is a statistical test that compares the means of two independent samples to determine if they are significantly different from each other.

98. The _________ distribution is a continuous probability distribution characterized by its bell-shaped curve, and it is symmetric about the mean.

99. The _________ distribution is used to model the number of events that occur in a fixed interval of time or space.

SECTION A – QUANTITATIVE METHODS

ANSWER KEY

1. Cash flow per period

2. Standard deviation

3. Type I

4. Sharpe

5. Correlation

6. Normal

7. Critical

8. Number of periods

9. Unsystematic

10. 68

11. Test statistic

12. Lower

13. Probability

14. Binomial

15. Mean

16. Compounding

17. Null

18. Adjusted

19. Bernoulli

20. Standard deviation

21. Zero

22. Strong

23. Mode

24. Scatter plot

25. Autocorrelation

26. Likelihood

27. Quantitative

84. Exponential

85. Skewness

86. t

87. Probability

88. Moving average

89. Likelihood

90. t

91. Mode

92. Sharpe

93. Alternative

94. Beta

95. Type I

96. Bayes'

97. Standard

98. Normal

99. Poisson

SECTION A – QUANTITATIVE METHODS

WORD SEARCH PUZZLES

Section A - Quiz 1

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C G G Q Q P S V X S L O P A C H S E Z Q
L E G R V N O F I Y J P N V K V W A K C
F Q P U C W W J K Q H H A A W K B J H X
J H D T F T N O I T A L E R R O C X K A
K N G Z A Z U N O M L S M I F R O G A R
I T M A F E O X Z W D H R A E F D O H P
A L B F D N F D Z W L A F N D C Y M A F
L S J K L Q T N E O N Q Y C I X C D H T
Z D X H L G D H N G S Z T E R M D K Z P
D E W C V C R B E S V P I H P W O O N M
D I S T R I B U T I O N L X D N C D Y G
I M E D I A N C U J L Y I G L U S Q E T
E D S F Q V B Y F J G Y B E O W K S K T
B N O I T A I V E D D R A D N A T S Y T
E M E C N A I R A V O C B X U A Y L H L
M X J V L E R L Q Q A E O Z F Y H R M D
M R Z F B C S X V W E P R A S U O Q V R
A Y P N N R E R B A U X P F G U Z V O O
L Y L Q K F M S B X H R U X L S G E Z B
M W Y A A Q N A R P E T I S S V F Z X F
```

1. mean **2.** median **3.** mode **4.** range **5.** variance **6.** standarddeviation

7. covariance **8.** correlation **9.** probability **10.** distribution

SECTION B – ECONOMICS

STUDY POINTS

S.No	Description
1.	Define **Elasticity?** **Elasticity** measures the sensitivity or responsiveness of one variable to another.
2.	State the **Main Forms of Elasticity?** There are three main forms of elasticity: • Price elasticity. • Income elasticity. • Cross-price elasticity.
3.	Explain **Price Elasticity?** **Price elasticity** of demand is a measure of how a product's demand changes in response to changes in its price. It is measured in percentage changes in each of the variables. Thus, we calculate the price elasticity of demand using the following: Edpx = %ΔQdx / %ΔPx Where: %ΔQdx = The percentage change in quantity demanded %ΔPx = The percentage change in price
4.	Mention the **Classifications of Price Elasticities of Demand?** • Demand is **elastic** if the price elasticity of demand (PED) is more significant than one. That is PED > 1. This implies that the quantity demanded changes by a larger proportion than the price. • Demand is **inelastic** if PED < 1. This implies that the quantity demanded changes by a smaller proportion than the price.

	• Demand has a unit elasticity if PED= -1. This implies that the percentage change in the quantity demanded and the price changes are equal and opposite.
5.	State the **Factors That Affect the Price Elasticity of Demand?** • **Availability of substitute goods**: More substitutes translate to a more elastic demand. • **Time horizon**: Generally, it is time-consuming to find alternative goods, and hence, the more time utilized, the more elastic the demand. The price elasticity of demand for a shorter period is always low, or it can even be inelastic. • **Income**: this is based on the proportion of income on a good. Intuitively, from the formulas, if the product makes a larger proportion of an individual's income, this translates to more elastic demand.
6.	Define **Income Elasticity?** The **income elasticity** is defined as the percentage change in quantity demanded divided by the percentage change in the customers' income ceteris paribus (holding all other things constant).
7.	**Equate** income elasticity? EdI = %ΔQdx/ %ΔI
8.	Discuss **Cross-price Elasticity?** Other than the price of a product and the income of the consumers, the prices of other products can also affect the demand for the product. The cross-price elasticity is defined on this basis. Here, we evaluate the effect of the percentage change in the prices of other products on the quantity of demand for a particular good.
9.	Formulate **Cross-price Elasticity?** EdPy = %ΔQdx/ %ΔPy

	Where Py represents the price of other products.
10.	Clarify the **Substitution Effect?** The **substitution effect** states that a good becomes more of a bargain relative to other goods as its price declines; therefore, it is substituted for other goods.
11.	Assess the **Income Effect?** The term **income effect**, in economics, refers to a change in the consumption of a good or service due to a change in income. It is important to note that the income effect mainly expresses how increased purchasing power affects consumption.
12.	What are **Normal Goods?** **Normal goods** are goods whose demand increases with increased consumer income. Note that the rate at which demand increases is lower than the rate at which income increases. The rate eventually slows down with further increments in income. Examples of goods are furniture, clothes, and automobiles.
13.	Explain **Inferior Goods?** **Inferior are goods** whose demand decreases when the consumers' income increases. Examples could be second-hand clothes, canned foods, public transportation, etc. This is because their demand falls with the availability of quality alternatives.
14.	Clarify **Giffen Goods?** **Giffen goods** are inferior goods whose negative income effect outweighs the substitution effect. That is, an increase in the prices of these goods causes an increase in the quantity consumed and less when the price decreases.
15.	Describe **Veblen Goods?** **Veblen goods** are primarily for prestige, i.e., they are ornamental. Their examples include such goods as expensive cars and high-end watches. Therefore, an increase in the prices of these goods causes an increase in the amount consumed and not less. An

	increase in the consumption of such goods is due to the belief that consuming more of the goods bestows a higher socio-economic status upon the consumer.
16.	State the **Law of Diminishing Marginal Returns?** The **law of diminishing marginal returns** states that the marginal return from an increased input, say labor, will decrease when this input is added continually to a fixed capital base.
17.	Explain the **Break-even Point of Production?** The **break-even point** can be defined as the production and sales levels of a given product at which the revenue generated from the sales is perfectly equal to the production cost. At this point, the company does not make any profit or loss; it breaks even.
18.	What are **Fixed Costs?** **Fixed costs** remain the same regardless of any company's production level. A good example is the cost of setting up production facilities, e.g., rent, fixed interest charges, and depreciation.
19.	Clarify **Variable Costs?** **Variable costs** are directly proportional to the volume produced, and examples include raw materials, wages paid, and other incurred expenses.
20.	How shall a **Company Set Its Selling Price**? A company must fix its selling price above the variable costs incurred per production unit. The difference between the selling price and the variable costs is referred to as the contribution to fixed costs and profits. An increase in sales causes a direct increase in contribution. At the sales break-even point, the contribution is strictly equal to the fixed cost. Therefore, companies will report losses and profits below this point.

	forward rate. Note that the two investment alternatives are risk-free because they are invested in risk-free assets. Since these investment alternatives are equal by considering the risk characteristics, the returns must also be equal. As such, we have the following relationship: 1 +id = Sf/d (1+if) (1/ Ff /d) Note that 1/Ff/d is the number of units of domestic currency for each unit of foreign sold forward. The relationship above can be rearranged to get the formula for a forward rate: Ff /d = Sf/d (1 + if/1 + id) This formula shows the relationship between the spot rate, the forward rate, and the interest rate in foreign and domestic countries.
285.	Identify the **International Fisher Effect in Spot vs. Forward Rates?** According to the Fisher effect, interest rate differences between two countries reflect the difference in the inflation rates of these two countries. High-interest rate countries experience higher inflation rates, so the same uninvested dollar today is worth much less in the future. Therefore, there is a need to have a higher interest rate to compensate for the loss of purchasing power.
286.	Explore the **Interest Rate Parity (IRP) in Spot vs. Forward?** Interest rate parity is a theory that states that the difference between the interest rates of two countries is the same as between the spot exchange rate and the forward exchange rate. This theory plays a major role in foreign exchange markets since it connects the dots between the interest rates, the spot exchange rates, and the foreign exchange rates.
287.	Assess the **Purchasing Power Parity (PPP) in Spot Rates vs. Forward Rates?**

	According to the purchasing power parity principle, a country's currency fluctuates as the inflation rate of another country fluctuates. Therefore, the depreciation rate in a currency is roughly equal to the excess inflation rate in the domestic country over another country's inflation rate.
288.	Describe **Exchange Rate Regimes?** Almost every exchange rate regime has its flaws, virtues, and particularities. Since exchange rates in different currencies fluctuate due to the influence of market forces, some nations peg their currencies on other currencies. Others still have market-determined floating rate regimes. That said, exchange rate arrangements are mainly used by developing economies. **Floating Exchange Rate** Most medium-sized nations have successfully managed to maintain floating exchange rate regimes over long periods. Floating exchange rate regimes use the market forces to dictate their currencies' exchange rates. **Pegged Currencies** The monetary systems of some nations, for example, China, use pegged exchange rate regimes. This means exchange rates are fixed to other currencies for a certain time. In the case of China, the yuan is pegged to the US dollar. The cornerstone of China's economic policy is managing the yuan exchange rate to benefit its exports. **Dollarization** Dollarization refers to a case in which a foreign currency is used as legal tender in addition to or instead of the domestic currency. This is primarily the case for small (and sometimes unstable) countries such as Ecuador, El Salvador, the British Virgin Islands, and Zimbabwe, which use the U.S. dollar instead of their currency. Other examples include Kosovo and Montenegro, which use the Euro without being part of the Eurozone.
289.	Analyze the **Effects of Exchange Rates on Countries' International Trade and Capital Flows?**

Changing Prices of Currencies

While shifts in supply and demand of products change the prices of those products, constant shifts in supply and demand for foreign currencies cause changes in the prices of currencies. Likewise, as prices of money change, demand for foreign currencies changes.

Increase in Demand for Imports

An increase in the demand for imported goods happens when products of a foreign nation sell at lower prices than domestic products. When domestic income rises, demand for imports rises. Moreover, in capital markets, individuals invest in other nations' securities when returns on a nation's investments are higher than the domestic interest rate.

Expensive Products

Many countries depend more on imports than domestically produced goods. This is because exchange rates play a significant role in determining the prices of imported products. If the domestic currency is weaker, consumers will have to pay higher prices in domestic currency for foreign goods.

They are Self-correcting over Time

When the supply of dollars in the international market grows, their values depreciate. As time passes, imports become unattractive, and exports become more attractive. The converse is also true.

Development of a Trade Deficit

If the interest rates of other nations are higher than those of dollar nations, the demand for foreign countries automatically falls. However, a stronger dollar means decreased exports because they seem expensive to foreign consumers. This leads to a trade deficit.

Effect on Standard of Living

	Differences in currency values affect our ability to buy imported goods and export domestic goods. A currency crisis significantly affects the lives and well-being of the citizenry. For example, take Argentina, which has defaulted on its debt five times in the last 200 years. Every time this happens, their currency gets devalued. As a result, Argentineans cannot import goods from other countries cheaply for a few years. This hurts Argentina's economic growth and its citizenry's standard of living.

SECTION B – ECONOMICS

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	There are three main forms of elasticity.	**TRUE. Elasticity** measures the sensitivity or responsiveness of one variable to another. Following are the three forms of elasticity: a. Price elasticity. b. Income elasticity. c. Cross-price elasticity.
2.	A company does not make any profit or loss at the break-even point.	**TRUE.** The **break-even point** can be defined as the production and sales levels of a given product at which the revenue generated from the sales is perfectly equal to the production cost.
3.	Inferior goods are goods whose demand increases with increased consumer income.	**FALSE. Normal goods** are goods whose demand increases with increased consumer income. Examples of goods are furniture, clothes, and automobiles.
4.	Rent, fixed interest charges, and depreciation are examples of fixed costs.	**TRUE. Fixed costs** are those that remain the same regardless of any company's production level.
5.	An optimal price is when a seller can make the lowest profit possible.	**FALSE.** An optimal price can be defined as the price at which a seller can make the **highest profit possible**; that is, the seller's price is maximized.
6.	The aggregate input of an economy is the value of all the goods and services produced within a predetermined time.	**FALSE.** The **aggregate output** of an economy is the value of all the goods and services produced within a predetermined time.
7.	The price elasticity of demand is measured in	**TRUE. Price elasticity of demand** is a measure of how a product's demand changes in response to changes in its price.

	percentage changes in each variable.	
8.	Investing is setting aside money for emergencies or future purchases.	**FALSE. Saving** is setting aside money for emergencies or future purchases.
9.	Second-hand clothes, canned foods, and public transportation are inferior goods.	**TRUE. Inferior goods** are those whose demand decreases when the consumers' income increases as their demand falls with the availability of quality alternatives.
10.	Market structure can be defined as the characteristics of a market, which can either be competitive or organizational.	**TRUE. Market structure** outlines the competition's nature and the market's pricing procedure. Therefore, it describes the number of entities producing similar goods and services in a market and whose structure is determined by the current competition in the market.
11.	The aggregate income refers to the economic value of all payments received by the customers of the factors of production of goods and services.	**FALSE.** The aggregate income refers to the economic value of all payments received by the **suppliers** of the factors of production of goods and services.
12.	Demand is inelastic if the price elasticity of demand is greater than one.	**FALSE.** Demand is **elastic** if the price elasticity of demand is greater than one. This implies that the quantity demanded changes by a larger proportion than the price.
13.	The law of diminishing marginal returns states that the marginal return from an increased input will also increase when this input is added	**FALSE.** The law of diminishing marginal returns states that the marginal return from an increased input will **decrease** when this input is added continually to a fixed capital base.

	continually to a fixed capital base.	
14.	The calculation of income elasticity is similar to price elasticity.	**TRUE.** The **income elasticity** is defined as the percentage change in quantity demanded divided by the percentage change in the customers' income ceteris paribus (holding all other things constant).
15.	The word inferior in inferior goods means substandard goods.	**FALSE.** The word inferior **does not mean** substandard goods. Instead, it relates to the affordability of such goods. As income increases, consumer demand for such goods falls because consumers might, for example, substitute secondhand clothes for new clothes. Consequently, the consumers view these goods as inferior.
16.	Variable costs include raw materials, wages paid, and other incurred expenses.	**TRUE. Variable costs** are directly proportional to the volume produced, and examples include raw materials, wages paid, and other incurred expenses.
17.	Marginal cost is equal to marginal revenue at the equilibrium point.	**TRUE.** A firm is said to be at **equilibrium** if the marginal cost is equal to marginal revenue, which is the profit-maximizing output level.
18.	Economies of scale refer to the cost advantage brought about by an increase in the output of a product.	**TRUE. Economies of scale** arise due to the inverse relationship between the per-unit fixed cost and the quantity produced; the greater the production, the lower the fixed costs per unit because the production costs have been spread out over many goods. As a result, synergies and operational efficiency cause a reduction in variable costs.
19.	Positive income elasticity means that an increase in income leads to a decrease in the quantity demanded.	**FALSE. Negative income** elasticity means that an increase in income leads to a decrease in the quantity demanded.

270.	Short-term controls have proved to be a part of a larger development goal for some profitable trading stories.	**FALSE. Long-term** controls have proved to be a part of a larger development goal for some profitable trading stories, whereas short-term controls deal with urgent matters (such as a recession).
271.	The growth rate of labor productivity can be described as the real Gross Domestic Product that a worker can produce per hour worked.	**FALSE. Labor productivity** refers to the real GDP that a worker can produce per hour worked. Hence, the **growth rate of labor productivity** refers to the **percentage increase or decrease in the real GDP produced by a worker per hour worked.**
272.	An economic union is a type of trade bloc.	**TRUE.** An **economic union** is a trade bloc of a common market with a customs union. The member nations have common policies regulating products, capital and labor, and freedom of movement of goods and services.
273.	Foreign exchange market participants consist of two broad categories.	**TRUE.** The **foreign exchange market** facilitates international trade between countries globally and is used to hedge foreign exchange rate risk. The following are the two broad categories: a. The **sell-side** consists of large foreign exchange trading banks, such as Citigroup. b. The **buy-side** consists of clients who use these banks to undertake transactions, such as corporate accounts, real money accounts, and government and central banks.
274.	The base currency trades at a forward discount when the forward rate exceeds the spot rate.	**FALSE.** The base currency is said to be trading at a **forward premium** when the forward rate is higher than the spot rate. Conversely, a forward discount is when the forward exchange rate is lower than the spot exchange rate.
275.	The interest rate parity states that	**FALSE.** The **international Fisher effect** states that the interest rate differences

	the interest rate differences between two countries reflect the difference in the inflation rates of these two countries.	between two countries reflect the difference in the inflation rates of these two countries. It suggests that the estimated appreciation or depreciation of two countries' currencies is proportional to the difference in their nominal interest rates.
276.	Free trade and economies of scale are advantages of trading blocs, common markets, and economic unions.	**TRUE.** The advantages of **trading blocs, common markets, and economic unions** are: • Free trade. • Economies of scale. • Job creation. • Protection. • Easy access to market and trade creation.
277.	The interest rate parity theory plays a major role in foreign exchange markets.	**TRUE.** The **interest rate parity** states that the difference between the interest rates of two countries is the same as between the spot exchange rate and the forward exchange rate. This theory plays a major role in foreign exchange markets since it connects the dots between the interest rates, the spot exchange rates, and the foreign exchange rates.
278.	Pegged currencies use market forces to dictate their currencies' exchange rates.	**FALSE. Floating exchange rate regimes** use market forces to dictate their currencies' exchange rates, whereas pegged currencies mean exchange rates are fixed to other currencies for a certain period.
279.	Dollarization is mostly the case for small and sometimes unstable countries.	**TRUE. Dollarization** refers to a case in which a foreign currency is used as legal tender in addition to or instead of the domestic currency.
280.	The exchange rates affect countries' international trade and capital flow.	**TRUE.** Most of the **effects of exchange rates** on countries' international trade and capital flow are listed below:

		• Changing prices of currencies. • Increase in demands for imports. • Expensive products. • They are self-correcting over time. • Development of a trade deficit. • Effect on the standard of living.

SECTION B - ECONOMICS

FILL IN THE BLANKS

Key Concepts and Definitions

1. The law of demand states that, ceteris paribus, as the price of a good increases, the quantity demanded _________.

2. _________ is a situation where the demand for a product exceeds its supply at a specific price.

3. A _________ curve shows the relationship between the quantity supplied and the price of a good, ceteris paribus.

4. The _________ effect refers to the change in consumption resulting from a change in the real income of consumers.

5. _________ occurs when the percentage change in quantity demanded is greater than the percentage change in price.

6. When a market reaches a state where quantity demanded equals quantity supplied, it is said to be in _________.

7. A good that can be used in place of another good is called a _________.

8. A _________ good is a product whose demand decreases as income increases.

9. The _________ cost of an item is the value of the next best alternative that is foregone.

10. _________ is the measure of the responsiveness of quantity demanded to a change in price.

11. In the context of international trade, _________ advantage refers to the ability of a country to produce a good at a lower opportunity cost than another country.

12. The _________ rate is the price of one currency in terms of another currency.

13. _________ is the economic principle that asserts that the marginal cost of producing an additional unit of output increases as more of a good is produced.

14. A market structure characterized by a large number of small firms, identical products, and easy entry and exit is known as _________ competition.

15. The _________ multiplier shows the change in equilibrium GDP due to a change in autonomous spending.

16. _________ inflation occurs when demand for goods and services exceeds their supply.

17. The _________ is the total market value of all final goods and services produced within a country in a given period.

18. _________ unemployment refers to unemployment resulting from changes in the structure of the economy.

SECTION B - ECONOMICS

ANSWER KEY

Key Concepts and Definitions

1. decreases
2. Shortage
3. supply
4. income
5. Elastic demand
6. equilibrium
7. substitute
8. inferior
9. opportunity
10. Price elasticity of demand
11. comparative
12. exchange

13. Diminishing returns

14. perfect

15. fiscal

16. Demand-pull

17. Gross Domestic Product (GDP)

18. Structural

19. progressive

20. Fiscal

21. Consumer Price Index (CPI)

22. Trade surplus

23. crowding-out

24. monopoly

25. Phillips

26. Production

SECTION B - ECONOMICS

WORD SEARCH PUZZLES

Section B - Quiz 1

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C P L O A L Z H W Z E R V B P
U U Q E J E Q W O M A R K E T
P O P J C Z A S R K V B X B S
B N X U A C F J U V U G L S E
Q S S P R I C E W P M T S R T
C S U L P R U S T D P R D Q U
D B K Z I O Q L E E I L T U T
A X A E Z N J G X M U X Y A I
A M Y F C A J K S A M D J N T
L O D V L K V E N N F G Q T S
P G U R H L N P D D S G G I B
E Q U I L I B R I U M M W T U
B E L A S T I C I T Y Y U Y S
I S T N E M E L P M O C N M Q
K R B B W E G S H O R T A G E
```

1. Demand **2.** Supply **3.** Equilibrium **4.** Price **5.** Quantity **6.** Elasticity

7. Substitutes **8.** Complements **9.** Market **10.** Shortage **11.** Surplus

SECTION C – FINANCIAL STATEMENT ANALYSIS

STUDY POINTS

S.No	Description
1.	Describe the **Role of Financial Reporting?** Financial reporting provides information on a company's performance, financial position, and changes in its financial position. It usually takes the form of interim financial statements, audited financial statements, management discussion and analysis, and additional disclosures as required by the company's regulators.
2.	Clarify the **Role of Financial Statement Analysis?** In financial statement analysis, a company's financial reports and other related pieces of information, such as financial notes and supplementary schedules, are assessed and used to evaluate the company's performance and financial position to make an investment or other economic decision. Financial statement analysis can be used to determine if a company is profitable, adequately capitalized, able to meet its long and short-term obligations, and able to generate positive cash flows continuously. It also allows analysts to project estimates or expectations of a company's future performance and financial position.
3.	Discuss **Statement of Financial Position?** The **statement of financial position** is also called the balance sheet or statement of financial condition. It gives a snapshot of a company's financial position at a particular time. It discloses: • The resources that a company has control over (its assets). • A company's obligations to its lenders and other creditors (liabilities).

	• The excess of a company's assets over its liabilities (its owner's equity, shareholder's equity, or owner's residual interest in the company's assets after deducting its liabilities).
4.	What is meant by **Accounting Equation**? The accounting equation, otherwise referred to as the balance sheet equation, expresses the relationship among the three components of the statement of financial position in the following form: Assets = Liabilities + Owner's equity The accounting equation must balance – a company's total assets must always equal the sum of its liabilities and owner's equity.
5.	Explain **Statement of Comprehensive Income?** The **statement of comprehensive income** can be presented as a single statement of comprehensive income. Alternatively, it can be given as two statements: an income statement and a statement of comprehensive income. In this instance, the statement commences with the profit or loss figure derived from the income statement and then presents components of other comprehensive income. The income statement is also known as the statement of operations or profit and loss (P&L). It provides information on the financial results of a company's business operations over a specified time. An income statement shows the revenue a company generated, how much expense it incurred, and how much profit or loss it made over time. The net profit or income is also called the bottom line or the income statement. The following **equation** may represent it: Net income = Revenue + Other income– Expenses Comprehensive income includes all items that can impact shareholder's equity but do not result from transactions with shareholders. Some items are included in net income computation, while others are included as other comprehensive income.

6.	Identify **Statement of Changes in Equity?** The **statement of changes in equity** is also referred to as the statement of changes in owners' equity or the statement of changes in shareholders' equity. It provides information on the changes in owners' investments in a company over time. Its main components are paid-in capital, retained earnings, and reserves. It is organized to present a beginning balance, ending balance, and increase or decrease during the reporting period for each component.
7.	Clarify **Statement of Cash Flows?** The **statement of cash flows** or cash flow statement discloses a company's sources and uses of cash. It can assist in evaluating the company's liquidity and solvency position. It classifies all a company's cash flows into operating, investing, or financing cash flows. **Cash flows from operating activities** include cash flows from a company's daily operations. Such cash flows determine net income. **Cash flows from investing activities** arise from acquiring and disposing of long-term assets, such as property and equipment. **Cash flows from financing activities** are derived from obtaining or repaying capital for the company.
8.	Describe the **Importance of Financial Statement Notes and Supplementary Information?** Notes to the financial statements provide essential disclosures such as the basis of preparation, the reporting currency, and the accounting policies. Besides, the notes reveal the methods and estimates that have been used in the preparation of the financial statements. These revelations are critical when comparing two or more companies based on their financial statements. The accounting standards (IFRS and US GAAP) afford companies a certain amount of flexibility when reporting certain transactions or activities. For example, two companies may report the

	expenses related to the purchase of the same item differently. Due to these differences in reporting choices that companies can make, it is often necessary for appropriate adjustments to be made to the figures reported in the financial statements before meaningful comparisons between companies' financial data can be made.
9.	Define **Management Commentary?** A **management commentary** or discussion and analysis report (MD&A) is usually included in public companies' annual reports. In this section, a company's management usually discusses matters of concern to the company, such as the results of its operations, risk strategies employed, planned capital expenditure, and future outlook. The MD&A is a helpful starting point for understanding the financial statements and can also provide key insights into a company's potential future performance.
10.	State the **Objectives of an Audit?** The two primary objectives of an audit are: • To provide reasonable (not absolute) assurance that financial statements are free from material misstatement. This, in essence, enables the independent auditor to express an opinion on whether or not the preparation of financial statements complied with a specified set of accounting standards. • To report on the financial statements following the auditor's findings as required by the International Standards for Auditing.
11.	What is an **Audit Report?** When an independent auditor provides a written opinion on a company's financial statements, it is called an **audit report.** The standard independent audit report usually has several paragraphs. To begin with, the first or "introductory" paragraph

	describes the financial statements and the responsibilities of management and the auditor. The second or "scope" paragraph describes the nature of the audit process and gives the basis for the auditor's expression about reasonable assurance. Lastly, the third or "opinion" paragraph expresses the auditor's opinion on the fairness of the financial statements.
12.	**How many** audit opinions are there? The audit opinion can take any one of the following **four forms**: 1) The **unqualified audit opinion** indicates that the financial statements are fairly presented under the accounting standards. 2) The **qualified audit opinion** indicates that there is some amount of limitation to the audit's scope or an exception to the accounting standards. 3) The **adverse audit opinion** indicates that the independent auditor has determined that the financial statements materially depart from accounting standards and are not fairly presented. 4) The **disclaimer of opinion** indicates that the auditor cannot issue an audit opinion for one reason or another.
13.	Demonstrate the **Importance of Effective Internal Controls?** Effective internal controls help to ensure that a company's process for financial reporting is sound. Having effective internal controls has become increasingly important. Under the Sarbanes-Oxley Act, the securities regulators in the United States now require the management of publicly traded companies to accept responsibility for the effectiveness of internal control processes, evaluate their effectiveness, have supporting evidence for the evaluation, and provide a report on internal controls.

	○ Accumulating cash for acquisitions or other financing decisions will be reflected in the company's balance sheet composition and computed financial ratios.
405.	How to **Forecast Future Net Income and Cash Flow?** Forecasting a company's future net income and cash flow begins with a forecast of its sales. A top-down approach is usually adopted. This involves forecasting industry sales based on their historical relationship with a macroeconomic indicator such as GDP growth. At the individual company level, sales forecasts could be based on past financial results and could be very detailed. Such a forecast could, for example, focus on each company's business segment. Assuming that the industry sales have been forecast, the next step would involve forecasting the company's market share based on its historical market share and a forward-looking assessment of its competitive position. The company's sales are then forecasted by multiplying its projected market share by total industry sales. After-sales forecast, a company's net income and cash flow are usually forecast. This is done based on a projection of profit margins (gross or operating) or expenses and the level of investment in working and fixed capital needed to support the forecast sales. Historically, operating profit margins tend to be less reliable than gross profit margins in projecting future margins for a new or relatively volatile business or one with significant fixed costs. Gross profit margins may also be detailed based on past results or forecast relationships. Forecasting future financial performance over multiple periods is needed in valuation models that estimate a company's or its equity's value by discounting future cash flows.
406.	Discuss the **Role of Financial Statement Analysis in Assessing Credit Quality?** A company's historical and projected financial statements are important sources of information for assessing credit quality. Whereas computation of a company's financial ratios is useful in determining its creditworthiness, the company's relative creditworthiness may be determined by comparing its financial

	ratios with those of its peers. Commonly used financial ratios include EBITDA/Average assets, Debt/EBITDA, Retained cash flow to debt, and Free cash flow to net debt. Credit analysis, or the evaluation of credit risk, also involves the projection of a company's period-by-period cash flows. It uses return measures related to operating cash flow because this represents the cash generated internally and available for the payment of creditors.
407.	State the **Groups of Quantitative Factors Used in Credit Analysis?** Four groups of quantitative factors are used in credit analysis: • **Scale and diversification** relate to a company's sensitivity to adverse events or economic conditions and other factors that may affect its debt-paying ability, such as access to capital markets. • **Tolerance for leverage** relates to a company's ability to service its indebtedness. • **Operational efficiency**: this relates to a company's cost structure. Companies that have lower costs are better positioned to deal with financial stress. • **Margin stability**: this relates to the past volatility of profit margins. Higher margin stability is associated with lower credit risk.
408.	Analyze the **Use of Financial Analysis in Screening Potential Equity Investments?** A simple stock screen may include the following criteria: • Valuation ratio, for example, the P/E ratio, which is less than a specified value. • A solvency ratio, for example, total debt/assets, which does not exceed a specified value. • Positive net income. • The dividend yield is greater than a specified value.

	These criteria are often not independent. Some criteria will check the results obtained from applying other criteria. Additionally, stock screen results can sometimes be concentrated in a subset of the sectors represented in the benchmark. Although they are most common among value investors, stock screens can be used by growth investors, value investors, and market-oriented investors. Growth screens typically feature criteria related to earnings growth and/or momentum. Value screens usually feature criteria that set upper limits for the value of one or more valuation ratios. Market-oriented screens do not strongly emphasize valuation or growth criteria.
409.	How **Back-Testing Can Be Useful?** Back-testing can be useful in evaluating how a portfolio based on a screen would have performed historically. It applies the portfolio selection rules to historical data and computes what returns would have been earned if a particular strategy had been used.
410.	Provide the **Limitations of Back-Testing**? Its relevance to investment success may be limited due to the following limitations: • **Survivorship bias**: if the back-testing database eliminates companies that cease to exist, the remaining companies will appear to have performed better collectively. • **Look-ahead bias**: if restated financial data is included in a database, there will be a mismatch between what investors would have known when they invested and the information used in the back-testing. • **Data-snooping bias** can occur when a model is built based on previous findings, and then the same database is used to test the model.
411.	Clarify the **Analyst Adjustments Related to Investments?** If two comparable companies have significant differences in the classification of investments, analyst adjustments may be useful in facilitating meaningful comparisons.

Management's intention will determine how investments in other companies' debt and equity securities are accounted for. When securities are classified as "financial assets measured at fair value through profit or loss," unrealized gains and losses will be reported in the income statement.

However, when securities are classified as "financial assets measured at fair value through other comprehensive income," unrealized gains and losses are not reported in the income statement. Instead, they are recognized in equity.

412. Elaborate on the **Analyst Adjustments Related to Inventory?**

The inventory method a company opts to use affects the value of inventory reflected on its balance sheet and the value of the cost of goods sold.

Companies that use the LIFO (last-in, first-out) method of inventory must also disclose the value of their inventory under the FIFO (first-in, first-out) method. To recast the inventory values for a company that uses the LIFO method on a FIFO basis, an analyst must add the ending balance of the LIFO reserve to the ending value of inventory under the LIFO method. Also, to adjust the cost of goods sold on a FIFO basis, an analyst must subtract the change in the LIFO reserve from the reported cost of goods sold under LIFO accounting.

413. Provide the **Analyst Adjustments Related to Property, Plant, and Equipment?**

The amount of reported depreciation expense depends on the choice of accounting method and the estimates used in the calculations. Companies can use straight-line, accelerated, or usage methods to depreciate fixed assets other than land. In addition to selecting a depreciation method, companies must estimate the asset's salvage value and useful life to compute depreciation expense.

Depreciation is related to the following accounts:

- The balance sheet accounts for gross property, plant and equipment (PPE), and accumulated depreciation.

	• The income statement amount for depreciation expense. • The statement of cash flows disclosure of capital expenditure (CAPEX) and asset disposals.
414.	Discuss the **Analyst Adjustments Related to Goodwill?** Goodwill comes from the company buying another at a price higher than the fair value of the net identifiable assets of the acquired company. For that matter, net identifiable assets include fixed assets, current assets, and specific intangible assets that need a valuation based on the combination of a business, such as brand and technology. Goodwill, the difference between the purchase price and the fair value of the acquired entity, is usually recorded as an asset. The goodwill is tested annually for impairment. If it is impaired, the acquiring company should reduce the amount of the asset and report a write-off resulting from the impairment. One challenge concerning goodwill is the comparative financial statement analysis. Ratios such as profitability ratios will generally differ among the companies due to differences in the accounting of the assets and income from intangible assets and goodwill.

SECTION C – FINANCIAL STATEMENT ANALYSIS

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	Financial reporting provides information on a company's performance, financial position, and changes in financial position.	**TRUE. Financial reporting** usually includes interim financial statements, audited financial statements, management discussion and analysis, and additional disclosures as required by the company's regulators.
2.	Liabilities refer to the resources a company owns or controls because of past events and from which future economic benefits are expected to flow.	**FALSE. Assets** refer to the resources a company owns or controls because of past events and from which future economic benefits are expected to flow.
3.	Cash and cash equivalents, inventories, trade, and other receivables are examples of current assets.	**TRUE. Current assets** are primarily held for trading or expected to be sold, used up, or otherwise realized in cash within the greater part of a year or one business operating cycle after the reporting period.
4.	Financial statements summarize a company's accounting information.	**TRUE.** The **statement of financial position, comprehensive income, changes in equity, and cash flows** represent a complete set of financial statements. These statements may be audited by independent external auditors who then provide an opinion on whether or not the financial statements give a fair representation of the company's performance and financial position under the accounting standards and principles.
5.	Liabilities refer to anything that the company owns.	**FALSE.** Liabilities refer to anything that the company **owes.** These obligations occur because of past events, and their settlement is expected to result in an outflow of economic benefits.

6.	Trade receivables refer to amounts owed to a business entity by its customers for products and services that have already been delivered.	**TRUE. Trade or accounts receivable** are usually reported at net realizable value, which approximates fair value based on collectability estimates. An allowance for doubtful accounts reflects an entity's estimate of amounts that will ultimately be uncollectible. Additions to this allowance in a particular period are reflected as bad debt expenses.
7.	Inventories are measured at the lower cost and market value under IFRS.	**FALSE.** Inventories are measured at the lower cost and **net realizable value** under IFRS, while under US GAAP, they are measured at the lower cost or market value.
8.	Financial statement analysis allows analysts to project estimates or expectations of a company's future performance and financial position.	**TRUE. Financial statement analysis** can be used to determine if a company is profitable, adequately capitalized, able to meet its long and short-term obligations, and able to generate positive cash flows continuously.
9.	Current assets are not expected to be sold or used up within the greater part of a year or one business operating cycle.	**FALSE. Non-current assets**, also called long-term or long-lived assets, are not expected to be sold or used up within the greater part of a year or one business operating cycle. Examples include property, plant and equipment, goodwill, intangible assets, and financial assets (with long maturities).
10.	Market value refers to the estimated selling price less the estimated completion costs and costs necessary to make the sale.	**FALSE. Net realizable value** refers to the estimated selling price less the estimated completion costs and costs necessary to make the sale.
11.	Other current assets are usually not sufficiently material to require a separate balance sheet line item.	**TRUE. Other current assets** include: a. **Prepaid expenses** are normal operating expenses that have been paid in advance.

		b. **Deferred tax assets** represent income taxes incurred before the income tax expense will be recognized on the income statement.
12.	Current liabilities include trade payables, financial liabilities, accrued expenses, and deferred income.	**TRUE. Current liabilities** are expected to be settled within a year or one business operating cycle after the reporting period.
13.	The statement of comprehensive income is also called a balance sheet.	**FALSE.** The **statement of financial position** is also referred to as the balance sheet or statement of financial condition that gives a snapshot of a company's financial position at a particular point in time and discloses the following: a. The resources that a company has control over. b. A company's obligations to its lenders and other creditors. c. The excess of a company's assets over its liabilities is called owner's equity.
14.	Market value is the current replacement cost, which can exceed the Net Realizable Value.	**FALSE.** Market value is the current replacement cost, which **cannot exceed the NRV and cannot be lower than the NRV, less a normal profit margin**. If the NRV or market value of inventory falls below the carrying amount, the company must write down the value of the inventory and reflect the loss in value in the profit and loss statement.
15.	Long-term financial liabilities and deferred tax liabilities are examples of non-current liabilities.	**TRUE. Non-current** or long-term liabilities refer to all other liabilities, including financial liabilities, which provide financing on a long-term basis.
16.	The income statement provides information on the	**TRUE.** The **income statement**, also known as the statement of operations or profit and loss, shows the amount of revenue a

	to issue low-quality financial reports.	• **Hiding poor performance** - usually arises from the desire to meet or beat market expectations as reflected in analysts' forecasts and/or management's forecasts or concerns over career and incentive compensation. • **Avoiding debt covenant violations** - can make managers inflate earnings and is particularly important for highly leveraged and unprofitable companies.
305.	An acquiring company may not use a target company's tax loss carry-forwards to offset its tax liabilities.	**FALSE.** An acquiring company **may use** a target company's tax loss carry-forwards to offset its tax liabilities.
306.	The decision to issue low-quality financial reports ultimately lies in the hands of individuals.	**TRUE.** The issuance of low-quality financial reports is subject to the following **three conditions**: a. **Opportunity** - may be presented either by internal or external factors. b. **Motivation** can result from pressure on a manager to meet certain criteria for personal gratification or corporate reasons. c. **Rationalization** - when concerned about whether or not a choice is correct, an individual will use rationalization to justify the choice to themselves.
307.	A deferred tax recognized in equity for revaluating property, plant and equipment, and intangible assets under US GAAP.	**FALSE.** A deferred tax recognized in equity for revaluating property, plant and equipment, and intangible assets **under IFRS**. This **does not apply to US GAAP** because revaluation is prohibited.
308.	Financial markets provide a mechanism for disciplining	**TRUE. Other mechanisms** that discipline financial reporting quality include the following:

	financial reporting quality.	a. **Market regulatory authorities** - regulations and regulatory authorities that administer them, establish and enforce rules, play a vital role in regulating financial reporting quality. b. **Auditors** - audit opinions assure the users of financial statements that the financial statements comply with relevant accounting standards and present the company's information objectively. c. **Private contracting** - loan agreements or investment contracts may also contain provisions that allow investors to recover all or part of their investment if certain financial triggers occur.
309.	Deferred tax assets and liabilities will increase if the income tax rate decreases.	**FALSE.** Deferred tax assets and liabilities will **decrease** if the income tax rate decreases; conversely, an increase in the income tax rate will trigger increments of deferred tax assets and liabilities.
310.	Companies may construct and report adjusted EBITDA by including additional items with net income.	**FALSE.** Companies may construct and report adjusted EBITDA by **excluding** additional items with net income.
311.	The FIFO cost assumption provides a more current picture of ending inventory value when prices change.	**TRUE.** When prices change, the **FIFO cost assumption** provides a more current picture of ending inventory value, and the balance sheet amounts will be more relevant to investors.
312.	The SEC prohibits a company from excluding charges or liabilities requiring cash settlement from non-GAAP liquidity measures other than EBIT and EBITDA.	**TRUE.** The Securities and Exchange Commission also **prohibits calculating a non-GAAP performance measure** intended to eliminate or smooth items tagged as non-recurring, infrequent, or unusual when such items are extremely likely to recur.

313.	Expense is the most frequently manipulated financial report item.	**FALSE. Revenue** is the most frequently manipulated financial report item. The following are the ways to examine revenue quality: • Examine the accounting policies note for a company's revenue recognition policies. • Look at revenue relationships. • Compare accounts receivable with revenues over several years. • Examine asset turnover.
314.	A company's management may capitalize on a payment to make the current period's financial position more attractive.	**TRUE.** A company management must determine the category in which a payment will fall when classifying payments. The payment can either **benefit the current period**, in which case it would be an expense, or **benefit future periods** and thereby be classified as a cost to be capitalized as an asset.
315.	The evaluation of a company's historical performance may only be carried out using the company's financial statements.	**FALSE.** The data used in evaluating a company's past financial performance **may come from several sources, including financial statements and corporate press releases, proxies, trade publications, and industry surveys**. The data may be processed by common-sizing the financial statements, computing financial ratios, and reviewing or computing industry-specific metrics.
316.	Forecasting a company's future net income and cash flow often begins with a projection of expenses.	**FALSE.** Forecasting a company's future net income and cash flow often begins with a **top-down sale forecast** in which industry sales and the company's market share are forecast.
317.	Credit analysis involves the projection of a company's period-by-period cash flows.	**TRUE.** Four groups of quantitative factors are used in **credit analysis**: a. **Scale and diversification** - relates to a company's sensitivity to adverse events

		or economic conditions and other factors that may affect its debt-paying ability. b. **Tolerance for leverage** - relates to a company's ability to service its indebtedness. c. **Operational efficiency** - relates to a company's cost structure. d. **Margin stability** - relates to the past volatility of profit margins.
318.	Back-testing can be useful in evaluating how a portfolio based on a screen would have performed historically.	**TRUE. Back-testing** applies the portfolio selection rules to historical data and computes what returns would have been earned if a particular strategy had been used. However, it may not provide a reliable indication of future stock performance because of survivorship bias, look-ahead bias, or data-snooping bias.

SECTION C – FINANCIAL STATEMENT ANALYSIS

FILL IN THE BLANKS

Concepts & Definitions

1. The _______ is a financial statement that provides a snapshot of a company’s financial position at a specific point in time.

2. The _______ reports a company’s revenues, expenses, and profits over a period of time.

3. The _______ details the cash inflows and outflows from operating, investing, and financing activities.

4. The _______ is a supplementary statement that shows changes in a company’s equity during a period.

5. _______ represents the owners' claims on the assets of a company.

6. _______ are obligations that a company owes to external parties.

7. _______ is the cost of goods sold divided by average inventory.

8. The _______ ratio measures the ability of a company to meet its short-term obligations with its most liquid assets.

9. _______ measures the percentage of sales that remains after all expenses have been deducted from revenues.

10. The _______ ratio shows the proportion of debt to equity in a company’s capital structure.

Key Ratios & Formulas

11. The _______ ratio is calculated as Current Assets divided by Current Liabilities.

12. _______ equals Total Liabilities divided by Total Assets.

13. _______ is calculated as (Net Income / Sales) * 100.

14. The _______ is calculated by dividing Gross Profit by Net Sales.

15. The _______ ratio is calculated as Net Income divided by Average Total Assets.

16. _______ equals (Net Income - Preferred Dividends) / Average Common Equity.

17. _______ is the ratio of Operating Income to Net Sales.

18. The _______ is calculated as Operating Cash Flow divided by Current Liabilities.

19. _______ equals (Net Sales - Cost of Goods Sold) / Average Inventory.

20. The _______ ratio measures how efficiently a company uses its assets to generate sales.

Cash Flow & Liquidity

21. _______ activities include cash flows related to the day-to-day operations of a business.

SECTION C – FINANCIAL STATEMENT ANALYSIS

ANSWER KEY

Concepts & Definitions

1. Balance Sheet

2. Income Statement

3. Cash Flow Statement

4. Statement of Changes in Equity

5. Equity

6. Liabilities

7. Inventory Turnover Ratio

8. Quick Ratio

9. Net Profit Margin

10. Debt-to-Equity Ratio

Key Ratios & Formulas

11. Current

12. Debt Ratio

13. Return on Sales (ROS) or Profit Margin

14. Gross Profit Margin

15. Return on Assets (ROA)

16. Return on Equity (ROE)

17. Operating Margin

18. Operating Cash Flow Ratio

19. Inventory Turnover Ratio

20. Asset Turnover Ratio

Cash Flow & Liquidity

21. Operating

22. Investing

23. Financing

24. Free Cash Flow

25. Cash Ratio

SECTION C – FINANCIAL STATEMENT ANALYSIS

WORD SEARCH PUZZLES

Section C - Quiz 1

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I R H D H C U R R E N T R M D
E U D G T Y Q U I C K A X I I
G T S P M I E M B N D R V Z R
A S G X V U F X A W J I C A C
R E N E E Y X O P R D Z X J X
E R I K Q L G V R E G I E K P
V E N E U F T R N P N I U L T
O T R C I R A D O C E S N B W
C N A I T E X S O S Y I E L D
S I E R Y V N M M W S D V S U
A Y Y P G O E M G Q Z Q E W Z
R E T U R N K A S S E T R E S
H M D E P R E C I A T I O N P
H T K V L U T O N T I N I F R
O F N O I T A Z I T R O M A W
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1. Current **2.** Quick **3.** Debt **4.** Return **5.** Equity **6.** Gross **7.** Margin **8.** Asset

9. Turnover **10.** Interest **11.** Coverage **12.** Price **13.** Earnings **14.** Dividend

15. Yield **16.** Revenue **17.** Expenses **18.** Profit **19.** Income **20.** Depreciation

21. Amortization **22.** Tax

SECTION D – CORPORATE ISSUERS

STUDY POINTS

S.No	Description
1.	Who **Initiates a Business**? A business is initiated by a founder(s) with significant know-how or expertise.
2.	What is a **Business Structure**? A **business structure** describes how a business is organized, influencing day-to-day operations.
3.	How a **Business Structure is Determined**? Four factors determine a business structure: • **Legal Identity**: Describes the legal relationship between the owner(s) and the business. • **Business Liability**: Indicates the level of liability an individual assumes due to a business' actions or debts. The levels of liability can be described as **limited** or **unlimited.** • **Owner-Operator Relationship**: Describes the relationship between a business's owner(s) and the management. • **Taxation**: Describes the tax regime applicable to a business structure.
4.	List the **Types of Business Structures?** • Sole proprietorship (or sole trader). • General partnership. • Limited partnership. • Corporation.
5.	Discuss **Sole Proprietorship?**

	In a proprietorship, the owner raises the business capital and fully controls business operations. Moreover, the owner benefits fully from financial returns and assumes all the business risks.
6.	State the **General Features of Sole Trader?** • The owner operates it. • The business is not a legal identity. Hence, it is considered as an extension of the owner. • The owner keeps all the financial returns and bears all risks. • Profits generated by the business are taxed as personal income. • It is simple and flexible to operate. • Financing is solely from the owner. • Business growth depends on the financial ability and risk appetite of the owner.
7.	Explain **General Partnership?** In the **general partnership** business structure, at least two owners (partners) whose roles and responsibilities are stated in the partnership agreement. Unlike a sole proprietorship, there are multiple sources of funds, and the returns and risks are shared.
8.	Mention the **General Features of a General Partnership?** • Partners operate the business. • It has no legal identity since ownership is set in the partnership agreement. • The partners share all business risks and business liabilities. • The partners share business returns, and profits are taxed like personal income. • Capital and expertise are provided to the partners.

	• Business growth depends on the ability of the partners to finance and the level of their risk appetite.
9.	Explore **Limited Partnership?** In a limited partnership, partners are divided into two: • At least one **general partner (GP) has unlimited liability** and is responsible for managing the business. • **Limited partners (LPs)** have, as the name suggests, limited liability. This means their losses are capped at their investment amounts in the limited partnership. The limited partners earn a share profit that is usually lower than that of the general partners. General partners earn more, given their managerial position in the business.
10.	Provide the **General Features of Limited Partnerships?** • The partners provide capital contributions and expertise. • The general partner manages the business and has unlimited liability. • Limited partners do not have business control and have limited liability. • All partners are entitled to a share in returns, where profits are taxed as personal income. • Business growth depends on the financial capabilities of the GPs and LPs, risk appetite, and the competence and integrity of the GP(s).
11.	Describe a **Corporation (or Limited Liability Company)?** A **corporation** is an advanced type of limited partnership in which the business owners have limited liability. Further, it is noteworthy that a corporation has higher access to capital and expertise. This facilitates its quick growth. As such, a corporation is the preferred business structure for larger companies. Indeed, a corporation is a

	dominant business structure globally regarding revenue and asset values. A corporation goes by different names depending on the jurisdiction within which it operates. For instance, it is referred to as a limited company in the UK and a limited liability company (LLC) in the US. In US LLCs, distributions to the owners and corporate profits are taxed.
12.	Clarify the **Types of Corporations?** **Nonprofit** Nonprofit corporations are established for such purposes as the promotion of public health, religious benefits, or charitable missions. They are usually not driven by profits. Nonprofits are the same as for-profits since they, too, have a board of directors and can have paid employees. In contrast to for-profit, nonprofits lack shareholders and do not pay dividends. Moreover, nonprofits do not pay taxes. Nonprofits can generate profits when they are managed well. In this case, all profits are reinvested into the corporation's mission. **Public and Private For-profits** The majority of corporations are formed to make profits for the owners. By engaging in any legal business, for-profit corporations are profit-motivated from inception. For-profit corporations can be **private** or **public**, depending on the number of shareholders (in some countries) and whether the corporation is listed on a stock exchange, among other factors. The shares of a public for-profit are usually listed on a stock exchange. In contrast, private for-profit companies do not trade on a stock exchange. Instead, they run on buyer-seller agreements and transactions.
13.	State the **General Features of a Corporation?** • A corporation is a separate legal entity. • Dividends (distributions) paid to the owners are taxed as personal income.

	• Corporations have unlimited opportunities to access capital, hence unlimited capital potential. • Owner-operator separation allows for higher and more diverse resourcing with significant risk control. • In some countries, corporations are tax disadvantaged due to double taxation of corporate profits. • Business liability is shared among multiple owners with limited liability.
14.	Compare **Public and Private Companies?** A corporation can either be regarded as private or public. The following factors determine this classification: • Issuance of shares. • Exchange listing and share transfer. • Registration and disclosure requirements.
15.	Discuss the **Issuance of Shares** of public and private companies? **Public Companies** Public companies may issue additional shares in the capital markets to raise huge amounts of capital from investors. The investors can then trade among themselves in the secondary market. **Private Companies** In contrast, private companies invite investors to purchase their shares through a private placement memorandum (PPM) (also called an offering memorandum). This document describes a business, the terms of the offering, and the risks involved in investing in the company. Private securities are typically unregulated. As such, only accredited investors may be invited to purchase the shares.

	cash flows and the risk of lending to or owning the company. Companies that use more fixed costs than variable costs in their cost structures will have a greater variation in net income as their revenues fluctuate. The greater the company's leverage, the more risk it has, and the higher the discount rate should be used in its valuation. Highly leveraged companies also have a greater chance of suffering serious losses during a downturn. Consequently, they are more prone to financial distress and bankruptcy.
193.	Explain **Financial Risk?** **Financial risk** is the risk associated with how a company finances its operations, i.e., whether it relies on equity or debt financing. Therefore, financial risk analysis considers the amount of leverage a company has. The higher the company's leverage, the higher the attendant financial risk for its stockholders.
194.	Describe **Business Risk?** **Business risk** is the risk that is inherent in a company's operations or operating earnings. It may also be considered a risk to a company's assets when no debt is used. Business risk reflects a company's sales risk as well as its operating risk. Companies operating in the same line of business will generally have similar risks.
195.	Assess **Sales Risk?** The uncertainty associated with the price and quantity of goods and services is called sales risk. It is affected by the demand for a company's product and the price per unit.
196.	Identify **Operating Risk?** **Operating risk** is the risk attributed to a company's operating cost structure, specifically concerning fixed costs. The greater the level

	of a company's fixed operating costs relative to its variable operating costs, the greater its operating risk.
197.	Explore the **Degree of Operating Leverage (DOL)?** The **degree of operating leverage** (DOL) assists a company in quantifying its operational risk, i.e., the risk arising from its mix of fixed and variable costs. DOL measures how sensitive a company's operating income is to changes in product demand, as measured by unit sales. The following **equation** can express the relationship: DOL = Percentage change in operating income/ Percentage change in units sold. Operating income is, however, equal to the difference between revenue and total operating costs (variable and fixed costs). Considering that fixed costs do not change, operating income will, therefore, change based on the contribution margin, i.e., the product of the quantity sold and the difference between the price per unit and the variable cost per unit. This simplifies the equation to: DOL = Q (P – V)/ Q (P – V) – F Where: Q = The number of units. P = The price per unit. V = The variable operating cost per unit. F = The fixed operating cost. P - V = The per unit contribution margin. Q (P – V) = The contribution margin.
198.	Describe the **Degree of Financial Leverage (DFL)?** The **degree of financial leverage** (DFL) assists a company in quantifying its financial risk, i.e., the risk relating to how it finances its operations.

	DFL refers to the sensitivity of the cash flows available to the owners of a company when operating income changes. The following **equation** can express the relationship: DFL = Percentage change in net income/ Percentage change in operating income. Alternatively, DFL = Q (P − V) − F/Q (P − V) − F − C DFL helps us to understand how changes in a company's operating income translate into changes in net income after interest and tax expenses have been factored in.
199.	Clarify the **Degree of Total Leverage (DTL)?** If we combine a company's degree of operating leverage with its financial leverage, we get the degree of total leverage. The **equation** below can express the relationship: DTL = Percentage change in net income/ Percentage change in the number of units sold. Alternatively, DT L = Q (P − V)/ Q (P − V) − F − C And, DTL = DOL ∗ DFL
200.	Analyze the **Effect of Financial Leverage on Net Income and Return on Equity?** A high degree of financial leverage implies that a company has high-interest payments, which could negatively impact its net income, bottom-line earnings per share, and return on equity (ROE). However, financial leverage increases the variability of a company's net income and its return on equity. This means that a company's net income and return on equity can either increase or

	decrease depending on other factors, such as the macroeconomic environment. An increase in a company's reliance on debt financing increases its risk of default. Besides, it increases the likelihood that the company's operating earnings, net income, and ROE will increase in good economic times. Ultimately, financial leverage increases the risk for a company's shareholders. It is noteworthy that financial leverage can magnify the returns shareholders receive from their shares. A company's net income and its ROE will increase upon attainment of what would be considered optimal financial leverage. However, if a company is financially over-leveraged, net income and return on equity could decrease.
201.	Define the **Breakeven Point or Quantity of Sales**? The **breakeven point or breakeven quantity of sales** refers to the number of units of a company's product produced and sold, at which point the company's net income becomes zero.
202.	How to **Calculate the Breakeven Quantity of Sales?** At the point where a company's net income is zero, its revenues equal its costs. However, a company's costs are made up of variable operating costs, fixed operating costs, and fixed financing costs. With this in mind, if Revenue = Costs, then: $P\,Q = V\,Q + F + C$ Where: P = Price per unit. Q = Number of units produced and sold. V = Variable cost per unit. F = Fixed operating costs. C = Fixed financing costs. Therefore, if QBE is the breakeven quantity of sales, then:

	P QBE = V QBE + F + C And, QBE = F + C/P − V In other words, a company's breakeven sales quantity equals the sum of its fixed operating and financing costs divided by its unit contribution margin or the difference between the price per unit and variable cost per unit.
203.	How to **Calculate Operating Breakeven Quantity of Sales?** At the operating breakeven quantity of sales, Revenues = Operating costs, so mathematically: P QOBE = V QOBE + F Where: P = Price per unit. QOBE = Number of units produced and sold. V = Variable cost per unit. F = Fixed operating costs. Therefore, rearranging the formula, the operating breakeven quantity of sales, QOBE = F/P − V In other words, a company's operating breakeven quantity of sales equals the company's fixed operating costs divided by the difference between the price per unit and variable cost per unit.

SECTION D – CORPORATE ISSUERS

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	Four factors determine a business structure.	**TRUE.** A **business structure** describes how a business is organized, influencing day-to-day operations, and is determined by four factors: a. **Legal identity** - Describes the legal relationship between the owner(s) and the business. b. **Business liability** - Indicates the level of liability (limited or unlimited) an individual assumes due to a business' actions or debts. c. **Owner-operator relationship** - Describes the relationship between a business's owner(s) and the management. d. **Taxation** - Describes the tax regime applicable to a business structure.
2.	A business model outlines how a business is organized to deliver value to its customers.	**TRUE.** A **business model** describes what a business is about, how it runs its operations and generates revenue and profits, and how it differs from its competitors.
3.	The owner raises the business capital and fully controls business operations in a general partnership.	**FALSE.** The owner raises the business capital and fully controls business operations in a **sole proprietorship.** Moreover, the owner benefits fully from financial returns and assumes all the business risks.
4.	The value chain to customers describes the target customers and how a business	**FALSE.** The **value proposition** to customers describes the target customers and how a business retains its customers and outlines the products or services a firm offers and pricing relative to its competitors.

	retains its customers.	
5.	A business model should outline the target customers of a firm.	**TRUE.** Key issues that should be seen in a business model include the geographies, market segments, and **customer segments** that a firm will serve. Concerning customer segments, a business model can be Business-to-Business (B2B) or Business-to-Consumer (B2C) markets.
6.	Profits generated by the business of a sole trader are taxed as personal income.	**TRUE.** The general features of a **sole trader** are: • Simple and flexible to operate. • The owner solely finances and operates it. • The business is not a legal identity; hence, it is considered an extension of the owner. • Business growth depends on the financial ability and risk appetite of the owner.
7.	There is only a single source of funds in a general partnership.	**FALSE.** There are **multiple sources of funds,** and the returns and risks are shared in a general partnership.
8.	A business model should define a firm's products or services offered.	**TRUE.** A business model should outline how the **products and services** differ from the firm's competitors based on the target customers' needs.
9.	Investment risk is higher for bondholders than equity holders.	**FALSE.** Investment risk is **higher for equity holders** since no contractual obligation is set to distribute risk to the shareholders or repay the capital investment. This implies that equity holders may lose entire investments if a company goes bankrupt.
10.	There is at least one partner in a general partnership.	**FALSE.** There are at **least two partners** whose roles and responsibilities are stated in the partnership agreement in the general partnership business structure.
11.	Channels strategy is responsible for	**TRUE.** The **channel strategy** of a firm involves different functions executed from assets that might be utilized. Further, it entails a firm's

	into operational profits after taxes.	**Invested capital** = Common equity + preferred equity + debt.
230.	Common equity is a short-term financing method.	**FALSE.** Common equity is a **long-term external option for financing**. It represents ownership in a company in which the shareholders receive dividends and are entitled to the residual value of its assets if the company goes out of business.
231.	The first approach of incorporating the flotation costs into the cost of capital is a correct and popular method.	**FALSE.** According to the first approach, a company can specify flotation costs as an amount per share or a percentage of the share price. Adjusting the cost of capital for flotation costs using this approach is **incorrect** since it adjusts the present value of the future cash flows of a project by a fixed percentage, which is not necessarily equal to the present value of the flotation costs. **Despite being incorrect, this approach is popular** as it is easier to demonstrate how financing costs change as a company exhausts its internally generated equity and switches to externally generated equity.
232.	Some degree of asymmetric information exists because investors never have as much information as managers.	**TRUE. Asymmetric information** arises from managers having more information about the company's future investment and current performance than outsiders.
233.	A company produces a higher return for investors than the required return if the return on invested capital is lower than the	**FALSE.** A company produces a higher return for investors than the required return if the return on invested capital is **higher** than its cost of capital, enhancing its (the firm's) value for shareholders. The return on invested capital for the issuer will often surpass the cost of capital, creating value for shareholders if management has made investments with a positive NPV and/or an IRR larger than its cost of capital.

	company's cost of capital.	
234.	Unsecured debt lenders are likely to recover more than secured lenders in case of default.	**FALSE. Secured debt lenders** are likely to recover more than unsecured lenders. Likewise, senior debt is likely to recover more than subordinated debt. Consequently, secured and senior debt lenders can tolerate management actions that increase leverage.
235.	The second approach of the incorporation of flotation costs is the recommended approach.	**TRUE.** The second approach is recommended, which adjusts cash flows in the valuation computation. In other words, instead of including flotation costs in the cost of capital, a company **should incorporate them into any valuation analysis** as additional project costs.
236.	Debt and equity capital providers demand higher returns for companies with low asymmetric information.	**FALSE.** Debt and equity capital providers demand higher returns for companies with **high asymmetric information** because they have a greater likelihood of agency costs.
237.	Options are financial derivatives that give buyers the right, but not the obligation, to buy or sell an underlying asset at an agreed-upon price and date.	**TRUE. Real options** are capital allocation options that allow managers the right, but not the obligation, to undertake certain business initiatives in the future. Such initiatives may include postponement, abandonment, or project expansion and can alter the value of today's investment decisions.
238.	When flotation costs are incorporated into the cost of capital, the adjusted cost of capital would be less than if flotation costs	**FALSE.** Adjusting the cost of capital to incorporate flotation costs results in a **larger denominator in the cost of equity** formula, leading to an overall **increase in the cost of equity value**.

	were not included.	
239.	The pecking order theory suggests that managers choose a method of financing starting with the most potential information content.	**FALSE.** The pecking order theory suggests that managers choose a method of financing starting with the **least** potential information content (internal funds) and lower preference for the form with the greatest potential information (debt and equity).
240.	There are four common approaches to evaluating capital projects with real options.	**TRUE.** Following are the **four common approaches** to **evaluating capital projects with real options**: a. Using the discounted cash flow without considering options - A firm goes ahead with an investment if the NPV is positive without considering real options. b. Using discounted cash flow, subtracting the real options' incremental costs and adding the real options value. c. Decision trees and option pricing models.
241.	A lower credit rating of a company's debt signifies lower risk for equity and debt investors.	**FALSE**. A lower credit rating signifies **higher risk** and, as such, higher returns that equity and debt investors demand.
242.	The break-even point is used to know the level of sales necessary to cover all costs.	**TRUE.** The **breakeven point or breakeven quantity of sales** refers to the number of units of a company's product produced and sold, at which point the company's net income becomes zero.

SECTION D – CORPORATE ISSUERS

FILL IN THE BLANKS

Corporate Governance

1. Corporate governance is the system of _____ by which a company is directed and controlled.

2. The primary objective of corporate governance is to _____ the interests of management and shareholders.

3. A board of directors is responsible for _____ the management of the company and ensuring that the company is run in the best interest of the shareholders.

4. The role of independent directors is to provide unbiased _____ of the company's management.

5. The _____ arises when there is a conflict of interest between management and shareholders.

6. An effective corporate governance system should include mechanisms to monitor and control _____ behavior.

7. The _____ is the principal in the principal-agent relationship in corporate governance.

8. A common method to align the interests of management with shareholders is through _____.

9. Dual-class shares allow some shareholders to have _____ than others.

10. _____ allows minority shareholders to have more influence in the election of directors.

Capital Structure

11. Capital structure refers to the mix of a company's _____ used to finance its operations.

12. The _____ states that in a perfect market, the value of a company is unaffected by how it is financed.

13. The _____ suggests that companies balance the benefits of debt, such as tax shields, with the costs, such as bankruptcy risk.

14. The _____ suggests that companies prefer to finance new projects using internal funds first, followed by debt, and finally equity as a last resort.

15. The _____ is the return that lenders demand for providing capital to the company.

16. The _____ is the return that equity investors require on their investment in the firm.

17. The _____ is the average rate of return a company is expected to pay its security holders to finance its assets.

18. The _____ ratio measures the relative proportion of shareholders' equity and debt used to finance a company's assets.

SECTION D – CORPORATE ISSUERS

ANSWER KEY

Corporate Governance

1. rules and practices

2. align

3. overseeing

4. oversight

5. agency problem

6. managerial

7. shareholder

8. stock options

9. more voting rights

10. Cumulative voting

Capital Structure

11. debt and equity

12. Modigliani-Miller theorem

13. trade-off theory

14. pecking order theory

15. cost of debt

16. cost of equity

17. weighted average cost of capital

18. debt-to-equity

19. interest coverage

20. financial leverage

Dividend Policy

21. dividend policy

22. residual dividend

23. stable dividend

24. payout ratio

SECTION D – CORPORATE ISSUERS

WORD SEARCH PUZZLES

Section D - Quiz 1

```
I T R A N S P A R E N C Y C F U I I M E
L N W Y R G E A R I N G O D S C P V A V
G D S R E G K M P X Y S P G R W H Y P N
P P H E T J C O O T M R T T E G D S G K
P N Y I E N N T G X Q I I F D P L K S G
R P M N N G B O A R D G M E L S E J T G
E H Q V T O A A W Q N Q A B O E I I A R
F Q B E I V T T L P R T L K H L Y X B O
E M N S O E E Q M A U R D W E U T D I W
R E O T N R S S F Q T N O U R S C N L T
E Z I M Z N Y R O W E I M Y A C P E I H
N R T E P A C O E I R Y P P H I Y D T A
C I U N U N I T G U T R T A S H E I Y T
E S B T V C L C A B G A T I C T N V I U
N K I J Y E O E R C G Z R C U E X I R M
U A R J F O P R E O Q C S W J Q Z D K V
C C T S E T L I V T R J V C X I E D H R
C X S U Q P H D E B E P M P T U O Y A P
A U I E Y P P V L E J C B D H T L Q L C
W J D Z W S D O C D N V I W G W P J X A
```

1. Leverage **2.** Debt **3.** Equity **4.** Capital **5.** Gearing **6.** WACC **7.** Optimal

8. Ratios **9.** Risk **10.** Return **11.** Dividend **12.** Payout **13.** Retention **14.** Yield

15. Stability **16.** Growth **17.** Distribution **18.** Reinvestment **19.** Preference **20.** Policy

21. Governance **22.** Board **23.** Directors **24.** Shareholders **25.** Ethics

26. Transparency

Section D - Quiz 2

```
N J F T P Y T I U Q E N Q L T L E C U O
V A L U A T I O N B R H N D J Q C J L I
L R N G U R D J B I F L J I Q J Q Y A X
E N R H N E G D T D S R B D O W V U E T
A J A Z B I R E B D H E L O X N T H B F
T N T T T Y C D T E W R L P P X Q A T O
R S O U Y W E N P R C I R E G R E M I S
E R G I R T O C A M U A A W C V H L D R
C W T N T N I R E N U P P V B A P R E E
E V G J I A O D K I I I T I L L Y C R V
I P N L Y T R V I I T F M W T G V A C O
V A I R G Q A G E U N J C E R A E S V E
A Y S I P C D R E R Q G T E R O L H I K
B A A G P P Q M X T R I N N Y P C F S A
L B E I N V L D O O N Y L B F S Y G S T
E L L C N N O I T I S I U Q C A C F U A
S E F I D K P R Y R O T N E V N I K A Z
R S D I V E S T I T U R E F E K D J N Y
M B O N D S Z Z D R P B N P U R G C C Y
M R G G E F F I C I E N C Y R D C W E Y
```

1. Merger **2.** Acquisition **3.** Synergy **4.** Takeover **5.** Valuation **6.** Bidder **7.** Target

8. Premium **9.** Integration **10.** Divestiture **11.** Working **12.** Capital **13.** Liquidity

14. Inventory **15.** Receivables **16.** Payables **17.** Cash **18.** Turnover **19.** Cycle

20. Efficiency **21.** Financing **22.** Debt **23.** Equity **24.** Leasing **25.** Bonds

26. Issuance **27.** Credit **28.** Ratings

Section D - Quiz 3

```
P N O I T A C I F I S R E V I D C Q M L
D A D V A N T A G E A M W F E R Y G C W
I Y U W E X P O S U R E B O W C N N G H
V I T H B G W X C Y J K Y U R K K I M N
E N T I S Z T Z L O C C C W Y A D N I U
R S H P L W F O N I M N A K I O T O E V
S U G T Y I C W W Y Q P E B E Y U I H A
I R N Y W C B T L S C U E V R P J T O Y
F A I J D O N A Y I A N I T L M R I H S
I N R Y F L R E T G Q M E D I O L S G Y
C C U I F E J G I I E U A V A T S O N Y
A E T N O A O C P C F T I R L T I P F T
T S C U N D Y W G M I O A D K O I V X I
I T U K I E I L N L E F R R I E S O E L
O K R S P R N T I K Y F F P T T T N N I
N H T I S S B A G W F O Z E J S Y G I T
U L S R C H Y U D D K I I J W L E D P A
M R E U A I R S E V I T A V I R E D H L
Y O R B G P G Q H Y C T P U R K N A B O
X J N O I T A V O N N I E J L D W I U V
```

1. Strategy **2.** SWOT **3.** Competitive **4.** Advantage **5.** Growth **6.** Diversification

7. Market **8.** Positioning **9.** Innovation **10.** Leadership **11.** Ratios **12.** Liquidity

13. Profitability **14.** Solvency **15.** Efficiency **16.** Risk **17.** Hedging **18.** Derivatives

19. Exposure **20.** Insurance **21.** Diversification **22.** Volatility **23.** Restructuring

24. Bankruptcy **25.** Insolvency **26.** Liquidation **27.** Buyout **28.** Spinoff

SECTION E – EQUITY INVESTMENTS

STUDY POINTS

S.No	Description
1.	Explain the **Main Functions of the Financial System?** **Saving** Both individuals and companies set aside money in the present to have more to spend in the future. Individuals typically save during their working years to withdraw money later on to fund their retirement. Corporations may save money collected from customers to repay suppliers or lenders, purchase new equipment, or acquire other companies. Money can be saved in various investment vehicles, from low-risk treasury bills to higher-risk corporate bonds and stocks. Investors who put money into riskier investments expect to be compensated with higher returns. **Borrowing** In contrast to saving, borrowing involves receiving money in the present that will be repaid in the future. People can borrow money through secured loans, like most car loans and mortgages; for example, the lender can sell the asset posted as collateral if the borrower defaults. On the other hand, student loans or credit card debt is typically unsecured. Since no collateral is to be recovered if the borrower fails to pay, lenders will typically charge a higher interest rate on unsecured loans to compensate for the greater downside risk. In addition, companies often utilize both debt and equity to fund current and future investments. Finally, governments borrow money to finance current spending by issuing bills, notes, and loans repaid with future taxes or revenue from government projects. **Raising Equity Capital** Companies use investment banks to assist in raising equity capital, where investors trade cash for a share of ownership in the company. While equities don't promise investors a fixed payment in the future, investors expect future payments in future dividends or capital gains. In addition, analysts help investors accurately value company shares, while regulatory reporting requirements

and accounting standards aim to ensure that a company's financial statements do not mislead investors.

Managing Risks

Investors use forward contracts, futures contracts, options contracts, insurance contracts, and other derivatives to manage risks. These contracts offset the effect of adverse price movements in assets that a party may need to buy or sell in the future. For example, an airline may enter into a forward contract to buy jet fuel at a certain price on some fixed date, hedging the risk of rising prices. The party on the other side of the trade may be the fuel supplier, hedging the risk of falling prices. While their risks are opposite, both parties achieve their purposes with a single transaction.

Exchanging Assets for Immediate Delivery (Spot Market Trading)

People and companies use the spot market to trade a currency to acquire other currencies or commodities, which will be delivered immediately when the transaction occurs.

Information-Motivated Trading

While investors and information-motivated traders ultimately try to buy low and sell high, information-motivated traders are different in that they expect to earn excess returns from their informational advantage, and the normal returns traditional investors earn for the risk of holding an asset over time. Information-motivated traders believe their information allows them to buy undervalued companies and sell overvalued companies, expecting to profit when the share prices more accurately reflect the intrinsic value of the companies.

Active investment managers participate in information-motivated trading to beat their benchmark or the return earned by "buy and hold" investors taking similar risks. In theory, active managers can gain an information edge over other market participants by hiring skilled professionals and conducting thorough research on potential investments. Investors are also information-motivated traders when they allocate funds with the expectation of earning conditional returns greater than the unconditional returns they would earn in the same asset class.

	Determining Appropriate Rates of Return Since savers are on the opposite end of transactions with borrowers and equity sellers, the rate of return must be set at the point where both parties are satisfied. The cost of moving money through time, or the equilibrium interest rate, is the rate at which aggregate supply for funds through savings equals aggregate demand. Savers won't supply capital if too low of a rate is offered, and borrowers won't demand capital if too high of a rate is offered. To determine the rate of return, the equilibrium interest rate must be adjusted depending on the security's risk characteristics, terms, and liquidity. **Efficient Capital Allocation** Efficient capital allocation allows the market's scarce capital to be allocated to only the most productive investments. A market is efficient when market participants have access to accurate information. When investors are thorough in their analysis of available information, they improve the efficiency of the market by simply acting in their own best interests. For example, well-informed investors will not make a loan to someone with poor credit without being appropriately compensated with a higher rate of return. They will not invest in projects unless the value of future cash flows exceeds the cost.
2.	Describe **Classifications of Assets?** • **Securities:** includes both debt and equity securities. Securities may be further classified as public or private securities, depending on if they are traded on a public exchange. • **Currencies:** monies issued by national monetary authorities. • **Contracts:** agreements to trade other assets in the future. • **Commodities:** precious metals, energy products, industrial metals, and agricultural products. • **Real Assets:** all tangible properties, such as real estate, airplanes, airports, machinery, timberland, and pipelines.

3.	Identify **Classifications of Markets?** • **Futures Markets/Forward Markets/Options Markets:** traded contracts require delivery at some date in the future. Contracts in the options markets are only delivered if the option is exercised. • **Spot Markets:** traded contracts require immediate delivery. • **Primary Market:** funds flow from the purchaser to the issuer. • **Secondary Market:** funds flow between traders. • **Money Markets:** trades debt instruments maturing in one year or less. • **Capital Markets:** trades instruments maturing over a year, such as bonds and equities. Corporations usually finance their operations through the capital markets. • **Traditional Investment Markets:** transactions involve only direct or indirect investments in publicly traded debts and equities. • **Alternative Investment Markets:** includes private markets investments, which are more difficult to trade and value. To compensate for the limited liquidity, investors in these markets expect to earn a greater risk-adjusted return than they would in traditional investments.
4.	Elaborate on **Fixed Income Investments**? **Fixed-income investments** include promises to repay borrowed money and a variety of other instruments with payment schedules. People, companies, and governments create fixed-income instruments when they borrow money. While there is no consensus definition on the exact cut-offs, fixed-income securities are often classified based on maturity date as short-term (less than one or two years), intermediate-term (two to five years), and long-term (greater than five years).
5.	What is **Included in Fixed Income Investments**?

	• **Notes:** fixed income instruments, usually with a maturity of ten years or less. • **Bonds:** fixed-income instruments, usually with more than ten years of maturity. • **Convertible bonds:** can be converted into the issuing corporation's stock by the holder after a specified amount of time. • **Bills/Securities of Deposit/Commercial Paper:** short-term securities, usually maturing in a year or less, issued by governments, banks, and corporations. • **Repurchase Agreements:** short-term lending instruments in which the borrower sells an instrument and promises to buy it back at a higher price. • **Money Market Instruments:** debt instruments maturing in one year or less, purchased by money market funds and corporations seeking a return on short-term cash balances.
6.	State the **Elements of Equities?** Equities represent ownership rights in companies and include: • **Common Stock:** shareholders have a right to company dividends if the board of directors declares a dividend, elects the company's board of directors, and a share of the proceeds if the company is liquidated. • **Preferred Stock:** shareholders have no voting rights but generally have the right to a regular dividend and have priority over common shareholders to liquidation proceeds. Cumulative preferred equity is preferred shares requiring a company to repay any preferred dividends before dividends are paid to common shareholders. • **Warrants** are securities issued by a corporation that allows holders to buy a company's securities (usually common stock) at the exercise price before the warrant expires.
7.	Discuss **Pooled Investments?**

127.	For which **Companies** the constant growth or a multistage dividend discount model is **Appropriate?** The Gordon (constant) growth dividend discount model is particularly useful for valuing the equity of dividend-paying companies insensitive to the business cycle and in a mature growth phase. On the other hand, multistage models are often used to model rapidly growing companies. The multistage DDM can be extended beyond two stages to however many stages are deemed appropriate. For instance, the valuation of a fairly young company may benefit from a three-stage DDM.
128.	Explain the **Rationale for Using Price Multiples to Value Equity?** The use of price multipliers to earnings, book value, and sales have all shown to have significant predictive value in determining relative future returns, implying that price multiples can be an effective tool for the valuation of companies. In addition, calculating the "justified value" (the value justified by fundamentals or a set of cash flow predictions) of certain multiples offers an alternative way of estimating intrinsic value.
129.	Assess the **Justified Price/Earnings Multiple?** Assuming a constant rate of growth, the justified forward price-to-earnings ratio can be found using the following equation: $\frac{P0}{E1} = \text{Justified forward P/E}$ $\frac{P0}{E1} = \frac{p}{r - g}$ Where: p = payout ratio r = required rate of return g = expected growth rate of dividends.

	The justified forward P/E is inversely related to the required rate of return and positively related to the growth rate. However, this relationship may not be true because a higher payout ratio may imply a slower growth rate due to the company retaining a lower proportion of earnings for reinvestment. These estimates may be highly sensitive to small changes in assumptions, so it may be useful to conduct a sensitivity analysis.
130.	Discuss the **Method of Comparables?** The economic rationale underlying the comparables method is the law of one price: identical assets should sell for the same price. Thus, if an appropriate benchmark multiplier representative of a peer group or industry can be set, an analyst can determine the current relative value of a given company. However, it is not always easy to determine comparable companies or industries due to other business lines and differing company sizes. For instance, it would be relatively hard to find a comparable company to Apple – one that sells over 200 million smartphones per year and millions of computers and tablets throughout the world.
131.	Define **Enterprise Value?** **Enterprise value** (EV), often viewed as the cost of a takeover, is most frequently determined as market capitalization plus the market value of preferred stock plus the market value of debt minus cash equivalents and short-term investments.
132.	Explore **Enterprise Value Multiples and their Use in Estimating Equity Value?** **EV/EBITDA** EBITDA (earnings before interest, taxes, depreciation, and amortization) can be viewed as a source of funds to pay off the financial stakeholders in the company (lenders, shareholders, the government, etc.). EV/EBITDA is arguably the most common EV multiple. The EV/EBITDA ratio for S&P 500 companies has averaged 13 over the

	past few years. As a general guideline, analysts commonly interpret an EV/EBITDA value below 10 as healthy. Since EBITDA is usually positive even when net income is negative, EV/EBITDA can be calculated when a price-to-earnings (P/E) multiple may not be available. **EV/Operating income** EV/Operating income can also be an alternative to EV/EBITDA. Analysts may have difficulty finding the market value of a company's debt, which may have to be estimated based on comparable bond values.
133.	Mention the **Advantages and Disadvantages of Multiplier Models?** **Advantage** It can consider comparable companies and may prove particularly useful in valuing companies with negative earnings. Limits the amount of projection necessary and ties the valuation into historical data. Price multiples have shown to be fairly good predictors of future performance. **Disadvantage** A comparable company analysis may be skewed by industry or peer group mispricing.
134.	Clarify **Asset-Based Valuation Models and their use in estimating equity value?** An **asset-based valuation** of a company uses estimates of the market or fair value of the company's assets and liabilities and, thus, is most appropriate for companies with a high proportion of current assets and current liabilities and few/insignificant intangible assets. Asset-based valuations are frequently combined with multiplier models to value private companies or supplement public companies' valuation. Not all companies own assets for which the fair value can be easily determined, and market values can differ significantly from

	carrying values. Furthermore, asset valuations may be of limited use in a hyper-inflationary environment.
135.	State the **Advantages and Disadvantages of Asset-Based Valuation Models?** **Advantage** Simple calculation and no projections necessary, particularly useful in valuing firms that are heavy in current assets and light in intangibles. It is also useful in supplementing other valuation methods. **Disadvantage** It is not usually the best stand-alone option for valuing firms as going concerns. The model doesn't consider current or expected cash flows, earnings, or growth rates.

SECTION E – EQUITY INVESTMENTS

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	Securities include both debt and equity securities.	**TRUE. Securities** may be classified as public or private securities, depending on if they are traded on a public exchange.
2.	Contracts are the monies issued by national monetary authorities.	**FALSE. Currencies** are the monies issued by national monetary authorities, whereas contracts are the agreements to trade other assets in the future.
3.	Corporations usually finance their operations through the capital markets.	**TRUE. Capital markets** trade instruments maturing over a year, such as bonds and equities.
4.	Traded contracts in the future market require immediate delivery.	**FALSE.** Traded contracts in the **spot markets** require immediate delivery.
5.	An investor having a short position means that the investor owns the asset or has borrowed money to purchase the asset.	**FALSE.** A position in an asset describes how much of the asset an investor owns. An investor having a **long position** means that the investor owns the asset or has borrowed money to purchase it, or having a short position means that the investor sold the asset without owning it.
6.	Block brokers, investment banks, exchanges, and dealers are some examples of financial intermediaries.	**TRUE. Financial intermediaries** help entities achieve their goals by providing products and services that help connect buyers and sellers. The key financial intermediaries are brokers, alternative trading systems, broker-dealers, depository institutions, insurance companies, arbitrageurs, and clearing houses.
7.	Funds flow from the purchaser to the issuer in the secondary market.	**FALSE.** Funds flow from the purchaser to the issuer in the **primary market**, whereas, funds flow between traders in the secondary market.

8.	People, companies, and governments create fixed-income instruments when they borrow money.	**TRUE. Fixed-income investments** include promises to repay borrowed money and a variety of other instruments with payment schedules.
9.	Reserve currencies include the primary and secondary reserve currencies.	**TRUE. Reserve currencies** are held by banks and other monetary authorities in large quantities. a. **Primary reserve currencies** include the US dollar and the euro. b. **Secondary reserve currencies** include the British pound, the Japanese yen, and the Swiss franc.
10.	Transactions involve only direct or indirect investments in publicly traded debts and equities in alternative investment markets.	**FALSE.** Transactions involve only direct or indirect investments in publicly traded debts and equities in **traditional investment markets.**
11.	Block brokers provide brokerage services to large traders.	**TRUE. Block brokers** serve to manage large orders so that their clients lose the least amount of money due to adverse movements in the market. Large orders typically cause the market to move against the trader: large buy orders trade at a premium, and large sell orders trade at a discount.
12.	Notes and bonds are the fixed income instruments, usually with a maturity of ten years or more.	**FALSE.** Notes are the fixed income instruments, usually with a maturity of ten years or **less,** while the bonds usually have a maturity of more than ten years.
13.	An investor generally has a long position in a contract if they are liable for the	**FALSE.** An investor generally has a long position in a contract if they **take physical delivery of the underlying asset or its cash equivalent** and a short position if they are liable for delivery of the asset.

	delivery of the asset.	
14.	Investors expect to earn a greater risk-adjusted return on the alternative investment market.	**TRUE. Alternative investment markets** include private markets investments, which are more difficult to trade and value. To compensate for the limited liquidity, investors in these markets expect to earn a greater risk-adjusted return than they would in traditional investments.
15.	Governments, banks, and corporations issue bills/securities of deposit/commercial paper.	**TRUE. Bills/securities of deposit/commercial paper** are short-term securities issued by governments, banks, and corporations, usually maturing in a year or less.
16.	Equities represent ownership rights in companies.	**TRUE.** Equities represent ownership rights in companies and include: a. **Common stock** - shareholders have a right to company dividends, elect the company's board of directors, and a share of the proceeds if the company is liquidated. b. **Preferred stock** - shareholders have no voting rights but generally have the right to a regular dividend and have priority over common shareholders to liquidation proceeds. c. **Warrants** are securities issued by a corporation that allows holders to buy a company's securities at the exercise price before the warrant expires.
17.	Repurchase agreements are short-term lending instruments.	**TRUE. Repurchase agreements** are short-term lending instruments in which the borrower sells an instrument and promises to buy it back at a higher price.
18.	Validity is defined as how to fill the order.	**FALSE. Execution** is defined as how to fill the order, whereas, validity is defined as when the order may be filled.

19.	Pooled investments are made up of open-ended and closed-ended funds.	**TRUE. Pooled investments** represent indirect ownership of assets held by an entity by purchasing shares, units, depository receipts, or limited partnership interests. They are typically used to gain access to skilled investment management and/or to efficiently diversify an investor's portfolio.
20.	Money market instruments are equity instruments.	**FALSE.** Money market instruments are **debt instruments** maturing in one year or less, purchased by money market funds and corporations seeking a return on short-term cash balances.
21.	Investment banks assist their clients with potential takeover targets.	**TRUE. Investment banks** help corporate clients issue common stock, preferred stock, notes, and bonds.
22.	The pooled investment includes mutual funds, exchange-traded funds, asset-backed securities, and hedge funds.	**TRUE.** The pooled investment includes: • **Mutual funds** - open-ended and closed-ended investment vehicles that pool money from many investors for investment in a portfolio of securities. • **Exchange-traded funds** - open-ended funds that investors can trade in the secondary markets. • **Asset-backed securities** - securities whose values and income payments are derived from a pool of assets. • **Hedge funds** - characteristics include using leverage to boost fund returns and a fee structure that charges a performance fee when positive returns are achieved in addition to the standard management fee.
23.	Direct investments in real assets are usually quite costly.	**TRUE. Real assets** are investments in tangible properties that the investors find attractive due to their potential income and tax benefits and low correlation to other asset classes. However, direct investments in real assets are usually quite costly as

		investors must either maintain the property themselves or hire a manager to do it for them.
24.	The limit order price must be no higher than a specified amount for sell orders and no lower than a specified amount for buy orders.	**FALSE.** The limit order price must be no higher than a specified amount for **buy orders** and no lower than a specified amount for **sell orders**.
25.	An investor can take a short position by borrowing shares of a stock from a securities lender and then selling them.	**TRUE. Securities lenders** require short sellers to post the proceeds from the stock sale as collateral for the loan. The proceeds are then invested in short-term securities, with interest being returned to the short seller at the rebate rate.
26.	Closed-ended funds issue new shares and redeem existing shares at the fund's net asset value, and investors are typically able to trade their shares directly with the fund.	**FALSE. Open-ended funds** issue new shares and redeem existing shares at the fund's net asset value (usually daily). Investors are typically able to trade their shares directly with the fund.
27.	Broker-dealers describe an entity that is both a broker and a dealer.	**TRUE. Broker-dealers** have an inherent conflict of interest in that a broker aims to acquire the best price for their clients, but a dealer maximizes their profit by buying from their clients at low prices and selling to their clients at high prices.
28.	There are five types of contracts.	**TRUE. Contracts** are agreements to trade other assets in the future, many of which are derivatives. Types of contracts include: a. Forward contracts. b. Futures contracts. c. Swap contracts.

164.	A holder-of-record is a person who currently holds the security.	**FALSE.** A holder-of-record is the name of the person who is **the registered security owner and** has the rights to the dividend.
165.	An industry has relatively low competition in the growth stage of life-cycle.	**TRUE.** The **growth stage** is characterized by rapidly increasing demand, improving profitability, falling prices, and relatively low competition (though a threat of new competitors is generally at its highest point in this stage).
166.	Overcapacity gives participants more pricing power.	**FALSE. Tight or limited** capacity gives participants more pricing power, whereas overcapacity leads to price-cutting and a competitive environment as excess supply chases demand.
167.	The repurchased shares do participate in subsequent voting or dividend issues.	**FALSE.** In a share repurchase, the company uses cash to buy back its shares. Once repurchased, the shares **do not participate** in subsequent voting or dividend issues and **are not considered** when computing the earnings per share.
168.	Investors who acquire shares on or after the ex-dividend date will also benefit from receiving the dividend.	**FALSE.** The ex-dividend date is the day a share will trade without its dividend. As a result, investors who owned shares before the ex-dividend date will receive a dividend once it is paid. In contrast, investors who acquire shares on or after the ex-dividend date **will not have the benefit** of receiving the dividend.
169.	Fragmented industries tend to be highly competitive.	**TRUE. Fragmented industries** tend to be highly competitive because: a. There are too many competitors to keep tabs on each member. b. Small gains in the market share of an industry participant may dramatically impact its profit. c. The fragmented players think individualistically instead of as members of a larger group.

170.	The mature stage of the industry life cycle is characterized by slowing growth and intense competition.	**FALSE.** The **shakeout stage** is characterized by slowing growth, intense competition, declining profitability, and a focus on cost reduction.
171.	The record date is typically one or two business days after the ex-dividend date.	**TRUE.** The **holder-of-record date**, also called the record date, is the business day on which a shareholder listed in the company's records is deemed to have ownership of the company's shares to decide who can and cannot receive a dividend when paid.
172.	The mature stage of the industry life cycle is characterized by negative growth.	**FALSE.** The mature stage of the industry life cycle is characterized **by little or no growth**, industry consolidation, and high barriers to entry. In contrast, the decline stage is characterized by negative growth, excess capacity, and intense competition.
173.	The dividend payment date does not have to be a business day.	**TRUE.** The **payment date**, or payable date, is when a company mails or transfers dividend payments to its shareholders on record. The payment date does not have to be a business day; it can occur on a weekend or holiday.
174.	Stable market shares indicate highly competitive industries.	**FALSE.** Stable market shares typically indicate **less competitive industries**, while unstable market shares often indicate highly competitive industries with limited pricing power.
175.	The Gordon growth dividend discount model assumes dividends grow indefinitely at a rising rate.	**FALSE.** The Gordon growth dividend discount model assumes dividends grow indefinitely at a **constant rate**.
176.	The multi-stage dividend discount	**TRUE.** The **Multi-Stage Dividend Discount Model** is a bit more complicated as it

	model is more complicated than the Gordon model.	involves using both a short-term and a long-term growth rate to estimate a company's current value. The two-stage DDM assumes that the company will pay dividends that grow at a constant rate at some point, but dividends are currently growing at an elevated and unsustainable rate.
177.	A decrease in the required rate of return will cause an increase in the justified forward Price/Earnings multiple.	**TRUE.** Due to the inverse relationship between the required rate of return and the justified P/E, **a decrease in the required return will justify a higher forward P/E.** This should make sense intuitively since investors are willing to pay a higher price for assets as they relax their return requirements.
178.	An asset-based valuation of a company is most appropriate for companies with a high proportion of intangible assets.	**FALSE.** An asset-based valuation of a company uses estimates of the market or fair value of the company's assets and liabilities and, thus, is most appropriate for companies with a **high proportion of current assets and current liabilities** and few/insignificant intangible assets.
179.	P/E, P/B, P/S, and P/CF are the price multiples that security analysts use.	**TRUE. Price multiples** that security analysts use include the following: • Price-to-earnings ratio (P/E). • Price-to-book ratio (P/B). • Price-to-sales ratio (P/S). • Price-to-cash-flow ratio (P/CF).
180.	A multiplier model would be the most helpful in valuing a startup firm.	**TRUE.** Multiplier models are particularly **useful in valuing companies with negative earnings**, which is often the case for startup companies. Also, there should be many comparable startups raising capital in the same geographical location, which makes the valuation easier.
181.	A low Price-to-Sales ratio is the least useful multiple for	**FALSE.** A low P/S multiple is the **most useful** multiple for predicting future returns.

	predicting future returns.	
182.	FCFE model aims to calculate a company's capacity to pay future dividends.	**TRUE.** The **free Cash Flow to Equity (FCFE) model** aims to calculate a company's capacity to pay future dividends, going beyond simply discounting expected dividends. This approach may provide a more useful valuation, especially when the company does not pay sporadic dividends.

SECTION E – EQUITY INVESTMENTS

FILL IN THE BLANKS

1. The primary purpose of the equity market is to provide _________ for companies.

2. Common stockholders have a _________ claim on the company's assets and income.

3. Preferred stock typically pays a _________ dividend, which is not guaranteed.

4. The equity market is also referred to as the _________ market.

5. _________ represents an ownership interest in a company.

6. The two main types of equity are _________ and preferred stock.

7. The price-to-earnings ratio (P/E) is calculated as the market price per share divided by _________ per share.

8. A stock split increases the number of shares outstanding while decreasing the _________ per share.

9. The _________ ratio measures the proportion of earnings paid out as dividends.

10. A company with a high P/E ratio is generally expected to have _________ growth.

11. The _________ model is used to estimate the value of a stock based on the present value of future dividends.

12. _________ is the risk that stock prices will decline.

13. The _________ represents the total market value of a company's outstanding shares.

14. The beta coefficient measures a stock's _________ relative to the overall market.

15. _________ is the cost of equity for a company.

16. The _________ model is a commonly used method to calculate the required rate of return on equity.

17. The intrinsic value of a stock is the present value of its expected future _________.

18. A _________ stock is expected to provide returns primarily in the form of capital gains.

19. A _________ stock is expected to provide returns primarily in the form of dividends.

20. The _________ is a measure of how much a company's stock price is expected to fluctuate in comparison to the market.

21. The P/E ratio is considered a _________ ratio.

SECTION E – EQUITY INVESTMENTS

ANSWER KEY

1. capital

2. residual

3. fixed

4. stock

5. Equity

6. common stock

7. earnings

8. price

9. dividend payout

10. higher

11. dividend discount

12. Market risk

13. market capitalization

14. volatility

15. Required return

16. Capital Asset Pricing Model (CAPM)

17. cash flows

18. growth

19. income

20. beta

21. valuation

22. speculative

23. constant

24. limited

25. price-to-book

26. CAPM

27. unsystematic risk

SECTION E – EQUITY INVESTMENTS

WORD SEARCH PUZZLES

Section E - Quiz 1

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X V P K J E B O F S K S A L P H A Y V M
G O X H L Q Y E S Y S T E M A T I C U A
O Y L N O E U M T M N Y H U I G P N H R
B F H K A B L Y V A I O R Q D D S D T K
B B X K B W T R F N Z M I K K Y P Y W E
L C L N B I L T T F N I P T S M J H O T
Z V H Q U T Q R E A E V V T A C J O R I
M B L Q E B I O J D K B E Y K U Q T G B
A Y E K H N R S I J Z M M P U U L K D D
F K R U S O O V B B A J A K Y A I A L I
U A B I U K I Q G T H G X Q W L U O V S
M S C R Y D N O I T A Z I L A T I P A C
A U T Y E X T C J P J B I M N D M E M O
I N L N T P B K R W U D D Q D W M T V U
C D D T Y U E F Z R H E E Y U J P A O N
E K N O I T A C I F I S R E V I D R O T
R Y V D J P P C K G F H L W N W A S A E
Q G E P H B L P R E M I U M W G M M T D
O D R F O M U E O B N Y H H J D T H T B
N W R Y D J H J S V L D T E D V F T C L
```

1. Dividend **2.** Growth **3.** Market **4.** Capitalization **5.** Beta **6.** Alpha **7.** Equity

8. Valuation **9.** Intrinsic **10.** Discounted **11.** Multiples **12.** Market **13.** Premium

14. Rate **15.** Systematic **16.** Unsystematic **17.** Diversification

SECTION F – FIXED INCOME

STUDY POINTS

S.No	Description
1.	Define a **Fixed-Income Security?** A **fixed-income security** is a financial obligation that pays a fixed amount of interest—in the form of coupon payments—to investors at specified points in the future. The payments are anchored on contractual guidelines and must be made. This is, in fact, the main distinguishing feature between fixed-income securities and stocks because payment of dividends is at the discretion of directors. Shareholders have no right to compel directors to pay dividends. In the field of finance, the terms "fixed-income securities," "debt securities," and "bonds" are often used interchangeably.
2.	Who is the **Issuer of Bonds?** Bonds can be issued by a host of institutions, including: • Supranational organizations (i.e., European Union, World Trade Organization, etc.). • Sovereign governments (Countries). • Non-sovereign governments (States, Provinces, and Municipalities). • Quasi-government entities (i.e., government-sponsored enterprises, agency-related nonprofit organizations). • Companies (i.e., Apple). • Special legal entities.
3.	List the **Sectors of the bond market**? The bond market is generally broken up into **three sectors:** • Government bonds.

	• Corporate bonds. • Structured finance (a sector of finance that was created to help transfer risk using complex legal and corporate entities).
4.	Contrast between **Investment and Non-investment Grade Bonds**? **Investment-grade bonds** are rated above BBB or Baa3 and have a relatively low risk of default. **Non-investment-grade bonds** are considered low quality to reflect a relatively higher probability of default.
5.	What is meant by **Maturity?** **Maturity** refers to the date when the issuer is obligated to redeem the bonds, while **tenor** is the time left before maturity. Tenor is important as it indicates the time the investor expects to receive payments and the amount of time left before the bond is redeemed. Almost all bonds have a maturity of 30 years or less. Those with less than a year of maturity are called money market instruments. They include treasury bills, commercial paper, and bankers' acceptances. Bonds with more than a year of maturity are called capital market instruments, e.g., debentures.
6.	Define **Par Value?** **Par value** is the amount that will be repaid at maturity. It is also called the principal or principal value. In this regard, it is important to note the following: • If the market price exceeds the par value, the bond trades at a **premium**. • If the market price is less than the par value, the bond trades at a **discount**. • If the market price equals the par value, the bond trades at **par**.
7.	Elaborate on **Coupon Rate and Frequency?**

	The coupon rate is the interest rate that the bondholder will receive. Interest rates can either be fixed over the life of the bond or floating, and the payment frequency can differ depending on the issuer. Annual, semi-annual, and monthly payments are common. Plain vanilla bonds pay a fixed coupon rate, while floating-rate bonds pay a floating interest rate. The floating rate is tied to a reference rate such as LIBOR (London Interbank Offered Rate).
8.	How **Yield Can Be Measured**? The yield is different from the coupon rate and can be measured in **two ways**: • The current yield is the annual coupon payment divided by the bond price. • Yield to maturity is the internal rate of return (IRR) that equates the present value (PV) of a bond's expected cash flows to its price.
9.	Do **Zero-coupon Bonds pay Interest**? Zero-coupon bonds **do not** pay interest but are issued at a discount to par value and then redeemed at par.
10.	Define **Eurobond?** An international bond denominated in a currency that does not belong to the country to which it is issued is called a **Eurobond**. However, it is important to note that Eurobonds are not always issued in Euros.
11.	Explain **Dual Currency Bonds?** **Dual currency bonds** pay interest in one currency, the US dollar, and principal in another, say the Euro.
12.	Describe a **Bond Indenture?** A **bond indenture** is a legal document that outlines all the parameters of the bond issue, such as the par amount, issuer, coupon rate, security pledge, and the rights of bondholders. When

	analyzing a bond, it is important to review the credit risk of the issuer – the entity legally obliged to repay the bondholders. The indenture will also describe the security ranking of the bonds, which can be secured or unsecured. Secured bonds grant bondholders a claim on a specific asset, while unsecured or debentures offer no collateral and are repaid after secured bonds.
13.	What are **Collaterals?** **Collaterals** are assets or guarantees that a lender accepts as security for a bond above and beyond the issuer's promise to pay. Collateral backing increases the credit quality of a bond and is one of the factors considered when determining the payable interest rate. Secured bonds or debts rank higher than unsecured debts in the event of a winding up. Some collateral items include physical equipment, mortgages, and stocks.
14.	Elaborate on the **Legal Identity of the Bond Issuer and its Legal Form?** It's important to identify the issuer of a bond by its legal name. The investor must understand who the issuer is. The issuer can be: • A subsidiary of a parent legal entity. • A holding company; or • A special purpose vehicle (SPV) with a separate legal entity set up to issue bonds. Such bonds are secured by assets hived off the entity's sponsor (i.e., removed from the sponsor's balance sheet). Thus, bondholders/trustees cannot go after the sponsor's other assets in the event of default. The SPV is also '**bankruptcy remote**' because it is immune to any financial strain suffered by its sponsor. Securities issued by the SPV are known as asset-backed securities.
15.	What are **Covered Bonds?** **Covered bonds** are securities issued by a bank or mortgage institution and collateralized against a pool of assets. Unlike asset-backed securities, covered bonds offer more protection to

- Committed bank lines.
- Debt coming due and committed capital expenditures.

197. Evaluate the **Factors that affects the Yield Spreads on all corporate bonds**?

Yield spreads on all corporate bonds can be affected by several factors, including:

- **Credit cycle:** as the credit cycle (expansion and contraction of access to credit over time) improves, the credit spread will be narrow. On the other hand, a weakening credit cycle will widen the credit spreads.
- **Broader economic conditions:** weakening economic conditions will make the investors demand a greater risk premium and hence a wider credit spread. On the opposite side, good economic conditions will tighten the credit spread because investors expect an improvement of credit measures which lowers default risk.
- **Availability of funding in the financial sector:** Since bonds primarily trade over the counter, investors require broker-dealers ' willingness to provide sufficient capital for market-making. When there is financial and regulatory stress, it may cause a reduction in capital available for market-making and willingness to buy or sell bonds with credit risk. Low funding availability will cause wider spreads.
- **Financial performance of the issuer:** corporate bond spreads are affected by the issuer's development news, such as good earnings news. Such affects the investor's view on its ability to service and repay debt. Favorable news increases the demand for the issuer's bonds, narrowing the credit spread. Bad news intuitively widens the credit spread.
- **General market supply and demand:** in periods of heavy new issue supply, credit spreads widen if there is insufficient demand for these new corporate bonds. When there is a high demand for bonds, credit spreads narrow.

198. What is **Included in High-Yield Analysis?**

	A high-yield analysis is typically more in-depth than an investment-grade analysis and has special considerations, including the following: • Greater focus on liquidity and cash flow. • Detailed financial projections. • Detailed analysis of debt structure. • Corporate Structure: parent vs. subsidiaries, cash movement from subsidiary to parent ("upstream") or vice versa ("downstream"). • Covenant analysis: Change of control put (right to require the issuer to buy back the debt), restricted payments, limitations on liens and additional indebtedness, restriction on subsidiaries, maintenance covenants. • Equity-like approach.
199.	Elaborate on **Sovereign Debt?** Government bonds in developed countries have traditionally been viewed as the default risk-free rate, after which all other bonds are priced. The most important considerations for evaluating debt in sovereign countries, according to S&P's methodology, are: • Institutional effectiveness and political risk. • Economic structure and growth prospects: income per capita, trend growth prospects, sources and stability of growth, size of the public sector relative to the private sector. • Flexibility and performance profile. • Monetary flexibility.
200.	Clarify **Non-Sovereign Government Debt?** Sub-sovereign or local governments and quasi-government entities such as states and cities that are created by central

	governments issue bonds as well. For example, the City of London has outstanding debt. These are often referred to as municipal bonds.
201.	Identify **General Obligation (GO) Bonds?** Majority of local government bonds are general obligation bonds. These are unsecured securities backed by the full faith of the issuing non-sovereign government. GO bonds cannot use monetary policy the way many sovereign governments can. Therefore, these securities carry another layer of risk. Analysts should pay particular attention to the volatility and variability of revenues these municipalities generate. Also, these municipalities or states might have unfunded pension and post-retirement obligations, which could weaken their creditworthiness.
202.	What are **Revenue Bonds?** Municipalities or states might also issue revenue bonds, which are issued to finance specific projects. However, they carry a higher degree of risk than GO bonds because they are dependent on a single source of revenue – the given project. Therefore, analysts should focus on the projected utilization of the project and its future cash flows. A key gauge of creditworthiness for revenue bonds is the debt-service-coverage (DSC) ratio, which measures how much revenue is available to cover debt payments after operating expenses.

SECTION F – FIXED INCOME

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	The payments of a fixed-income security are anchored on contractual guidelines and must be made.	**TRUE.** A **fixed-income security** is a financial obligation that pays a fixed amount of interest in coupon payments to investors at specified points in the future.
2.	Bonds are broken into two categories based on credit risk.	**TRUE.** Bonds are broken into the following two categories based on **credit risk:** a. Investment-grade. b. Non-investment grade.
3.	Non-investment-grade bonds are rated above BBB or Baa3 and have a relatively low risk of default.	**FALSE. Investment-grade** bonds are rated above BBB or Baa3 and have a relatively low risk of default. Conversely, non-investment-grade bonds are considered low quality to reflect a relatively higher probability of default.
4.	Sovereign and non-sovereign governments can issue bonds.	**TRUE. Bonds can be issued by** a host of institutions, including: • Supranational organizations. • Quasi-government entities. • Companies. • Special legal entities.
5.	Tenor refers to when the issuer is obligated to redeem the bonds.	**FALSE. Maturity** refers to the date when the issuer is obligated to redeem the bonds, while tenor is the time left before maturity.
6.	The pool of securitized assets from which cash flows are generated is	**TRUE. Securitization** involves transferring ownership of assets from original owners (usually individual borrowers) to a special legal entity. This entity, in return, issues asset-backed securities backed by these transferred assets.

	known as collateral.	
7.	The bond market is broken up into three sectors.	**TRUE.** The **bond market** is generally broken up into the following **three sectors:** a. Government bonds. b. Corporate bonds. c. Structured finance.
8.	Money market instruments are those with a maturity of more than a year.	**FALSE.** Money market instruments are those with a maturity of **less than a year**, including treasury bills, commercial paper, and bankers' acceptances. Whereas, bonds with more than a year of maturity are called capital market instruments, e.g., debentures.
9.	Par value is also called principal value.	**TRUE. Par value**, also called principal or principal value, is the amount that will be repaid at maturity.
10.	Providing lower interest rate risk is a major benefit of securitization.	**FALSE.** Securitization **does not reduce** interest rate risk but helps investors access securities matching their risk, return, and maturity needs.
11.	The coupon rate is the interest rate that the bondholder will receive.	**TRUE. Interest rates** can either be fixed over the life of the bond or floating, and the payment frequency can differ depending on the issuer. Annual, semi-annual, and monthly payments are common.
12.	The yield can be measured in two ways.	**TRUE.** The yield is different from the coupon rate and can be measured in two ways: a. The **current yield** is the annual coupon payment divided by the bond price. b. **Yield-to-maturity** is the internal rate of return that equates the present value of a bond's expected cash flows to its price.
13.	The bond trades at a discount if the market price exceeds the par value.	**FALSE.** The bond trades at a **premium** if the market price exceeds the par value. In contrast, the bond trades at a discount if the market price is less than the par value. Further, the

		bond trades at par if the market price equals the par value.
14.	Zero-coupon bonds pay interest, are issued at a premium to par value, and then redeemed at par.	**FALSE.** Zero-coupon bonds **do not pay interest** but are issued at a **discount** to par value and then redeemed at par.
15.	A bond indenture is a legal document that outlines all the parameters of the bond issue.	**TRUE.** A **bond indenture** is a legal document that outlines all the parameters of the bond issue, such as the par amount, issuer, coupon rate, security pledge, security ranking of the bonds, and the rights of bondholders.
16.	Floating-rate bonds pay a fixed coupon rate.	**FALSE. Plain vanilla bonds** pay a fixed coupon rate, while floating-rate bonds pay a floating interest rate tied to a reference rate such as LIBOR (London Interbank Offered Rate).
17.	Securitization increases the efficiency of financial markets.	**TRUE.** Securitization allows for the **creation of tradable securities with much liquidity** and results in more efficient financial markets.
18.	Eurobonds are denominated in the currency in which they are issued.	**FALSE.** Eurobonds are denominated in a currency **other than the country's home currency or market** in which they are issued. They are less regulated, usually unsecured, and underwritten by an international syndicate.
19.	Bond losses are first absorbed by the most junior (lower) classes.	**TRUE.** Depending on how they share any losses resulting from borrowers' defaults, there could be more than one bond class. Bond classes can be categorized as **senior tranches** or **subordinated (junior) tranches,** and the losses are first absorbed by the most junior (lower) classes.

200.	Leverage ratios can perform key credit analysis.	**TRUE. Leverage ratios** include: • **Debt/Capital** - A higher ratio implies more leverage and, thus, higher credit risk. • **Debt/EBITDA** - A higher ratio implies more leverage and, thus, higher credit risk. • **FFO/Debt** - A higher ratio means a greater ability to pay debts since debt is in the denominator here.
201.	Coverage ratios measure the issuer's ability to meet or cover interest payments.	**TRUE. Coverage ratios** include: • EBITDA/Interest expense. • EBIT/Interest expense.
202.	EBITDA /Interest expense is a conservative coverage measure in credit analysis.	**FALSE. EBIT/Interest expense** is a conservative measure of interest coverage since it sometimes does not account for the repayment of capital and excludes depreciation and amortization.
203.	The credit spread will widen as the credit cycle improves.	**FALSE.** As the credit cycle improves, the credit spread **will be narrow**. On the other hand, a weakening credit cycle will widen the credit spreads.
204.	Issuer's liquidity can be assessed by net working capital and operating cash flow.	**TRUE.** While assessing an **issuer's liquidity**, credit analysts tend to also look at the following: • Cash on the balance sheet. • Committed bank lines. • Debt coming due and committed capital expenditures.
205.	Weakening economic conditions will tighten the credit spread.	**FALSE.** Weakening economic conditions will make the investors demand a greater risk premium, hence a **wider credit spread**. On the opposite side, good economic conditions will tighten the credit spread because investors expect improved credit measures, lowering default risk.

206.	The majority of local government bonds are general obligation bonds.	**TRUE. General obligation bonds** are unsecured securities backed by the full faith of the issuing non-sovereign government.
207.	A high demand for bonds widens the credit spreads.	**FALSE.** Credit spreads **narrow** when there is a high demand for bonds, whereas credit spreads widen if there is insufficient demand for new corporate bonds.
208.	Special considerations are important when evaluating the creditworthiness of debt issuers in three market segments.	**TRUE.** Special considerations are important when evaluating the **creditworthiness of debt issuers** in the following three market segments: a. **High-yield corporate bonds** - those rated below Baa3/BBB by the major rating agencies. b. **Sovereign bonds** – are government bonds in developed countries traditionally viewed as the default risk-free rate, after which all other bonds are priced. c. **Non-sovereign government debt** - municipal bonds issued by sub-sovereign or local governments and quasi-government entities.
209.	Non-sovereign governments can utilize monetary policy.	**FALSE.** Non-sovereign governments **cannot** utilize monetary policy and typically must balance their operating budgets. This obligation makes it an important criterion for analysts when choosing between non-sovereign government and sovereign debt.
210.	Revenue bonds carry a lower risk than general obligation bonds.	**FALSE.** Revenue bonds carry a **higher** risk than general obligation bonds because they depend on a single source of revenue, the given project. Therefore, analysts should focus on the project's projected utilization and future cash flows.

SECTION F – FIXED INCOME

FILL IN THE BLANKS

1. A bond's coupon rate is the annual interest payment divided by the _________.

2. The yield to maturity (YTM) is the discount rate that makes the present value of the bond's cash flows equal to its _________.

3. A bond selling at a price above its face value is said to be selling at a _________.

4. A bond with a coupon rate lower than its YTM will be priced at a _________.

5. A zero-coupon bond pays _________ interest payments during its life.

6. The duration of a bond is a measure of its _________ to interest rate changes.

7. The convexity of a bond is a measure of the curvature in the relationship between bond prices and _________.

8. The _________ curve shows the relationship between yield and maturity for bonds of the same credit quality.

9. A bond's _________ is the date on which the principal amount is repaid to investors.

10. Credit risk is the risk that a bond issuer will _________ on its payments.

11. The _________ rate is the rate at which a borrower can refinance a bond if it is callable.

12. When a bond is _________, the issuer can repurchase it before maturity at a specified price.

13. A _________ bond can be converted into a specified number of shares of the issuer's stock.

14. The _________ value of a bond is the amount that the issuer agrees to pay the bondholder at maturity.

15. A _________ bond pays interest only when the issuer has sufficient earnings.

16. Interest rate risk is the risk that a bond's price will decline due to an increase in _________.

17. The _________ is the difference between the yield on a corporate bond and a government bond of similar maturity.

18. A bond's _________ represents the investor's required rate of return.

19. A bond that is trading at par has a price equal to its _________.

20. The _________ is the amount of interest that has accumulated on a bond since the last interest payment.

21. The _________ of a bond is calculated by dividing the annual coupon payment by the current market price of the bond.

SECTION F – FIXED INCOME

ANSWER KEY

1. face value

2. market price

3. premium

4. discount

5. no

6. sensitivity

7. yields

8. yield

9. maturity

10. default

11. call

12. callable

13. convertible

14. face

15. income

16. interest rates

17. credit spread

18. yield to maturity

19. face value

20. accrued interest

21. current yield

22. credit

23. present

24. higher

25. higher

26. longer

27. cash flows

SECTION F – FIXED INCOME

WORD SEARCH PUZZLES

Section F - Quiz 1

```
U V J W P T T F C O L Z C M Z C K N M W
D T J U N K Y C A L L A B L E Y I E L D
E O Z O F L O A T I N G H H N O P U O C
F P M X V E U M P L A P I C N I R P P Y
A S Y E N Y X P T H Y Z Q J F C K J S W
U F M O D I F I E D H E D G I N G U O M
L F X H C Z H E F K L N V M M B N E Q C
T A L H R F T B F I S Y C Q K O O R O P
X V A F E M B O E F C O N V E X I T Y S
A O P D U P G M C Y N T H C D Y O N Q E
M L I U S H J T T T R F C M Z Y X F K N
W A C R S K Z M I H N U C F H N X Z E S
M T I A I D L E V D G V S B E C N O T I
A I N T J U D C E T V S Q A O W I K N T
T L U I O I A O M O T T R T E G H H J I
U I M O T N E M T S E V N I E R Q W Y V
R T B N F E R N O I T A L F N I T A C I
I Y F D B Z P A Z J R Q E A Q X C G W T
T L S B O Z S K Y H P S X R O A B J V Y
Y Z B X H O H D M N A I Q N L M D S K C
```

1. Coupon **2.** Maturity **3.** Principal **4.** Issuer **5.** Yield **6.** Spot **7.** Duration

8. Convexity **9.** Spread **10.** Reinvestment **11.** Default **12.** Modified **13.** Effective

14. Volatility **15.** Sensitivity **16.** Hedging **17.** Treasury **18.** Municipal **19.** Floating

20. Callable **21.** Inflation **22.** Junk

SECTION G – DERIVATIVES

STUDY POINTS

S.No	Description
1.	What is a **Derivative?** A **derivative** is a financial instrument that derives (obtains) its value from the performance of an underlying. The underlying may be a single asset, a group of assets, or variables such as interest rates.
2.	How a **Derivative is Created**? A derivative is created in the form of a derivative contract involving two counterparties: a buyer and a seller. A derivative contract is a legal agreement between counterparties that defines the rights and obligations of each party. It contains specific maturity or settlement. The buyer is the **long** or the **holder**, owns the derivative, and is said to hold a long position. On the other hand, the seller is referred to as **short** and holds a short position.
3.	Define **Stand-alone and Embedded Derivatives**? Stand-alone derivatives are distinct. An example is a derivative on a bond. Embedded derivatives are derivatives within an underlying, for example, callable, puttable, and convertible bonds.
4.	Classify **Derivatives?** Derivatives can be classified into either one of two categories: **Firm Commitment (Forward Commitment)** In firm commitment, an amount is pre-determined, and the parties involved make an agreement to exchange it at a future date. Firm commitments include **forward contracts**, **futures contracts**, and **swaps**.

	Contingent Claim In a contingent claim, the settlement of the trade depends on one of the counterparties. **Options** are the primary contingent claims.
5.	State the **Benefits of Using Derivatives?** Investors can get access to broad opportunities by creating or modifying exposures in the following ways: • Earning profits by short selling an underlying whose value is expected to decline. • Diversifying a portfolio. • Offsetting the financial market exposure that comes with a commercial transaction. • Creating large exposures to an underlying using relatively low amounts of cash. • Either increasing or decreasing financial market exposure. For instance, hedging uses derivatives to offset (neutralize) existing (anticipated) exposure to an underlying.
6.	Discuss the **Uses of Derivatives?** • **Hedging**: Reduce or eliminate certain forms of risk. • **Speculation**: Derivatives have, as an inherent feature, a high degree of leverage. This means that investors only commit small amounts of money to a derivative position relative to the equivalent position in the underlying asset. Small movements in the underlying can lead to large movements in the derivative – both positive and negative. • **Arbitrage**: Simultaneous buying and selling to take advantage of varying prices for the same asset to earn a riskless profit.
7.	Assess **Derivative Underlyings?** One way of classifying a derivative is by using the underlying from which the derivative derives its value. Commonly used underlyings

	are equities, fixed income, interest rates, currencies, commodities, and credit.
8.	Explore **Equities?** Derivatives that use equities as the underlying may reference a single stock, a group of stocks, or a stock index. Options are predominantly associated with individual stocks, while index derivatives are mostly traded as options, futures, forwards, and swaps. In index or equity swaps, an investor can receive a return on one index or interest rate and pay the return on one stock index. Investment managers can also use index swaps to increase or decrease exposure to an equity market without trading in individual shares. Investors trade options on individual stocks. Also, issuers may use stock options to compensate their executives and employees as a motivation for greater corporate performance, which leads to higher stock prices. Besides, companies may issue warrants. Warrants are stock options that give their holders the right to purchase shares at a fixed price directly from the issuer in the future.
9.	Describe **Fixed-Income Instruments and Interest Rates?** Fixed income instruments mostly use bonds as the underlying. Associated derivatives include futures, swaps, and options. Interest rate is a fixed income underlying used by interest rate derivatives, such as forwards, futures, and options. Note that interest rate is not considered an asset. Interest rate swaps are usually used to convert from a fixed interest rate to a floating interest rate exposure – or vice versa – over a certain period. Interest rate swaps mostly use a **market reference rate** (MRR) as the underlying. The most common market reference rate is the secured overnight financing rate (SOFR).
10.	Identify **Currencies?**

	Derivatives can be used to hedge foreign exchange risk in commercial and financial transactions. For example, exporters may use forward contracts to sell domestic currency and buy foreign currency in a way that coincides with the delivery of goods or services in a foreign country.
11.	Assess **Commodities?** Commodities are classified into hard or soft commodities. **Hard commodities** are natural resources such as crude oil. **Soft commodities** are agricultural products such as crops and cattle. Derivatives on commodities are usually used to manage the price risk of an individual commodity or a commodity index separate from physical delivery.
12.	Describe **Credit?** Derivative contracts that use credit as an underlying are based on the default risk of either a single or a group of issuers. For instance, Credit Default Swaps (CDS) help manage the risk of loss if a borrower defaults.
13.	What is **Included in Others?** Others include weather, cryptocurrencies, and longevity. Derivatives that use such underlying are less common, and their pricing is challenging.
14.	Elaborate on **Over-the-Counter (OTC) Derivative Markets?** OTC derivative markets can be formal institutions such as NASDAQ or an information connection of parties who buy from and sell to one another. In OTC derivative markets, **derivatives end-users** enter contracts with **dealers** or a financial intermediary such as a bank. The dealers (also regarded as the market markers) engage in bilateral transactions to transfer risk to other parties.

	Terms of OTC can be modified to match a desired risk exposure profile. This is a beneficial feature to derivative end users who want to hedge existing or expected exposure.
15.	Describe **Exchange-Traded Derivative (ETD) Markets?** In ETD markets, derivatives are traded in more formal and standardized contracts, promoting higher liquidity and transparency. Such derivatives include futures, options, and other financial contracts at the exchange. The exchange determines the terms and conditions, including the size of each contract, type, quality, and location of the underlying. The exchange members consist of dealers (market markers) who are prepared to buy at one price and sell at a higher price. If they cannot find counterparties to trade, risk takers such as speculators may be willing to assume an exposure in the underlying. Exchange-traded derivatives have standardized terms and conditions. As such, clearing and settlement are done efficiently. Exchange-traded derivatives demand collateral on deposit upon initiation and during the life of a contract to reduce counterparty risk. The deposit is paid through a financial intermediary, which assures counterparty default.
16.	Contrast **Clearing and Settlement?** **Clearing** is a process where the exchange/central counterparty verifies the execution of a transaction, exchange of payments, and records of the participants. **Settlement** refers to the payment of final amounts and/or delivery of securities or physical commodities between the counterparties based upon exchange rules.
17.	Analyze the **Differences between Over-the-Counter (OTC) Derivative and Exchange-Traded Derivative (ETD) Markets?** • OTC offers more flexibility and customizability.

As a rule of thumb, at time t = 0, a company's market value, V0, is equivalent to the present value of its outstanding debt obligations, P V (D), and equity, E0, where the borrowed funds are in zero-coupon debt with a face value of D.

In an equation, we can express this relationship as:

$$V_0 = E_0 + PV(D)$$

112. Assess the **Situation** in the **Event of Debt Maturity**?

In the event of debt maturity at t = T, the assets and debts of the company will be split between debtholders and shareholders, with two possible outcomes based on the company's value at that given time:

Solvency (VT > D)

Solvency refers to a company's ability to meet its financial obligations and long-term debt. If, at time T, a firm's value (VT) is greater than the face value of the debt (VT > D), the firm is solvent and thus able to return capital to both the shareholders and the debtholders.

Debt holders come first when distributing capital returns. As such, they receive the debt repayments (D) in full. On the other hand, the shareholders receive what remains. That is ET = VT − D.

In summary, we've established that shareholders benefit if a company can meet its debt obligation and maintain solvency. On the other hand, debt holders benefit when a company is solvent and meets its debt obligations.

Insolvency (VT < D)

Insolvency refers to a company's inability to meet its financial obligations and long-term debts. This occurs if, at debt maturity (T), a company's value is less than the debt's face value, VT < D.

When a firm is insolvent, the shareholders receive the residual equal to zero (ET = 0), and the debtholders are owed more than the firm's total assets. As such, the debtholders receive VT < D to cover the debt of D at time T.

113. Explain how to value a **Derivative using a One-Period Binomial Model?**

The idea behind the binomial model is that at maturity, an asset's spot price, S0, can either increase to Su1 or decrease to Sd1. We do not need to know the asset's future price in advance since it depends on a random variable's outcome. The asset price movements, from S0 to Su1 or Sd1, can be seen as the outcome of a Bernoulli trial.

Let us denote q as the probability of an increase in the asset's price. Because of the presence of only two possibilities, i.e., an increase or a decrease in the asset's price, we can denote the probability of a decrease in its price as 1-q, such that the two probabilities will add up to 1.

The gross return when the asset price increases will be:

$$\mathrm{Ru} = \frac{\mathrm{Su1}}{\mathrm{S0}} > 1$$

When the asset price decreases, the gross return will be:

$$\mathrm{Rd} = \frac{\mathrm{Sd1}}{\mathrm{S0}} < 1$$

Su1(or Ru So) and Sd1 (or RdS0) differ in the spread of possible future price outcomes.

114. How to **Price a European Call Option using a One-Period Binomial Model?**

Consider a one-year call option with an underlying price of S0 and an exercise price of X. Also, assume that Sd 1 < X < Su and that the one-period binomial model is equivalent to the time of expiration of one year.

The one-period binomial model gives the underlying asset values in one year, where the option value is defined as a function of the underlying value.

	At time t=0: The value of the call option is c0. **At time t=1:** After one year, the option expires. At this time, the value of the option will either be cu1 = max(0, Su 1 − X) if the underlying price rises to Su 1 or cd1 = max(0, Su 1 − X) if the underlying price falls to Sd1. Intuitively, for the up movement, the call option is in the money, and for the down direction, the option is out of the money.
115.	Describe the concept of **Risk Neutrality in derivatives pricing?** The **risk-neutral probability** is the computed probability used in binomial option pricing that equates the discounted weighted sum of the expected values of the underlying to the option's current price. It is calculated using the risk-free rate and the assumed up-and- down gross returns of the underlying, as follows: $$\pi = \frac{(1 + r) - Rd}{Ru - Rd}$$ Risk-neutral pricing determines the risk-neutral probability (used to calculate the present value of future cash flows) using only the expected volatilities, i.e. (Ru and Rd) and the risk-free rate.

SECTION G – DERIVATIVES

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	A derivative is a financial instrument that derives its value from the performance of an underlying.	**TRUE.** A **derivative** is a financial instrument that derives (obtains) its value from the performance of an underlying. The underlying may be a single asset, a group of assets, or variables such as interest rates.
2.	Derivatives can be stand-alone or embedded.	**TRUE. Stand-alone derivatives** are distinct, for example, a derivative on a bond. Alternatively, **embedded derivatives** are derivatives within an underlying, for example, callable, puttable, and convertible bonds.
3.	When multiple traders buy low-priced assets and sell high-priced assets simultaneously, it increases the demand and price for the latter.	**FALSE.** When multiple traders buy low-priced assets and sell high-priced assets simultaneously, it **increases the demand and price for the former** while **decreasing the demand and price for the latter**.
4.	A derivative is created in the form of a derivative contract involving two counterparties.	**TRUE.** A **derivative contract** is a legal agreement between counterparties that defines the rights and obligations of each part and contains specific maturity or settlement.
5.	Arbitrage opportunities disappear quickly.	**TRUE. Arbitrage** refers to buying an asset in a cheaper market and selling it in a more expensive market to make a risk-free profit.
6.	The seller is the long or the holder who owns the derivative.	**FALSE.** The **buyer** is the long or the holder, owns the derivative, and is said to hold a long position. On the other hand, the seller is referred to as short and holds a short position.

7.	Derivatives can be classified into firm commitment or contingent claim.	**TRUE.** Derivatives can be classified into either one of two categories: a. In **firm commitment**, an amount is pre-determined, and the parties agree to exchange it at a future date. It includes forward contracts, futures contracts, and swaps. b. In a **contingent claim**, the trade settlement depends on one of the counterparties. Options are the primary contingent claims.
8.	Derivatives can be used for three purposes.	**TRUE. Uses of derivatives:** a. **Hedging** - Reduce or eliminate certain forms of risk. b. **Speculation** - Derivatives have a high degree of leverage, meaning that investors only commit small amounts of money to a derivative position relative to the equivalent position in the underlying asset. Small movements in the underlying can lead to large movements in the derivative, both positive and negative. c. **Arbitrage** - Simultaneous buying and selling to take advantage of varying prices for the same asset to earn a riskless profit.
9.	Index derivatives are mostly traded as options, futures, forwards, and swaps.	**TRUE.** Derivatives that use **equities** as the underlying may reference a single stock, a group of stocks, or a stock index. Options are predominantly associated with individual stocks, while index derivatives are mostly traded as options, futures, forwards, and swaps.
10.	Companies may issue warrants.	**TRUE. Warrants** are stock options that give their holders the right to purchase shares at a fixed price directly from the issuer in the future.

11.	The future price is the price an investor must pay immediately to acquire the asset.	**FALSE.** The **spot price** is the price an investor must pay immediately to acquire the asset. In other words, it is the asset's current value or the amount that sellers and buyers agree it is worth. In contrast, the future price refers to the projected price of an asset at a later date.
12.	Arbitrage opportunity results in two ways.	**TRUE.** The value of derivative contracts is derived from future cash flows linked to the underlying assets. As such, **arbitrage opportunity** results in the following ways: a. **Case 1** - Two assets with identical future cash flow trade at different prices. b. **Case 2** - An asset with a definite future price does not trade at the present value of its future price, calculated at an appropriate discount rate.
13.	Interest rate swaps mostly use a market reference rate as the underlying.	**TRUE. Interest rate swaps** are usually converted from a fixed interest rate to a floating interest rate exposure or vice versa over a certain period. The most common market reference rate is the secured overnight financing rate.
14.	Derivatives cannot be used to hedge foreign exchange risk.	**FALSE.** Derivatives **can be used** to hedge foreign exchange risk in commercial and financial transactions.
15.	Derivatives can be used on commodities.	**TRUE. Derivatives on commodities** are usually used to manage the price risk of an individual commodity or a commodity index separate from physical delivery.
16.	Replication mirrors or offsets a derivative position.	**TRUE. Replication** mirrors or offsets a derivative position, given that the law of one price holds and arbitrage does not exist. It implies that a trader can take opposing positions in a derivative and the underlying, creating a default risk-free hedge portfolio and replicating the payoff of a risk-free asset.

		current price compared with the expected future settlement price.
79.	A call option is out of the money if the underlying spot price exceeds the exercise price.	**FALSE.** A call option is **in the money** if the underlying spot price exceeds the exercise price. Alternatively, a call option is out of the money if the spot price exceeds the current exercise price.
80.	Exchange-traded derivatives are associated with high counterparty risk.	**FALSE.** Exchange-traded derivatives are associated with **low counterparty risk** due to the mark-to-market process and margining procedures.
81.	Derivative instruments are exposed to systemic risk.	**TRUE.** Systemic risk occurs due to **extensive risk-taking and leverage in derivative markets**. As such, financial market supervisory has increased, with a heightened focus on the effect of financial innovation and financial conditions necessary for market stability.
82.	A put option is in the money if the spot price exceeds the exercise price.	**FALSE.** A put option is in the money if the spot price is **less than** the exercise price. In contrast, a put option is out of the money if the spot price is higher than the current exercise price.
83.	Certain factors affect the value of an option.	**TRUE.** The **factors that affect the value of an option** are: • Value of the underlying. • Exercise price. • Time to maturity. • Risk-free rate. • Volatility. • Income or cost associated with the underlying.
84.	An option is considered deep out of the money if it is highly exercisable.	**FALSE.** An option is considered **deep in the money** if it is highly exercisable, which means the stock price is much higher than the exercise price. Conversely, an option is considered deep out of the money if it is unlikely to be exercised.

85.	The value of the underlying has a direct impact on the right to exercise an option.	**TRUE.** The value of the call option (and long forward) appreciates when the **spot price of the underlying increases**. In contrast, the put option (and short forward) appreciates when the **spot price of the underlying declines.**
86.	A higher exercise price will increase the likelihood of exercise and settlement value if it is in the money for a call option.	**FALSE.** The exercise price determines whether an option buyer will exercise the option at the expiration. For a call option, a **lower** exercise price will increase the likelihood of exercise and settlement value if it is in the money.
87.	A longer time to maturity increases the likelihood of the option finishing in the money only for a call option.	**FALSE.** The time value of an option represents the likelihood that favorable changes to the underlying price will increase the profitability of the exercise. A longer time to maturity increases the likelihood of the option finishing in the money for **both call and put options**, thus increasing the option's value.
88.	Lower volatility decreases the time value of both put and call options.	**TRUE. Volatility** measures the expected dispersion of future movements of an underlying asset. **Higher volatility** of the underlying asset increases the chances of call and put options finishing in the money without affecting the downside case – the option expires worthless.
89.	A higher risk-free rate decreases the value of the call option.	**FALSE.** A higher risk-free rate **increases** the value of the call option, as a higher risk-free rate lowers the present value of the exercise price, provided the option is in the money. Intuitively, a higher risk-free rate decreases the exercise value of a put option.
90.	Put-call parity is a no-arbitrage concept.	**TRUE. Put-call parity** involves a combination of cash and derivative instruments in a portfolio and allows pricing and valuation of these positions without directly modeling them using non-arbitrage conditions.

91.	Income increases the value of a call option and decreases the value of a put option.	**FALSE.** The present value of the income or benefits is subtracted from the underlying price. As such, income **decreases** the value of a call option and **increases** the value of a put option.
92.	The risk-free bond replicating individual positions under put call parity is a short underlying, long call option, and short put option.	**FALSE.** The risk-free bond replicating individual positions under put call parity is a **long underlying**, **short call** option and **long put** option.
93.	Cost decreases the call option's value and increases the put option's value.	**FALSE.** If the asset owner incurs costs, compensation is done to cover the added costs. As such, the present value of the costs is added to the underlying price. Therefore, cost **increases** the value of the call option and **decreases** the value of the put option.
94.	The risk-neutral probability is calculated using the risk-free rate and the assumed up-and-down gross returns of the underlying.	**TRUE.** The **risk-neutral probability** is the computed probability used in binomial option pricing that equates the discounted weighted sum of the expected values of the underlying to the option's current price.

SECTION G – DERIVATIVES

FILL IN THE BLANKS

1. A derivative is a financial instrument whose value is derived from the value of an underlying _____.

2. The most common types of derivatives include forwards, futures, options, and _____.

3. A forward contract is an agreement to buy or sell an asset at a specified future date for a price that is _____ today.

4. The party that agrees to buy the underlying asset in a forward contract is taking a _____ position.

5. The party that agrees to sell the underlying asset in a forward contract is taking a _____ position.

6. A futures contract is similar to a forward contract but is traded on a(n) _____.

7. In a futures contract, the exchange requires both parties to post _____ as collateral.

8. The daily settlement process in futures contracts is known as _____.

9. An option gives the holder the right, but not the _____, to buy or sell an asset at a specified price.

10. A call option gives the holder the right to _____ an asset at a specified strike price.

11. A put option gives the holder the right to _____ an asset at a specified strike price.

12. The price paid by the buyer of an option is called the _____.

13. The strike price of an option is also known as the _____ price.

14. An option that can be exercised at any time before expiration is called a(n) _____ option.

15. An option that can only be exercised at expiration is called a(n) _____ option.

16. A swap is an agreement between two parties to exchange sequences of cash flows for a specified period of _____.

17. In an interest rate swap, one party typically pays a fixed rate while the other pays a(n) _____ rate.

18. A credit default swap (CDS) is a type of derivative that provides protection against the risk of _____.

19. The Black-Scholes model is used to price European _____.

20. The key inputs to the Black-Scholes model include the stock price, strike price, time to expiration, risk-free rate, and _____.

21. The Greeks are a set of measures used to assess the sensitivity of an option's price to various factors; Delta measures sensitivity to the underlying asset's _____.

SECTION G – DERIVATIVES

ANSWER KEY

1. asset

2. swaps

3. agreed upon

4. long

5. short

6. exchange

7. margin

8. marking to market

9. obligation

10. buy

11. sell

12. premium

13. exercise

14. American

15. European

16. time

17. floating

18. default

19. options

20. volatility

21. price

22. volatility

23. time

24. interest

25. delta

26. straight line

27. strike price

SECTION G – DERIVATIVES
WORD SEARCH PUZZLES

Section G - Quiz 1

```
G N O I T A R I P X E E Q B R
S T S T C V B O P T I O N U J
F L G N I Y L R E D N U J L X
O O F F O Y A P D M I W M L L
R N I F S Y Q M S E R U T U F
W G N U T T R T R O H S K E X
A Y I D M S R E Z F F A R G C
R C T E V I T A V I R E D A S
D I I F Z J Z R D I B E A R T
E X A H D E Z G I D L M F T R
F C L P D P A W S K L E V I A
M A I N T E N A N C E E D B N
D T N E M E L T T E S Y M R G
H J F M G N I G D E H U T A L
S P E C U L A T I O N N I Y E
```

1. Derivative **2.** Underlying **3.** Option **4.** Futures **5.** Forward **6.** Swap **7.** Payoff **8.** Expiration **9.** Strike **10.** Long **11.** Short **12.** Bull **13.** Bear **14.** Straddle **15.** Strangle **16.** Initial **17.** Maintenance **18.** Settlement **19.** Hedging **20.** Speculation **21.** Arbitrage **22.** Delivery

Section G - Quiz 2

```
G Y D W H O H R O V U V J G C P F L T P
O V A L U A T I O N N Z I I L A Y V D R
A E T G A M M A G O I P U E B Y A S Z H
Y G H U P U P S I T K P G O Q M T O L J
T A E W O S E T T L E M E N T E L U Z M
R R T B R F P L A I M O N I B N E M P Z
A E A U T O L A P I C N I R P T D C R V
P V E E F C M D E X I F N C M A I G I O
R E V G O A R B I T R A G E H S R N C L
E L Y H L T R A N S P A R E N C Y I I A
T E R M I N A T I O N T Y I T Y B R N T
N U I R O M A Q L Y V O R C G S V A G I
U U L X Q I F J I F U T I D N U T E L L
O R G N I T A O L F N A I T P E Q L G I
C E T T E B D P L I T C P M A J R C Q T
T L C N O T I O N A L V A J G L V R U Y
W A C E X C H A N G E L W E B T U W U D
D E V E G A I S H Q R K S V W H T G N C
E D I S C O U N T I N G S W L G E M E I
B S E N S I T I V I T Y H U U G L J B R
```

1. Swap **2.** Currency **3.** Notional **4.** Principal **5.** Fixed **6.** Floating **7.** Dealer
8. Counterparty **9.** Termination **10.** Payment **11.** Pricing **12.** Valuation
13. Discounting **14.** Binomial **15.** Intrinsic **16.** Volatility **17.** Arbitrage **18.** Delta
19. Gamma **20.** Theta **21.** Vega **22.** Rho **23.** Sensitivity **24.** Option **25.** Portfolio
26. Exchange **27.** Clearing **28.** Leverage **29.** Transparency **30.** Regulation
31. Settlement

SECTION H – ALTERNATIVE INVESTMENTS

STUDY POINTS

S.No	Description
1.	What is **Included in Traditional Investments**? • Long-only, publicly traded investments in stocks. • Long investments in publicly traded bonds. • Cash.
2.	What is **Included in Alternative Investments**? • Private capital. • Real assets. • Hedge funds.
3.	Why **Consider Alternative Investments?** An investor should consider alternative investments for several reasons. To begin with, alternative investments offer a range of low correlations with traditional investments. In addition, alternative investments allow investors to enhance the risk-return characteristics of their portfolios and increase their returns. Last but not least, alternative investments improve an investor's return potential thanks to their higher yields.
4.	List the **Characteristics of Alternative Investments?** • Investment managers limit their specialization to alternative investments. • Alternative investments have a relatively low correlation of returns to traditional investments. • Alternative investments have less transparency and regulation compared to traditional investments.

	• Alternative investments have inadequate historical risk and return data. • Alternative investments have unique tax and legal issues. • Alternative investments have higher fees, usually made of performance (incentive) and management fees. • Alternative investments are typically concentrated portfolios. • Alternative investments usually have limitations on redemptions, such as lockups and gates.
5.	Provide the **Categories of Alternative Investments?** Alternative investments are divided into broad categories: • Private capital. • Real assets. • Hedge funds. • Other.
6.	What is **Included in Private Capital?** • **Private equity** investors might participate directly or indirectly through private equity funds. Private equity funds make investments in both new and established businesses. Note that these firms may not be listed on a public exchange. Alternatively, private equity funds may invest in public companies intending to privatize them. • **Private debt** is primarily provided to private entities. Examples of private debt include direct lending, mezzanine loans, venture debt, and distressed debt.
7.	Elaborate on the **Items Included in Real Assets**? • **Real estate investments** are made in buildings or land, either directly or indirectly. Such investments consist of private commercial real estate debt and private commercial real estate

	equity, such as the ownership of an office building, public real estate equity, and public real estate debt. • **Infrastructure** assets are long-lived, capital-intensive real estate resources such as airports and power plants. An increasingly popular method of investing in infrastructure is the public-private partnership (PPP) strategy, in which case both public authorities and private investors have an interest. • **Natural resources** include: ○ **Commodities**: These investments may be made in physical commodity products, such as grains, metals, and crude oil. ○ **Farmland/agricultural land**: Agriculture involves the rearing of animals or plants. Investing in agricultural property, therefore, encompasses a variety of methods, such as leasing farmland or receiving income from the sale of agricultural produce such as corn, milk, or meat. ○ **Timberland**: Investing in timberland often entails putting money into managed or natural tree plantations to reap a profit when the trees are harvested.
8.	Clarify **Hedge Funds?** **Hedge funds** employ numerous strategies to offer investors optimal returns. These private funds are actively managed—often by "star investors"—and apply aggressive strategies across asset classes. Most hedge funds use derivatives, high leverage, arbitrage, and other investment strategies unavailable to traditional fund managers. Redemptions are typically restricted in some way, such as lockup periods, notice periods, and redemption fees.
9.	What is **Included in Other Alternative Investments**? **Other alternative investments** include collectibles and other investments that don't fall into the traditional categories. Collectors have been around for centuries, and the investments are as varied as the collectors themselves. Traditional collectibles such as stamps, cars, furniture, gems, fine art, and others still

	dominate the market. However, new collectibles, such as baseball cards, action figures, and almost anything you can imagine, offer investors new ways to invest their money outside traditional financial markets. There are indices used to track the values of certain collectible markets, such as stamps, and these collectibles are auctioned in small markets. It is worth noting that some auctions are open to the public while others are not.
10.	List the **Methods of Investing in alternative investments?** Methods of investing in alternative investments include: • Fund investing. • Co-investing. • Direct investing.
11.	Discuss **Fund Investing?** In **fund investing**, investors contribute capital to a fund, and then the fund identifies, chooses, and makes investments on behalf of the investors. Investors pay a fee based on the value of the fund managers' assets. In addition, they pay a performance fee if the fund manager delivers a return above the benchmark. Fund investing can be termed as indirect investing in alternative assets. Fund investors' decisions are based on whether to invest in a fund. Furthermore, investors do not influence a fund's investments. Fund investing applies to all types of alternative investments.
12.	Mention the **Advantages of Fund Investing?** • Fund investing requires less investor involvement compared to direct and co-investing. • The alternative investment option is accessible to anyone, regardless of their expertise.

	• Diversification benefits come from the multiple investments found in a single fund. • It requires a low minimum capital compared to the other investments.
13.	Provide the **Disadvantages of Fund Investing?** • It is costly since an investor must pay management and performance fees. • An investor is expected to conduct due diligence when selecting the appropriate fund. • There are exit restrictions due to lockups and similar limitations.
14.	Explore **Co-investing?** In **co-investing**, an investor indirectly invests in assets through a fund. Nevertheless, the investor owns the rights (co-investment rights) to invest in the same assets directly. Intuitively, co-investing allows the investor to make parallel investments when the fund identifies a lucrative deal.
15.	State the **Advantages of Co-investing?** • An investor can learn from the fund's expertise and improve at direct investing. • Investors co-invest an additional amount into an investment, often without paying management fees on the capital they used for direct investments. • Co-investing allows investors to be more actively involved in managing their portfolios than fund investing.
16.	Mention the **Disadvantages of Co-investing?** • Co-investors have limited control over the investment selection process compared to direct investing.

	• It may be subject to adverse selection. A fund may offer less attractive investment opportunities to the co-investor while allocating capital to more appealing deals. • Co-investing requires an investor to be more actively involved since they must evaluate both investment opportunities and the fund manager.
17.	Explore **Direct Investing?** In **direct investment**, an investor directly invests in an asset without using intermediaries. Direct investing, therefore, gives investors more control and flexibility when selecting investments, financing methods, and approaches. However, investing directly in alternative investments is most popular among established private equity and real estate investors. Pension and sovereign wealth funds may also invest in infrastructure and natural resources.
18.	Provide the **Advantages of Direct Investing?** • An investor avoids paying ongoing management fees to an external manager. • Direct investing allows an investor to create a portfolio of investments that suits their needs. • Direct investing gives investors the utmost flexibility and control over their investments.
19.	Assess the **Disadvantages of Direct Investing?** • Direct investing requires a greater level of investment expertise. • A direct investor won't enjoy the diversification benefits of fund investing. • Direct investing requires more significant levels of due diligence because of the absence of a fund manager. • Compared to fund investing, it requires a higher minimum capital.

Besides an appraisal, academics, practitioners, and industry associations develop repeat sales indices based on the transaction. The repeat sales index tracks the price change when a repeat sale is executed. However, sample selection bias affects repeat sale indexes.

For efficient use of indexes, investors should learn how to construct them. In addition, investors should know the indexes' limitations.

113. Clarify the **Risk Characteristics of Real Estate?**

Like any other investment, real estate might fail to meet performance expectations. This is because property values are affected by national and global economic factors, interest rate levels, and local factors. Such factors include the ability to fund the management team and the owners' or agents' ability to operate the real estate's underlying properties.

There is a greater risk in investing in distressed property and property development than in well-funded properties or steady operations. In property development, risks include regulation problems (such as environmental regulations), delays in construction, and the cost of unexpected changes in the budget (cost overrun) and financing risk.

Real estate is also characterized by using leverage to maximize investor returns. Leverage can amplify both the losses and profits for equity investors, bringing about the risk of an investor's failure to service debt at maturity.

114. Explore the **Diversification Benefits of Real Estate?**

Real estate is arguably a lucrative investment since it creates a high and stable current income. In real estate, income generation is a more prevalent source of income than capital appreciation. For this reason, real estate is associated with lower risks because it is linked to long-term leases. The credit quality of the tenants determines the reliability of the rent.

The correlation of real estate with other assets makes it a perfect investment for diversification.

115.	Describe **Infrastructure Investments**? **Infrastructure investments** consist of real, long-lived, and capital-intensive assets. They provide essential services to the public. Examples include airports and healthcare facilities.
116.	List the **Characteristics of Infrastructure Investments?** • They have a highly leveraged financial structure. The use of leverage improves investor returns. • Infrastructure investments are usually strategically significant. • Investments in infrastructure are usually monopolistic and regulated. Consideration of regulations such as ESG (Environmental, Social, and Governance) is common. • They have steady long-term cashflows, adjusted to reflect economic growth and inflation. • Infrastructure investment involves significant capital investments. • By definition, infrastructure investments have long operating lives. • They have well-defined risks.
117.	State the **Types of Infrastructure Investments?** Infrastructure investments are classified based on the underlying assets. From a broader perspective, infrastructure investments are classified as follows: • Social infrastructure assets. • Economic infrastructure assets.
118.	Discuss **Social Infrastructures?** **Social infrastructures** focus on human activities such as education, healthcare, and correctional facilities. Social

	infrastructures are usually administered by a public authority or private institution established by the same public authority.
119.	Identify **Economic Infrastructure?** **Economic infrastructure** assets reinforce economic activity and include three types of assets: • **Transportation assets:** such as roads, bridges, tunnels, and airports. • **ICT assets include** databases, broadcasting systems, and information-transmitting assets. • **Utility and energy assets** that generate power and produce potable water.
120.	Contrast between **Greenfield and Brownfield Investment**? **Greenfield investment** defines investments in infrastructure assets that are yet to be built. **Brownfield investing** is used when investing in existing infrastructure assets.
121.	Elaborate on **Forms of Infrastructure Investments?** Infrastructure investments can assume several forms, impacting liquidity, cash flow, and income streams. Investment in infrastructure can take the form of direct or indirect investments. **Investing directly** in the underlying assets provides oversight. It, however, involves huge capital investment and attracts concentration and liquidity risks, given that assets ought to be managed and operated. **Indirect infrastructure investment** involves investment vehicles such as infrastructure funds, infrastructure ETFs, and company shares.
122.	State the **Return Characteristics of infrastructure investment**?

	There is an imminent risk in infrastructure development that the revenues may diverge from expectations. Moreover, investment in infrastructure usually involves leverage. This occasions financial risk. Low-risk infrastructure investments usually yield steady cashflows and higher dividend payout ratios. The downside is that infrastructure investments have less growth potential and are expected to provide lower returns.
123.	Provide the **Risk Characteristics of infrastructure investment**? In infrastructure investment, there is an imminent risk that the revenues may diverge from expectations. Moreover, investment in infrastructure usually involves leverage, therefore, financial risk. There are **three main risk-return profiles** of infrastructure: • **Early-stage investments**: These are investments in greenfield projects. While they have a high-risk profile, they also have higher return prospects. • **Mid-stage investments**: These are investments in existing projects. They also have high return prospects. However, compared to early-stage investments, their return prospects are lower. • **Late-stage investments**: These are investments in projects that have existed for a few years. Their returns are mainly from yield.
124.	State the **Types of Risks in infrastructure investments**? There are two main types of risks in infrastructure investments: • Performance risks. • Structural risks.
125.	Elaborate on **Performance Risk and its Management?** • **Demand/volume risk**: Payments are based on the availability of an asset instead of its use.

	• **Operational risk**: Engage respected contractors in operation and maintenance contracts and give them incentives to meet agreed-upon performance requirements. • **Construction risk**: To reduce construction cost overruns and delays, enter into fixed-price time-certain contracts with credible and competent contractors.
126.	Analyze **Structural Risk and its Management?** • **Financing/interest rate risk**: Use interest rate swaps. • **Regulatory risk**: Perform proper due diligence and ensure clear contract provisions govern this risk. • **Political risk**: Invest in political risk insurance to be protected, when necessary, from occurrences such as expropriation, political unrest, and sovereign debt default. • **Currency risk**: Make that agreement include suitable mechanisms to allow tariff adjustments in case of significant currency fluctuations. • **Tax/profit repatriation risk:** Make sure that agreements have proper mechanisms in place to allow for a tariff adjustment and/or other compensation in the event of major changes to the legislation or regulations affecting tax and profit repatriation.
127.	Clarify the **Diversification Benefits of Infrastructure Investment?** One of the aims of infrastructure investors is to earn steady long-term cashflows that reflect the effects of economic growth and inflation. Moreover, they may be seeking capital appreciation. As a result, infrastructure investments generate additional income streams and increase portfolio diversification since they have a low correlation with existing investments.

SECTION H – ALTERNATIVE INVESTMENTS

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	Traditional is not a synonym for common.	**TRUE. Traditional investments** are: • Long-only, publicly traded investments in stocks. • Long investments in publicly traded bonds. • Cash.
2.	Alternative is not a synonym for unusual.	**TRUE. Alternative investments** often supplement traditional investments in a portfolio and include assets such as derivatives, real estate, short-selling stocks, commodities, and anything outside the traditional pool of investment classes.
3.	Partnerships in alternative investments consist of a GP and some LPs.	**TRUE.** Partnerships in alternative investments typically consist of a **General Partner (GP)** and some **Limited Partners (LPs)** where the GP is the fund manager while the LPs are the accredited investors.
4.	Alternative investments have a relatively high correlation of returns and more transparency and regulation than traditional investments.	**FALSE.** Alternative investments have a relatively **low** correlation of returns and **less** transparency and regulation than traditional investments.
5.	Alternative investments have inadequate historical risk and return data.	**TRUE. Characteristics of alternative investments** are: • Narrow specialization of the investment managers. • Unique tax and legal issues. • Higher fees are usually made of performance fees.

		• Usually concentrated portfolios. • Limitations on redemptions.
6.	Limited partners bear unlimited liability and can manage multiple funds.	**FALSE.** Limited partners usually assume the **investment risks,** while the **general partner** manages the business and bears unlimited liability given any unfavorable eventuality and can manage multiple funds.
7.	Hedge funds are a category of alternative investments.	**TRUE. Hedge funds** employ numerous strategies to offer their investors absolute returns, are actively managed, often by star investors, and employ aggressive strategies across asset classes. Many will use derivatives, high leverage, arbitrage, and other investment strategies unavailable to traditional fund managers.
8.	Private capital funds employ alternative investment strategies quite different from those of a typical mutual fund.	**TRUE. Private capital funds** are private investment vehicles, typically managed by the general partner on behalf of the limited partners. Common strategies include leveraged buyouts, venture capital, development capital, and distressed investing.
9.	Limited partners perform active roles and are part of the fund management.	**FALSE.** Limited partners perform **passive roles** and **are not part** of the fund management; therefore, the general partner makes all decisions and runs the fund.
10.	Private capital can be in the form of two forms.	**TRUE.** The **two forms of private capital** are: a. **Private equity** investors might participate directly or indirectly through private equity funds and invest in new and established businesses. b. **Private debt** is primarily provided to private entities. Examples include direct lending, mezzanine, venture, and distressed debt.

11.	Mortgages and construction financing are public debt investments.	**FALSE.** Real estate is typically owned via a title, often registered by the local government in the land registry. Mortgages and construction financing are **private debt investments**, while MBS and collateralized obligations are public debts available to investors.
12.	An LPA governs the relationship between the general partners and the limited partners.	**TRUE.** A **Limited Partnership Agreement** (LPA) is a legal document that defines the rules of the partnership and institutes the structure guides for the entire fund's operations.
13.	Fund investing can be termed as direct investing in alternative assets.	**FALSE.** Fund investing can be termed as **indirect investing** in alternative assets because fund investors' decisions are based on whether to invest in the fund.
14.	Direct lending involves private debt investors offering capital to borrowers directly.	**TRUE. Direct lending** involves private debt investors offering capital to borrowers directly. In return, the private debt investors receive interest, the original principal, and other required repayments.
15.	There are five primary valuation techniques in real estate and private equity.	**TRUE. Primary valuation techniques** in real estate and private equity include: a. The Internal Rate of Return (IRR). b. The Multiple Invested Capital (MOIC). c. Quartile ranking. d. The cape rate. e. The capital loss ratio.
16.	Direct real estate investments usually have short holding periods and low	**FALSE.** Direct real estate investments usually have **long holding periods**, **high transaction fees**, and diverse landlord-related responsibilities, including environmental assessments and other legal concerns.

	listed investment vehicles.	
116.	Infrastructure investments consist of real, long-lived, and capital-intensive assets.	**TRUE. Infrastructure investments** consist of real, long-lived, and capital-intensive assets and provide essential services to the public. Examples include airports and healthcare facilities.
117.	Repeat sales indices are developed by academics and industry associations based on the transaction in question instead of an appraisal.	**TRUE.** The **repeat sales index** tracks the price change when a repeat sale is executed, but repeat sale indexes are affected by sample selection bias. For efficient use of indexes, investors should learn how to construct them and their corresponding limitations.
118.	Capital appreciation is a more prevalent source of income than income generation in real estate.	**FALSE.** In real estate, **income generation** is a more prevalent source of income than capital appreciation. For this reason, real estate is associated with lower risks because it is linked to long-term leases. The credit quality of the tenants determines the reliability of the rent.
119.	Low correlation with other assets is a disadvantage of direct investment in real estate.	**FALSE.** Real estate's low correlations with other assets are **not** a disadvantage **but a key reason** for investing in real estate.
120.	Infrastructure investments are classified as social and economic	**TRUE.** Infrastructure investments are classified based on the underlying assets. From a broader perspective, infrastructure investments are classified as follows:

	infrastructure assets.	a. **Social infrastructure assets** - focus on human activities such as education, healthcare, and correctional facilities. b. **Economic infrastructure assets** reinforce economic activity, including transportation, ICT, utility, and energy.
121.	Many investors are investing in infrastructure as part of supply-side growth.	**FALSE.** Many investors are investing in infrastructure as part of **demand-side growth.** Meanwhile, government investment opportunities create supply-side growth. This is done by increasing infrastructure financing as well as the privatization of some services.
122.	Brownfield investment defines investments in infrastructure assets that are yet to be built.	**FALSE. Greenfield investment** defines investments in infrastructure assets yet to be built, whereas brownfield investing is used when investing in existing infrastructure assets.
123.	Investment in infrastructure usually involves leverage.	**TRUE. Financial, operational, and construction risks** are always present in infrastructure investment. Other risks include regulatory, currency, political, and profit-repatriation risks.
124.	High-risk infrastructure investments usually yield steady cashflows and higher dividend payout ratios.	**FALSE.** Infrastructure **low-risk investments** usually yield steady cashflows and higher dividend payout ratios. The downside is that infrastructure investments have less growth potential and are expected to provide lower returns.
125.	Infrastructure investment has a highly leveraged financial structure.	**TRUE. Characteristics of infrastructure investments** include: • Defined risks. • Long operational lives. • Significant capital investments. • Stable long-term cash flows. • Monopolistic and regulated. • Strategically important.

SECTION H – ALTERNATIVE INVESTMENTS

FILL IN THE BLANKS

1. Real estate investments often include ___ as an income component.

2. Hedge funds typically charge a performance fee, which is often ___ of the profits.

3. Private equity investments are generally ___ compared to public equity investments.

4. The two main types of hedge fund strategies are ___ and ___ strategies.

5. Commodities are often used as a ___ against inflation.

6. The main advantage of investing in private equity is the potential for ___ returns.

7. Real estate investment trusts (REITs) are required to distribute at least ___ of their taxable income to shareholders.

8. ___ are financial contracts that derive their value from the performance of an underlying asset.

9. A ___ is an agreement to buy or sell an asset at a future date for a predetermined price.

10. Private equity funds usually have a lock-up period of ___ years.

11. A ___ strategy in hedge funds aims to profit from price differences between related securities.

12. The ___ ratio is commonly used to evaluate the risk-adjusted return of hedge funds.

13. Infrastructure investments typically have a ___ duration and generate stable cash flows.

14. ___ is a measure used to assess the volatility of hedge fund returns.

15. The ___ method is often used to value private companies in private equity.

16. Commodities are often categorized into ___ and ___ commodities.

17. ___ is the main risk associated with commodity investments.

18. ___ is a common real estate investment strategy that involves improving a property to increase its value.

19. ___ hedge funds take both long and short positions in equities.

20. ___ is the term used to describe the practice of using borrowed money to increase the potential return of an investment.

21. A ___ is a private equity investment in a new or young company with high growth potential.

22. ___ refers to the process of exiting a private equity investment, often through an IPO or sale.

SECTION H – ALTERNATIVE INVESTMENTS

ANSWER KEY

1. Rental income

2. 20%

3. Illiquid

4. Absolute return, relative value

5. Hedge

6. Higher

7. 90%

8. Derivatives

9. Futures contract

10. 7-10

11. Arbitrage

12. Sharpe

13. Long

14. Standard deviation

15. Discounted cash flow (DCF)

16. Hard, soft

17. Price volatility

18. Value-add

19. Long/short equity

20. Leverage

21. Venture capital

22. Exit strategy

23. Market-neutral

24. Residential, commercial

25. Futures contracts

26. Limited partnership

27. Real Estate Investment Trust (REIT)

SECTION H – ALTERNATIVE INVESTMENTS

WORD SEARCH PUZZLES

Section H - Quiz 1

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R F K W J A B N O I T A I C E R P E D L
E Q L A B O L G K S L Z P O O V T Q E Z
N A N B V U A O X W C P L F W L U A N K
E V O P L T G J B A Z F Y G Q G S P S Y
W U I I K A X V D L E I F N E E R G O Q
A X T H P C R W Y G E T A R T S E I S W
B Q A S V P E T U L O S B A M T L S E J
L A D R E R E N U N N W T E A O O P I I
E R R E G O B O K E E A Z T F K C R T H
N B A N A J L N C L N Z S T S F S E I M
E I W T R E D A G C A E R W L U P C D A
V T K R E C I N S N U O O X I T E I O N
E R C A V T I Z I I P P A X E U C O M A
N A A P E L T N P Y A Y A R T R U U M G
T G B V L D E K Z W T R U N R E L S O E
R E O E O F G Q J T K T P N C S A J C M
V N S D C P L P O L N I Y P Z Y T G Y E
R K M A C R O P O E A N W T A O I O Z N
N E V D G F S O V D L E I F N W O R B T
V A L U A T I O N N O I S S E C N O C J
```

1. Absolute **2.** Leverage **3.** Neutral **4.** Event **5.** Global **6.** Macro **7.** Arbitrage

8. Selling **9.** Venture **10.** Mezzanine **11.** Management **12.** Portfolio **13.** Strategy

14. Occupancy **15.** Depreciation **16.** Lease **17.** Appraisal **18.** Valuation **19.** Estate

20. Spot **21.** Futures **22.** Backwardation **23.** Speculation **24.** Precious

25. Commodities **26.** Partnership **27.** Renewable **28.** Project **29.** Greenfield

30. Brownfield **31.** Concession

SECTION I – PORTFOLIO MANAGEMENT

STUDY POINTS

S.No	Description
1.	Describe the **Portfolio Approach to investing?** Investors have to ensure their investments **achieve their future needs**. Irrespective of the future financial goals, a portfolio approach to investment decision-making is important. It enables an investor to create a **diversified** investment portfolio.
2.	Clarify **Portfolio Diversification?** The benefits of a diversified portfolio are best summarized by the anecdotal wisdom of “don’t put all your eggs in one basket.” If a portfolio is too heavily allocated to one individual security, and that security fails for some reason, an investment portfolio can be reduced to zero. Diversification allows investors to spread some of the downside risk associated with any one investment position without necessarily decreasing the expected rate of return.
3.	Assess **Risk associated with Diversified Portfolios**? Diversified portfolios have a lower portfolio risk or volatility (as measured by standard deviation) than any individual position within the portfolio. The statistical relationship or interactions between individual portfolio positions (correlation) lowers the overall volatility by including multiple positions within a portfolio.
4.	Explain **Downside Protection?** Portfolio diversification is not an absolute failsafe for investors. A diversified portfolio may not protect an investor from losses during market turmoil. The reduction in risk as a result of portfolio diversification comes about due to the uncorrelated nature of the individual portfolio holdings. However, this correlation is not fixed, and when markets are stressed, previously uncorrelated assets can become correlated.
5.	Elaborate on the **Modern Portfolio Theory?** Although adopting a portfolio approach to investing seems intuitive, a theory exists behind it. The diversification concept

	follows the work of Harry Markowitz's 1952 publication and is known as Modern Portfolio Theory (MPT). The principle concept is that investors should not only hold portfolios but also focus on the relationship among the individual securities within the portfolio. MPT has its limitations but continues to be a cornerstone for portfolio managers.

6.	List the **Types of Investors?**
	
	The needs of investment clients vary widely, but we can group investors into two broad categories:
	
	• Individual investors.
	
	• Institutional investors.
	  ○ Endowments and foundations.
	  ○ Banks.
	  ○ Insurance companies.
	  ○ Investment companies.
	  ○ Sovereign wealth funds.

7.	Describe **Individual Investors?** **Individual investors** may be investing either for short-term or long-term goals. A short-term investment goal may be their children's education or purchasing a house. Long-term investment goals revolve around providing income for retirement. The implication is that some investors are focused on capital growth and look for investments with potential for capital appreciation while retirees want income-producing assets. Organizing a portfolio for an investor will also depend on their financial circumstances, such as home ownership, employment prospects, and other financial obligations.
8.	Explore **Institutional Investors?** There are many different types of institutional investors. Indeed, institutional assets constitute a major portion of the investment market.
9.	Discuss **Endowments and Foundations?**

	The typical objective of an endowment or foundation is to maintain the real (inflation-adjusted) capital value of the fund in perpetuity, as well as generate income to provide financial support for its beneficiaries.
10.	Identify **Banks?** **Banks** hold deposits and make loans, which can lead to excess reserves – in this case, a bank holds more deposits than the loans it has extended. Banks can invest these reserves, which typically have to be held in conservative and liquid assets like fixed-income and money market instruments. The bank's objective is to earn a rate of return in excess of its interest rate on its deposits.
11.	Assess **Insurance Companies?** **Insurance companies** receive premiums from the insurance policies they write. They need to invest these premiums to ensure sufficient funds are available to pay for insurance claims when these arise. Their investments are often conservative and aware of the time frame over which claims may arise.
12.	Clarify **Investment Companies?** **Investment companies** manage mutual funds, which are pooled investment vehicles. Mutual funds are seen as an efficient way for individual investors to gain access to a diversified portfolio and benefit from the skills of a professional investment manager. Mutual funds are managed according to the limits and restrictions of their investment mandates.
13.	What are **Sovereign Wealth Funds?** SWFs are government-owned investment funds. Some governments operate to invest the revenues from the natural resources of the country (e.g., oil) for the benefit of future generations, while others manage the assets of the state.
14.	Elaborate on the **Defined Contribution (DC) Pension Plan?** A **defined contribution pension plan** is an investment vehicle in which the amounts invested, or the contributions the employee

	makes to the plan, are defined or specified, but the benefits are not predetermined. The objective of the pension plan is to accumulate wealth by investing a portion of wages while working to provide income during retirement. A DC plan places the investment risk on the employee. The employee is responsible for ensuring their contributions and investment growth are sufficient to provide the desired income upon retirement. The final retirement income in a DC plan depends on the contributions made, investment performance, and the choices made by the employee regarding investment options within the plan.
15.	Determine a **Defined Benefit Pension Plan?** In a **defined benefit pension plan**, the employer must provide certain benefits to employees when they retire. The future benefit is specified or defined. The investment management of DB plans has to consider the timing of its future liabilities or cash flows by assessing the age of its plan members. If, for example, a DB pension plan has a lot of young members, the investment time horizon of the plan may be quite long. The plan investment manager may try to match the cash flow requirements of the plan with cash-flow-producing assets such as bonds. This checks the capacity of portfolio assets to offset portfolio liabilities.
16.	List the **Steps in the Portfolio Management Process?** • The planning step. • The execution step. • The feedback step.
17.	Elaborate on the **Planning Step?** Once the client's objectives and constraints have been established, an investment policy statement (IPS) must be developed. This written document spells out all the investment objectives and constraints that apply to a client's portfolio. It may also contain a

	reference to a benchmark. A benchmark can be used to assess investment performance and evaluate whether the objectives have been achieved.
18.	Discuss the **Execution Step?** The execution step has three stages: **a. Asset Allocation** The analyst or portfolio manager will form a view of the economic and capital market expectations for various available asset classes. This analysis may be top-down, starting with considering the macroeconomics or industry environment and evaluating the asset classes expected to perform well given the environment. Or, the analysis may be bottom-up. Instead of looking at macroeconomics or industry data, this analysis focuses on company-specific factors. A decision will be made on allocating assets to the available asset classes. Asset classes can include equities, bonds, cash, real estate, commodities, hedge funds, and private equity. **b. Security Analysis** Top-down and bottom-up views can be combined in selecting individual securities to assess the level of returns and risk. This informs the assignment of a valuation to securities being considered for portfolio inclusion. **c. Portfolio Construction** A diversified portfolio can be constructed using the investment policy statement (IPS), the desired asset allocation, and security analysis. Besides achieving investment performance, risk management is an important focus of the portfolio construction process. The IPS will outline a client's risk tolerance, and the portfolio manager must ensure the portfolio is aligned with this risk profile. The trades will be implemented once the portfolio manager has chosen securities to buy and in which quantities. Often, this trade is carried out by a specialized trade execution team or external stockbroker.
19.	Clarify the **Feedback Step?**

	• **Variety:** The number of types/formats of data. The data could be structured (e.g., SQL tables or CSV files), semi-structured (e.g., HTML code), or unstructured (e.g., video messages).
261.	Distinguish **Structured Data and Unstructured Data?** **Structured data** refers to information with a high degree of organization. Items can be organized in tables and stored in a database where each field represents the same type of information. **Unstructured data** refers to information with a low degree of organization. Items are unorganized and cannot be presented in tabular form, such as text messages, tweets, and emails.
262.	List the **Sources of Data?** • Financial markets: equity, swaps, futures, options, and other derivatives. • Businesses: financial statements, credit card purchases, and commercial transactions. • Governments: payroll, economic, trade, employment data, etc. • Individuals: product reviews, credit card purchases, social media posts, etc. • Sensors: shipping cargo information, traffic data, satellite imagery. • The Internet of Things: data generated by ‘smart‘ buildings through fittings such as CCT V cameras, vehicles, home appliances, etc.
263.	Clarify **Artificial Intelligence?** In broad terms, **artificial intelligence** refers to machines that can perform tasks in " intelligent ways.” It has much to do with developing computer systems that exhibit cognitive and decision-making abilities comparable or superior to humans. It is the broader concept of machines being able to carry out tasks in a way

	that we would consider "smart." AI can take the form of "if-then" statements or complex statistical models that map raw sensory data to symbolic categories.
264.	Explore **Machine Learning?** **Machine learning** is a current application of AI. It revolves around the idea that we should give machines access to data and let them learn independently without further human intervention. It's the idea that when exposed to more data, machines can make changes independently and come up with solutions to problems without relying on human expertise. Machines improve their performance over time.
265.	Describe the **Types of Machine Learning?** **Supervised Learning** Under supervised learning, computers learn to model data based on labeled training data containing inputs and the desired outputs. Each training example has one or more inputs and a desired output. Trying to predict the performance of a stock (up, down, or level) during the next business day can be modeled through supervised learning. **Unsupervised Learning** Under unsupervised learning, computers are only given input data and tasked with describing the data, for instance, by grouping or clustering data points. In this instance, computers learn from data that has not been labeled or categorized. The computers then "react" based on the data's presence or absence of commonalities. Trying to group companies based on their financial and not geographical or industrial characteristics would be a good example of unsupervised learning.
266.	Evaluate the **Applications of Fintech in the Investment Industry?** **Analysis of Loads of Data**

We now have crowd-sourced content services that analyze large datasets of security prices, financial statements, economic indicators, and qualitative bits of information that a portfolio manager can integrate into the investment decision-making process.

Complex algorithms have been developed to scour social media and sensor networks in search of consumer sentiments and product performance data that can be integrated into a manager's buy or sell decisions.

Automated Trading

Artificial intelligence has enabled the analysis of extremely large datasets that humans could only have analyzed just a couple of years back. For instance, we now have systems built to identify systematic investment strategies and automatically execute multiple trades over several financial markets worldwide. As a result, investment banking departments that used to accommodate hundreds of traders have now been reduced to just a few individuals supported by computer engineers.

Analytical Tools

We have machine learning algorithms built to sift through enormous amounts of financial data — company filings, earning calls, profit warnings, etc. These algorithms can then unearth trends and identify buy or sell stocks.

Automated Advice

Robo-advisors — internet-based intelligence models that provide investment advice with minimal human intervention – have revolutionized wealth management. These models cost much less than traditional adviser models and human resources.

267. Define a **Distributed Ledger**?

A distributed ledger is a database held and updated independently by each participant (or node) in a large network. Rather than having a central authority, records are independently constructed and held by every node (computer).

	Each node can process a transaction and come up with a conclusion. All the nodes then "vote" on the conclusion. If the majority agrees with the conclusion, the transaction is completed successfully, and all nodes maintain their identical copy of the ledger. There is no need for a centralized databank as in the case of a traditional ledger.
268.	What makes **DLT So Good?** DLT has several features that make it a favorite among investment managers: **Cryptography** This refers to algorithmic encryption of data such that it is inaccessible to an unauthorized party. Before any transaction can be approved, some computers on the network must solve a cryptographic problem. As a result, DLT has a high level of security and integrity. **Smart Contracts** These are computer programs that self-execute based on pre-specified terms and conditions agreed to by the parties to a contract. In the derivatives market, for instance, the system could be seen such that daily settlements are made automatically in line with the day's market movements. If a counterparty defaults on a payment, collateral can be instantaneously transferred to the relevant party. **Blockchain** Blockchain is a digital ledger in which information, such as changes in ownership of an asset, is recorded sequentially within blocks that are then linked together and secured using cryptography. Each block is made up of a group of transactions that are linked to a previous block. Blockchain enables the distribution of information while at the same time ensuring that it's not copied.
269.	Clarify the **Applications of Distributed Ledger Technology to Investment Management?** **Cryptocurrencies**

A cryptocurrency is an electronic currency that enables user payments without passing through a central authority, such as a bank or payment gateway. Cryptocurrencies allow near-real-time transactions between parties, with verification based on cryptography.

Most cryptocurrencies use open DLT systems where a decentralized distributed ledger records and verifies all transactions. Although most central banks worldwide have been reluctant to integrate cryptocurrencies into the mainstream monetary transaction structure, there are signs that this may change.

Tokenization

Tokenization is converting rights to an asset, say stocks, bonds, or even a building, into a digital token on a blockchain. DLT streamlines this process by creating a single digital record of ownership to verify ownership title and authenticity, including all historical transactions related to the asset.

Post-trade Clearing and Settlement

Post-trade transactions are known to be complex and time-consuming since they require multiple interactions between counterparties and financial intermediaries. DLT can streamline the process by providing near-real-time trade verification, reconciliation, and settlement.

Seamless compliance

In the face of ever-increasing rules and regulations in the field of investment, DLT can help firms ensure compliance by enabling near-real-time review of transactions. This eliminates the need for large post-trade monitoring teams and creates operational efficiency.

SECTION I – PORTFOLIO MANAGEMENT

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	Investors can be grouped into two broad categories.	**TRUE.** Different investors will have varying investment time horizons, tolerance for portfolio risk, income, and liquidity needs. The following are the **broad categories of investors:** a. Individual investors. b. Institutional investors.
2.	Diversified portfolios have a higher portfolio risk or volatility than any individual position within the portfolio.	**FALSE.** Diversified portfolios have a **lower** portfolio risk or volatility than any individual position within the portfolio. Due to the statistical relationship or interactions between individual portfolio positions, the overall volatility is lowered by including multiple positions within a portfolio.
3.	Modern portfolio theory continues to be a cornerstone for portfolio managers.	**TRUE. Modern portfolio theory** states that investors should not only hold portfolios but also focus on the relationship among the individual securities within the portfolio.
4.	Institutional investors are of many different types.	**TRUE.** There are many different types of institutional investors with different financial objectives. Some of the **institutional investors** are: • Pension funds. • Endowments. • Charities. • Banks. • Insurance companies. • Investment funds. • Sovereign wealth funds.
5.	A portfolio approach to investing protects	**FALSE.** A diversified portfolio **reduces risk without compromising investment returns** but does not completely protect an investor

	against investment losses.	against investment losses during market turmoil.
6.	Individual investors may be investing either for short-term or long-term goals.	**TRUE.** A **short-term investment goal** may be their children's education or purchasing a house. **Long-term investment goals** revolve around providing income for retirement.
7.	A bank holds less deposits than the loans it has extended.	**FALSE.** Banks hold deposits and make loans, which can lead to excess reserves; in this case, a bank holds **more** deposits than the loans it has extended.
8.	The bank's objective is to earn a rate of return less than the rate of interest it pays on its deposits.	**FALSE.** The bank's objective is to earn a rate of return **in excess of** its interest rate on its deposits.
9.	An endowment or foundation aims to generate income to support its beneficiaries financially.	**TRUE.** The **objective of an endowment or foundation** is to maintain the real (inflation-adjusted) capital value of the fund in perpetuity as well as generate income to provide financial support for its beneficiaries.
10.	Excess reserves held by banking institutions are usually invested in real estate and other tangible assets.	**FALSE.** Banks invest their excess reserves in conservative, liquid assets such as **fixed-income and money-market instruments**.
11.	The investments of insurance companies are often conservative and aware of the investment time frame over which claims may arise.	**TRUE.** Insurance companies receive **premiums from the insurance policies** they write and need to invest these premiums to ensure sufficient funds are available to pay for insurance claims when they arise.
12.	Investment companies manage mutual funds, which are	**TRUE. Mutual funds** are seen as an efficient way for individual investors to gain access to a diversified portfolio and benefit from the skills of a professional investment manager.

	pooled investment vehicles.	These funds are managed according to the limits and restrictions of their investment mandates.
13.	Contributions and benefits are specified in the defined contribution pension plan.	**FALSE.** A defined contribution pension plan is an investment vehicle in which the amounts invested or the contributions the employee makes to the plan are defined or specified, **but the benefits are not**.
14.	Sovereign wealth funds are government-owned investment funds.	**TRUE. Sovereign wealth funds** are government-owned investment funds. Some governments operate to invest the revenues from the natural resources of the country for the benefit of future generations, while others manage the assets of the state.
15.	The employer assumes the investment risk in a defined contribution pension plan.	**FALSE.** Since the benefits of a defined contribution plan are not defined, the **employee** assumes the investment risk and needs to ensure the contributions and investment growth are sufficient to provide the required income upon retirement.
16.	There are three steps in a portfolio management process.	**TRUE.** Once the type of investment client has been determined along with their financial goals, the following **series of steps** should be followed to ensure those goals and needs are met: a. The planning steps. b. The execution steps. c. The feedback steps.
17.	The employer and the employee share the investment risk in a defined benefit pension plan.	**FALSE.** In a defined benefit pension plan, the employer must provide certain benefits to employees when they retire; therefore, the **employer** must ensure sufficient assets to match the future defined benefits payable to employees upon retirement.
18.	An investment policy statement must be developed	**TRUE.** An **investment policy statement** is a written document that spells out all the investment objectives and constraints that apply to a client's portfolio. It may also

	algorithmic trading.	automatically and this has been their biggest selling point.
344.	Robo-advisors are internet-based intelligence models.	**TRUE. Robo-advisors** are internet-based intelligence models that provide investment advice with minimal human intervention and have revolutionized wealth management.
345.	A distributed ledger is a database held and updated independently by each participant in a large network.	**TRUE.** A **distributed ledger** is a database held and updated independently by each participant (or node) in a large network. Records are independently constructed and held by every node (computer). Each node can process a transaction and come up with a conclusion. All the nodes then vote on the conclusion. If the majority agrees, the transaction is completed successfully, and all nodes maintain identical copies of the ledger.
346.	Robo-advisors are costlier than traditional adviser models and human resources.	**FALSE.** Robo-advisor models **cost much less than** traditional adviser models and human resources.
347.	Distributed ledger technology has several features that make it a favorite among investment managers.	**TRUE. Distributed ledger technology** has several features that make it a favorite among investment managers: • Cryptography. • Smart contracts. • Blockchain.
348.	Betterment is one of the oldest Robo-advisors.	**TRUE. Betterment** is one of the oldest Robo-advisors. Once signed up, the investor completes a short survey about their investment needs and risk tolerance. The information is then used to develop an automatic investment plan. Built-in systems automatically adjust the investor's portfolio to maintain an optimal level of risk.
349.	Wealthfront is a Robo-advisor that builds investors'	**FALSE.** Wealthfront builds investors' **free financial plans** and **automates the investment process at a fee**. Sophisticated

	financial plans at a fee.	models combine investor data with relevant external data to build a portfolio that reflects the investor's risk appetite.
350.	Cryptography refers to the algorithmic encryption of data.	**TRUE. Cryptography** refers to algorithmic encryption of data such that it is inaccessible to an unauthorized party. Before any transaction can be approved, some computers on the network must solve a cryptographic problem. As a result, distributed ledger technology has high security and integrity.
351.	Distributed ledger technology enables the recording of value interactions without a central coordinating entity.	**TRUE.** Distributed ledger technology uses **independent computers** to record, share, and sync transactions in their respective electronic ledgers. This has eliminated the need for a centralized databank, as in a traditional ledger.
352.	Smart contracts are computer programs that self-execute based on post-specified terms and conditions.	**FALSE.** Smart contracts are computer programs that self-execute based on **pre-specified** terms and conditions agreed to by the parties to a contract. In the derivatives market, the system could be seen such that daily settlements are made automatically in line with the day's market movements. If a counterparty defaults on a payment, collateral can be instantaneously transferred to the relevant party.
353.	Blockchain enables the distribution of information while at the same time ensuring that it's not copied.	**TRUE. Blockchain** is a digital ledger in which information is recorded sequentially within blocks linked together and secured using cryptography. Each block is made up of a group of transactions that are linked to a previous block.
354.	Cryptocurrencies allow near-real-time transactions between parties, with verification	**TRUE.** A **cryptocurrency** is an electronic currency that enables user payments without passing through a central authority, such as a bank or payment gateway. Most cryptocurrencies use open distributed ledger

	based on cryptography.	technology systems where a decentralized ledger records and verifies all transactions.
355.	Distributed ledger technology streamlines the tokenization process.	**TRUE. Tokenization** is converting rights to an asset, say stocks, bonds, or even a building, into a digital token on a blockchain. Distributed ledger technology streamlines this process by creating a single digital record of ownership to verify ownership title and authenticity, including all historical transactions related to the asset.
356.	Distributed ledger technology has the ability to streamline the post-trade clearing and settlement process.	**TRUE. Post-trade transactions** are known to be complex and time-consuming since they require multiple interactions between counterparties and financial intermediaries. Distributed ledger technology can streamline the entire process by providing near-real-time trade verification, reconciliation, and settlement.
357.	Distributed ledger technology can enable seamless compliance.	**TRUE.** Distributed ledger technology can help firms ensure **compliance by enabling near-real-time review of transactions**. This eliminates the need for large post-trade monitoring teams and creates operational efficiency.

SECTION I - PORTFOLIO MANAGEMENT

FILL IN THE BLANKS

1. The main objective of portfolio management is to achieve a balance between ______ and ______.

2. The Capital Asset Pricing Model (CAPM) defines the relationship between expected return and ______ risk.

3. According to the CAPM, the expected return on a security is equal to the risk-free rate plus a ______ premium.

4. Systematic risk is also known as ______ risk.

5. Unsystematic risk is ______ through diversification.

6. The risk that can be diversified away in a portfolio is known as ______ risk.

7. Beta measures the sensitivity of a stock's returns to ______ returns.

8. A portfolio with a beta greater than 1 is considered to be ______ risky than the market.

9. A portfolio with a beta of 1 has the ______ level of risk as the market.

10. The efficient frontier represents the set of portfolios that offer the ______ level of return for a given level of risk.

11. The capital market line (CML) represents the risk-return trade-off for ______ portfolios.

12. The Security Market Line (SML) represents the expected return of assets based on their ______.

13. The ______ diversification principle states that adding more assets to a portfolio generally reduces risk.

14. Modern portfolio theory (MPT) was developed by ______.

15. The Sharpe ratio measures the ______ return of a portfolio per unit of risk.

16. The formula for the Sharpe ratio is (Portfolio Return - Risk-Free Rate) divided by ______.

17. The ______ ratio is used to measure the risk-adjusted return of a portfolio.

18. The Treynor ratio is similar to the Sharpe ratio but uses ______ in the denominator.

19. The Jensen's alpha measures the ______ return of a portfolio compared to its expected return based on CAPM.

20. If a portfolio has a positive alpha, it has ______ the market.

21. The risk-free rate is typically represented by the return on ______ securities.

22. Asset allocation refers to the process of dividing investments among different ______.

23. Strategic asset allocation is a ______-term investment strategy.

24. Tactical asset allocation involves adjusting the portfolio based on ______ market conditions.

25. Diversification helps in reducing ______ risk.

26. Correlation measures the degree to which two assets move in ______.

27. If two assets have a correlation of +1, they move ______.

28. If two assets have a correlation of -1, they move ______.

29. A correlation of 0 between two assets indicates that their returns are ______.

30. The Markowitz efficient frontier includes all portfolios that provide the ______ level of return for a given level of risk.

31. Risk aversion refers to the preference for ______ risk.

32. A risk-averse investor prefers portfolios that lie on the ______ of the efficient frontier.

33. An indifference curve represents combinations of risk and return that provide the same level of ______ to an investor.

SECTION I - PORTFOLIO MANAGEMENT

ANSWER KEY

1. risk, return

2. systematic

3. risk

4. market

5. reduced or eliminated

6. unsystematic

7. market

8. more

9. same

10. highest

11. efficient

12. beta

13. principle of

14. Harry Markowitz

15. excess

16. standard deviation

17. Sharpe

18. beta

19. excess

20. outperformed

21. government

22. asset classes

23. long

24. short-term

25. unsystematic

26. relation to each other

27. perfectly together

SECTION I - PORTFOLIO MANAGEMENT

WORD SEARCH PUZZLES

Section I - Quiz 1

W	C	E	T	G	V	Z	Y	E	D	D	Z	N	M	C	E	L	Y	T	S
G	K	Q	N	R	O	J	E	K	J	P	B	Q	O	H	W	U	J	E	J
M	C	U	O	F	E	I	X	H	B	S	C	E	N	A	R	I	O	E	Z
O	P	B	I	S	G	Y	X	S	D	Y	N	A	M	I	C	P	S	I	B
P	O	Q	T	Y	G	J	N	N	E	O	Q	S	A	E	A	C	I	N	S
W	T	I	A	W	U	B	O	O	I	M	T	T	C	R	O	W	A	N	Y
V	A	O	L	P	I	E	D	S	R	R	C	N	I	M	P	E	F	O	A
P	C	T	E	A	C	Y	R	L	E	T	A	T	P	S	C	G	J	I	G
Z	T	P	R	E	Q	E	I	S	P	I	Y	A	S	N	N	N	G	T	N
N	I	Q	R	F	V	A	S	E	R	X	R	T	A	I	Z	N	I	A	I
N	C	E	O	A	T	R	Q	A	A	I	R	R	K	Y	S	F	X	Z	C
O	A	U	C	K	I	Q	V	T	S	A	E	C	O	I	P	W	A	I	N
Z	L	Q	V	X	U	A	A	O	T	L	A	I	K	H	A	R	A	M	A
I	E	I	R	E	R	X	N	E	O	R	L	P	S	G	Z	I	W	I	L
R	F	N	T	P	I	F	G	T	T	O	L	Z	B	U	Y	T	Q	T	A
O	U	U	Z	R	Z	I	S	J	F	G	I	V	H	W	D	A	A	P	B
H	H	W	B	A	C	J	W	T	H	M	O	Z	Z	P	Q	N	L	O	E
L	B	W	W	H	G	E	R	A	H	U	P	X	J	L	A	G	E	L	R
U	C	W	H	S	N	O	I	T	A	V	R	E	S	E	R	P	W	D	A
X	Y	Q	Y	E	P	G	I	J	P	X	L	W	M	S	T	J	M	V	L

1. Portfolio **2.** Sharpe **3.** Correlation **4.** Variance **5.** Optimization **6.** Tolerance

7. Horizon **8.** Tax **9.** Legal **10.** Unique **11.** Aversion **12.** Preservation **13.** Strategic

14. Tactical **15.** Dynamic **16.** Rebalancing **17.** Parity **18.** Tail **19.** Stress

20. Scenario **21.** Treynor **22.** Comparison **23.** Tracking **24.** Style

Section I - Quiz 2

J H F D O P J L O A W G K F H J W Q I V
E R A V M T J E D D Y C K F A P J Y E C
D E A L S O I Y R T P A S S I V E S W G
D R Y N A L A H I S D L O H S E R H T N
C Z I E C Z R L J D T C E P S O R P Z I
I A S F A H I N A K E V I T I N G O C D
R N P P T T O I N D I F F E R E N C E R
J E D I U Y I R Y M S G U Z P C V S P E
E Y D E T T L Q I E U N H G N N E C F H
R P L X X A V Q V N E I O N O E R Z R N
U D O G G E L W E B G K I I I D I D A O
S E C N A N O S S I D C L C T I F F M I
O E C N A I L P M O C A O N A F I A I S
L L O P T L K Q Q N H R F A C N C R N R
C Q A B G E N L V W X T T L I O A A G E
S M H M A S V O F W E Q R A F C T Q T V
I K W D I O C I Z T T X O B I R I P N A
D D E S A T Z Q T K F K P E R E O W U X
B B J E J N P V R C B N L R E V N Y G N
N C R E I T N O R F A M F Y V O Y W T Y

1. Frontier **2.** Optimal **3.** Portfolio **4.** Utility **5.** Indifference **6.** Capital
7. Overconfidence **8.** Aversion **9.** Herding **10.** Prospect **11.** Anchoring **12.** Framing
13. Cognitive **14.** Dissonance **15.** Active **16.** Passive **17.** Index **18.** Tracking
19. Rebalancing **20.** Drift **21.** Thresholds **22.** Compliance **23.** Verification
24. Disclosure **25.** Verification

SECTION J – ETHICS

STUDY POINTS

S.NO	Description
1.	Define **Ethics?** For the investment industry, **ethics** are defined as a standard of conduct valued by the financial sector. These can be expressed via concrete rules of behavior as defined by law or through abstract concepts meant to define the spirit of organizational conduct.
2.	Provide the **Components of the Code of Ethics?** Members of the CFA Institute must: a. Act with integrity, competence, diligence, respect, and ethically with the public, clients, prospective clients, employers, employees, colleagues in the investment profession, and other participants in the global capital markets. b. Place the integrity of the investment profession and clients' interests above their interests. c. Use reasonable care and exercise independent professional judgment when conducting investment analysis, making investment recommendations, taking investment actions, and engaging in other professional activities. d. Practice and encourage others to practice professionally and ethically in a way that will reflect credit on themselves and the profession. e. Promote the integrity of, and uphold the rules governing the capital market. f. Maintain and improve their professional competence and strive to maintain and improve the competence of other investment professionals.
3.	Describe the **Role of a Code of Ethics in Defining a Profession?** The role of ethics within the investment profession is to promote the integrity and viability of global capital markets for the ultimate

	benefit of society. By learning and committing to that code of ethics, each professional contributes to a universal protocol of acceptable behavior. Within the financial sector, there are both Global Investment Practice Standards (GIPS) and CFA Institute Standards of Practice.
4.	Define **Profession?** A **profession** can be defined as an occupational group based on the unique education, specialist knowledge, and framework of practice and behavior that establishes community trust, respect, and recognition.
5.	Clarify the **Establishment of Trust by Professions?** A credible profession is characterized by a strong trust from both the clients and the society at large. Profession builds through the following ways: **Provision of Community Service** The provision of community services creates confidence, professional pride, and professional acceptance. Consequently, a trustworthy professional is substantially flexible and independent of government regulatory bodies when carrying out operations. **Making a Profession Client-based** The trust of a profession is built if it puts its integrity and the interest of clients above their interests. A client-focused profession is one whose actions portray a high level of care, skill, and diligence while prioritizing the interests of the clients. In summary, a profession gains trust if it shows fiduciary duty and a commitment to high-quality care when acting for the benefit of another party. **Normalizing the Practitioner's Behavior** A profession is trusted if it is grounded on codes and standards recognized by regulators and the government under which it is established. The regulatory bodies should understand the profession’s codes, standards, and enforcement. **Developing High Entry Standards for a Profession**

A profession should develop an elaborate entry requirement into a profession since membership in a profession is a sign to the consumers that the professional will deliver high-quality service. Such requirements include expertise, knowledge, technical skills, and ethics.

Possessing a Body of Expert Knowledge

Experienced and skilled practitioners should make available useful knowledge to their members to work resourcefully and ethically based on best practices.

Continuous Education

Having qualified into a profession, its members need ongoing education to accommodate the ever-changing knowledge and technical skills, technology, standards of ethical behavior, and legal and business environment where professional services are needed. Most professional regulatory bodies make it mandatory for the members to continue professional development - undergoing specified new learning each year.

Monitoring Professional Misconduct

Each professional found liable for professional misconduct should be held accountable. This is necessary to maintain the integrity and the reputation of the profession and, hence, trust.

Making a Profession Collegial

Although members of a profession do compete, they should respect each other rights, autonomy, and dignity.

Establishing Profession Oversight Authorities

Although each professional is responsible for maintaining a high level of professional standards and competency, an oversight body is established to make this happen. Moreover, the oversight body is mandated to provide continuous educational resources and information on professional changes as time passes.

Engagement of the Professional Members

	Professional members may help to protect the future of professional values by acting as educators to peers. They achieve this by volunteering to mentor and inspire other young professionals or those who wish to join a profession to develop expertise and ethics. In the long term, the future trust of a profession is protected.
6.	Analyze why **Investment management is an upcoming profession**? • The public understanding of its practice and codes is still growing. • There is a lack of recognition by the regulators and the employers, which prevents the establishment of a profession in investment management. • Not all participants in investment management are professionals since the participants have not engaged in specified training or are members of any recognized authority. This makes it hard for investment management to gain trust.
7.	What is meant by an **Ethical Dilemma**? An **ethical dilemma** occurs whenever there are two or more choices. There will always be the best choice in this given circumstance, even when each option appears to have negative consequences.
8.	State the **Major Areas** in which **ethical dilemmas occur**? • Misrepresentation. • Misconduct. • Fairness. • Loyalty, prudence, and care.
9.	Evaluate the **Difference Between Ethical and Legal Standards?** Often, people equate ethical behavior with legal choices. However, numerous choices may be legal but have no moral standing. It is

	important to remember that all laws stem from a place where the moral covenant has broken down. How control is maintained in these circumstances is to set laws and stipulate punishment. Therefore, the law is reactive. By contrast, ethical conduct is proactive, and standards of practice maintain a high level of morality beyond legality. One may adhere to the law's letter but ignore the law's spirit. For example, IRS regulations repeatedly single out actions with “no legitimate business purpose.” If an investment professional makes choices with no legitimate business purpose to avoid short-term loss (i.e., taxation, fees), then you may be staying within the legal parameters but with no purpose other than to get around the law.
10.	List the **Causes for an Investigation?** CFA Institute members may be investigated based on the following: • Violation of the CFA Institute Articles of Incorporation, Bylaws, Code of Ethics, Standards of Professional Conduct, Rules of Procedure, or other applicable rules. • Sanction or injunction imposed by a government or judicial agency or public or private self-regulatory organization with jurisdiction over investment-related activities. • Conviction of a felony, or (if the jurisdiction does not define a felony) any crime punishable by more than one year in prison. • Ban or restriction (permanently or indefinitely) from registration under a jurisdiction's securities laws or any restriction related to any investment-related professional activity. • Failure to complete, sign, and return the Professional Conduct Statement. • Falsifying information on a candidate or society membership application. • Failure to cooperate with the CFA Institute's inquiry and investigation into one's conduct or the conduct of another member.

- GIPS primarily benefits investment firms and the clients of those firms. Compliance with GIPS ensures that a firm's historical results are presented consistently and fairly.

- Conformity allows GIPS-compliant firms to bid against other GIPS-compliant firms throughout the world. Investors can easily compare the results of compliant firms, thereby adding credibility to GIPS-compliant performance data.

- Compliance with the GIPS strengthens the firm's internal controls over performance-related processes and procedures.

- Compliance with the GIPS enhances the credibility of investment management firms.

97. Describe the **Key Concepts of the GIPS for Firms?**

The following are the key concepts of the GIPS that apply to firms:

- GIPS are ethical standards for an investment performance presentation to ensure fair representation and full disclosure of investment performance. To claim compliance, firms must adhere to the requirements included in the GIPS standards.

- Meeting the objectives of fair representation and full disclosure will likely require more than simply adhering to the minimum requirements of the GIPS standards. Firms should also adhere to the recommendations to achieve best practices in calculating and presenting performance.

- The GIPS standards require firms to include all actual, discretionary, fee-paying portfolios in at least one composite defined by investment mandate, objective, or strategy to prevent firms from cherry-picking their best performance.

- GIPS relies on the integrity of input data. The accuracy of input data is critical to the accuracy of the performance presentation. The underlying valuations of portfolio holdings drive the portfolio's performance. These and other inputs need to be accurate. GIPS standards require firms to adhere to certain calculation methodologies and make specific disclosures about the firm's performance.

	• Firms must comply with all requirements of the GIPS standards, including any updates, guidance statements, interpretations, questions & answers (Q&As), and clarifications published by the CFA Institute and the GIPS Executive Committee, which are available on the GIPS website as well as in the GIPS Handbook.
98.	Explain the **Construction and Purpose of Composites in performance reporting?** A primary aspect of GIPS is to require composite aggregate portfolios managed with a similar investment approach. Therefore, if a firm presents its record for a particular equity classification, all portfolios meeting pre-established criteria for that class must be represented. A firm cannot include some and exclude others at will. In addition, composite guidelines prevent firms from using only their best-performing portfolios, thereby skewing presentation results. Non-discretionary portfolios must not be included in a firm's composites. A discretionary portfolio is a portfolio manager making buy and sell decisions. On the other hand, the manager of a non-discretionary portfolio does not have full discretion over how the money is invested.
99.	State the **Important Issues** that must **be considered by a firm when it complies with the GIPS?** Two important issues must be considered by a firm when it complies with the GIPS: • Definition of the firm. • Definition of discretionary.
100	Explore the **Definition of Firm under GIPS?** The GIPS standards state, "The firm should adopt the broadest, most meaningful definition of the firm. The scope of this definition should include all geographical (country, regional, etc.) offices operating under the same brand name, regardless of the actual name of the individual investment management company."

	This implies that: • Firms must be defined as investment firms, subsidiaries, or divisions held out to clients or prospective clients as a distinct business entity. • Total firm assets must aggregate the fair value of all discretionary and nondiscretionary assets under management within the defined firm. • Firms must include the performance of assets assigned to a sub-advisor in a composite, provided that the firm has discretion over selecting the sub-advisor. • Changes in a firm's organization are insufficient grounds for alternating historical composite results.
101	Clarify the **Definition of Discretionary under GIPS?** The GIPS states, "If documented client-imposed restrictions interfere with the implementation of the intended strategy to the extent that the portfolio is no longer representative of the strategy, the firm may determine that the portfolio is non-discretionary." Non-discretionary portfolios must not be included in a firm's composites.
102	Discuss the **Concept of Independent Verification?** Verification refers to the process by which a firm claiming compliance with GIPS requirements hires an independent third party to evaluate its processes and corroborate the firm's assertions. Although verification is a recommendation rather than a requirement for firms to claim compliance, it increases investor confidence. Verification must include a firm's practices, not just a small grouping of composites. Authentication should include information on the proper formulation of composite information and company policies

	established to calculate and present GIPS-compliant performance. Verification does not validate specific results of historical data.

SECTION J – ETHICS

TRUE FALSE QUESTIONS AND ANSWERS

S.No	Questions	Answers
1.	There are six components of a code of ethics.	**TRUE.** The following are the **six components of the code of ethics**: a. **Act with integrity, competence, diligence, respect, and ethics** with the public, clients, employers, employees, colleagues, and other participants in the global capital markets. b. **Place the integrity of the investment profession** and clients' interests above their interests. c. Use **reasonable care and exercise independent judgment** when conducting investment analysis, making recommendations, taking actions, and engaging in other professional activities. d. **Practice and encourage others** to practice professionally and ethically. e. **Promote** the **integrity of and uphold the rules governing** the capital market. f. **Maintain and improve their professional competence** and strive to maintain and improve the competence of other investment professionals.
2.	A large proportion of professions outline the importance of ethics.	**TRUE.** A **profession** can be defined as an occupational group based on the unique education, specialist knowledge, and framework of practice and behavior that establishes community trust, respect, and recognition.
3.	A credible profession is	**TRUE. Profession builds** through the following ways:

	characterized by a strong trust from both the clients and the society at large.	• Provision of community service. • Making a profession client-based. • Normalizing of the practitioner's behavior. • Developing high entry standards for a profession. • Possessing a body of expert knowledge. • Continuous education. • Monitoring professional misconduct. • Making a profession collegial. • Establishing professional oversight authorities. • Engagement of the professional members.
4.	A law is proactive.	**FALSE.** All laws stem from a place in which moral covenant has broken down. How control is maintained in these circumstances is to set laws and stipulate punishment; therefore, the law is **reactive,** and ethical conduct is proactive.
5.	There are some major areas where ethical dilemmas can occur.	**TRUE.** An **ethical dilemma** occurs whenever there are two or more choices. There will always be the best choice in this given circumstance, even when each option appears to have negative consequences. Major areas in which ethical dilemmas occur are: a. Misrepresentation. b. Misconduct. c. Fairness. d. Loyalty. e. Prudence f. Care.
6.	The CFA Institute's Professional Standards and Policy Committee reviews violations	**TRUE.** The **Professional Standards and Policy Committee** (PSPC) is authorized to conduct investigations and impose penalties. Within the PSPC, there are two subcommittees:

	of the CFA codes and standards.	a. The **Disciplinary Review Committee** (DRC) reviews potential violations and enforces the code of ethics. b. The **Standards of Practice Council** (SPC) is charged with editing, deleting, or adding standards and distributing updated information to CFA members.
7.	One must be held accountable concerning standards of conduct in seven areas.	**TRUE.** Concerning **standards of conduct**, the following are the seven areas by which one must be held accountable: a. **Professionalism** - law knowledge, independence and objectivity, misrepresentation, misconduct. b. **The integrity of capital markets** - material non-public information, market manipulation. c. **Duties to clients and prospective clients** - loyalty and prudence, fair dealing, suitability, and performance presentation. d. **Duties to employers** – loyalty, additional compensation arrangements, responsibilities of supervisors. e. **Investment analysis, recommendations, and action** - diligence, reasonable basis, communication with prospective clients, and records retention. f. **Conflicts of interest** – disclosure of conflicts, priority of transactions, referral fees. g. **Responsibilities as a CFA institute member or CFA candidate** - conduct as participants in CFA institute programs, reference to CFA institute, CFA designation, and CFA program.

	firms using the CFA institute name and logo.	institute name and logo by individuals and firms. Caution must be taken not to exaggerate the CFA institute's or its programs' meaning when used by members and candidates.
70.	Factual statements concerning the benefits of CFA program membership are prohibited.	**FALSE.** Factual statements concerning the benefits of CFA program membership are **permitted**, whereas inflammatory statements regarding the superior performance of CFA charter-holders are prohibited.
71.	Any statement referenced to historical fact must be framed as an opinion by the statement-maker.	**FALSE.** Any statement that **cannot be referenced** to historical fact must be framed as an opinion by the statement-maker. This includes communication via any medium, including electronic, speech, and print.
72.	CFA institute recognizes its designation, and uninformed firms often misuse marks.	**TRUE.** Members and candidates should provide **information related to standard VII(B)** to appropriate legal, compliance, and public relations contacts within their respective organizations.
73.	The GIPS standards provide a basis for including all relevant data and avoiding misrepresentations.	**TRUE.** The **Global Investment Performance Standards** establish a standardized set of ethical practices that guide practitioners in analyzing and presenting historical data as a basis for comparing investment results.
74.	Like-for-like performance comparison was a challenge before GIPS creation.	**TRUE.** Before GIPS creation, **like-for-like performance comparison** was a challenge because of the presence of the following misleading practices: • **Representative accounts** - Firms use the top-performing portfolio to represent the firm's overall investment results. • **Survivorship bias** - Firms exclude poor-performing portfolios in presenting their performance and present an average performance history. • **Varying time periods** - Firms hand-pick only the time periods during which the

		mandate outperformed the specified benchmark.
75.	Clients and prospective clients can benefit from compliance.	**TRUE. Clients and prospective clients** can have a degree of confidence in the numbers they are using to compare managers and that the numbers are representative of the experience of those managers and not conveniently crafted records that distort what happened.
76.	Software developers and vendors can claim compliance with global investment performance standards.	**FALSE.** Any investment firm can comply, and only management firms that manage assets can claim compliance. Parties that **cannot claim compliance** are as follows: • **Software developers.** • **Software vendors.** • Plan sponsors. • Consultants who do not manage assets.
77.	One of the objectives of GIPS is to promote investor interests and instill investor confidence.	**TRUE.** The **objectives** of the **Global Investment Performance Standards** are: • Ensure accurate and consistent data. • Promote fair, global competition among investment firms. • Promote industry self-regulation on a global basis. • Obtain worldwide acceptance of a single standard for calculating and presenting performance.
78.	A firm can comply with one or a limited portion of global investment performance standards.	**FALSE.** To claim compliance, a firm **must comply with all** global investment performance standards. Simple compliance with one or a limited portion of parameters is **insufficient**.
79.	Compliance is seen as proof that the	**TRUE. Investment management firms** in compliance with GIPS requirements enjoy a greater level of credibility in the hands of

	firm upholds high ethical behavior.	investors. By adopting GIPS, firms in countries with few to no standards can compete with countries with more developed standards.
80.	GIPS relies on the integrity of input data.	**TRUE.** The **accuracy of input data is critical** to the accuracy of the performance presentation. GIPS standards require firms to adhere to certain calculation methodologies and make specific disclosures about the firm's performance.
81.	A primary aspect of GIPS is to require composites.	**TRUE.** A primary aspect of GIPS is to require composite **aggregate portfolios managed with a similar investment approach**. Therefore, if a firm presents its record for a particular equity classification, all portfolios meeting pre-established criteria for that class must be represented.
82.	A non-discretionary portfolio is a portfolio manager making one in which buy and sell decisions.	**FALSE.** A **discretionary portfolio** is a portfolio manager making buy and sell decisions. On the other hand, the manager of a non-discretionary portfolio does not have full discretion over how the money is invested.
83.	Verification does not validate specific results of historical data.	**TRUE. Verification** refers to the process by which a firm claiming compliance with GIPS requirements hires an independent third party to evaluate its processes and corroborate the firm's assertions. Although verification is a recommendation rather than a requirement for firms to claim compliance, it increases investor confidence.

SECTION J – ETHICS

FILL IN THE BLANKS

1. The CFA Institute's Code of Ethics emphasizes that members must act with ______ and maintain the highest ethical standards.

2. The Standards of Professional Conduct include ______ with Laws and Regulations.

3. The principle of ______ requires CFA members to place their clients' interests before their own.

4. Members must use reasonable care and exercise ______ when dealing with clients.

5. Misrepresentation involves knowingly making a false statement or omitting ______ information.

6. The ______ Standard prohibits engaging in any conduct that compromises the reputation of the CFA Institute or the integrity of the capital markets.

7. The duty to disclose ______ conflicts of interest is essential to maintaining trust with clients.

8. Members and candidates are required to keep information about current, former, and prospective clients ______ unless permitted by law or the client.

9. The Standard on ______ prohibits the use of non-public information to gain an advantage in the market.

10. Members must make fair and ______ presentations of investment performance.

11. A ______ is any activity that could potentially influence a member's or candidate's ability to act impartially in their professional duties.

12. ______ refers to the expectation that CFA members will always act in a way that reflects well on the profession.

13. When managing client accounts, CFA members must act with diligence and in the best interest of the ______.

14. The ______ Standard requires CFA members to act with competence and diligence in all professional activities.

15. ______ is the act of deliberately misleading clients or employers for personal gain.

16. The ______ Standard prohibits any conduct that undermines the integrity or reputation of the CFA Institute.

17. ______ standards require CFA members to understand and comply with all applicable laws, rules, and regulations.

18. Members must refrain from engaging in any activity that conflicts with the interest of their ______.

19. The ______ rule in the CFA Code of Ethics states that members must act with integrity, competence, diligence, and respect.

SECTION J – ETHICS

ANSWER KEY

1. integrity

2. compliance

3. loyalty

4. professional judgment

5. material

6. Professionalism

7. potential

8. confidential

9. Material Nonpublic Information

10. accurate

11. conflict of interest

12. Professionalism

13. client

14. Knowledge and Understanding

15. Misconduct

16. Misconduct

17. Compliance

18. employer

19. Fundamental Responsibilities

20. conflicts of interest

21. Misconduct

22. transparency

23. Insider trading

24. gifts

25. information

26. Diligence and Reasonable Basis

27. fairness

28. market integrity

29. seven

30. Conflicts of Interest

31. misleading

32. conflicts of interest

33. Objectivity

34. reputation

35. Insider trading

36. conflict

37. Diligence and Reasonable Basis

38. first

39. Misrepresentation

40. local

41. financial

42. clients

43. competence

44. Confidentiality

45. disrepute

46. Misrepresentation

47. compromises

48. fairness

49. financial

50. Conflicts of Interest

51. accurate

52. Disclosure of Referral Fees

53. clients

54. integrity

55. non-public

56. Misconduct

57. information

58. Knowledge and Understanding

59. conflict

60. Misrepresentation

61. basis

62. Professionalism

63. financial

64. Conflicts of Interest

65. clients

66. Confidentiality

67. clients

68. Misrepresentation

69. disrepute

70. Diligence and Reasonable Basis

71. clients

72. Conflicts of Interest

73. non-public

74. Conflicts of Interest

75. accurate

76. Disclosure of Referral Fees

77. financial

78. Misrepresentation

79. disrepute

80. Confidentiality

81. conflict

82. Professionalism

83. information

84. Misrepresentation

85. financial

86. Conflicts of Interest

87. basis

88. Professionalism

89. financial

90. Disclosure of Referral Fees

91. conflict

92. Confidentiality

93. disrepute

94. Diligence and Reasonable Basis

95. clients

96. Misrepresentation

97. non-public

98. Conflicts of Interest

99. accurate

100. Professionalism

SECTION J – ETHICS

WORD SEARCH PUZZLES

Section J - Quiz 1

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J L I X Z M S C T N E M E C R O F N E U
B Y M N Q R W Y Y S E D P G C N C W C G
Y T D T O Y V T N C Y R E B I R B L P L
Y I C P N I L P D E Y V I O L A T I O N
K L J Z Z A T Z M O C Z B W S R A N B L
R A A Q Y L L A W T S N X Q A N O S R H
X I Y O I Y O G T C C L A N R S M I Z O
Q T L T N O P G O N S I S I E R C D R G
L N S C D H B A A Z E P L R L O E E S Y
A E E E E D D J E E A S U F M P A R T F
C D T P P J F K E R C S E P N A M S L K
I I P S E E N K E C O N E R K O E O R I
H F W E N B Q N Y L T T E U P N C X C I
T N D R D B C M C V E I B G O E T G S B
E O E B E Y N S G N T M V H I B R H V B
Y C M X N G I I C J P A M I Y L E S H D
F Y W I C D S E B W Y I Q Q T P I U I A
R F J A E L N F A I R N E S S Y A D G M
A E Y T I R G E T N I T Z D N X L X T X
D M S I L A N O I S S E F O R P J U T N
```

1. Integrity **2.** Diligence **3.** Professionalism **4.** Competence **5.** Ethical

6. Transparency **7.** Honesty **8.** Loyalty **9.** Respect **10.** Misrepresentation **11.** Conflict

12. Confidentiality **13.** Fairness **14.** Independence **15.** Objectivity **16.** Insider

17. Disclosures **18.** Bribery **19.** Compliance **20.** Violation **21.** Enforcement

ABOUT AUTHOR

Get ready to meet a titan in the accounting world! Muhammad Zain, a master in the art of numbers and a beacon of knowledge in finance and accounting, is transforming the industry with his expertise and dedication.

In February 2018, Zain achieved a significant milestone by conquering the Uniform Certified Public Accountant (CPA) exams of the prestigious American Institute of Certified Public Accountants (AICPA), US. But his journey of excellence began earlier. In March 2014, he demonstrated his skills and passion by acing both the Certified Management Accountant (CMA) exams from the Institute of Management Accountants (IMA), US, and the Certified Internal Auditor (CIA) exams from the Institute of Internal Auditors (IIA), US.

Zain's educational foundation is robust, with a Masters of Business Administration (MBA) completed in March 2010 and a Bachelor of Commerce (BCom) completed in November 2007, both from the esteemed University of Karachi, Pakistan.

His professional journey includes five years at EY Ford Rhodes, Pakistan, a member firm of the iconic Ernst & Young Global Limited (Big 4), and nine years contributing his unique insights to the industry.

On 27 February 2017, Zain founded Zain Academy, driven by the mission "Knowledge for ALL" and a vision to revolutionize the working environment. Zain Academy is dedicated to creating superior and accessible exam preparation materials for CIA, CMA, CFA, CPA, and FRM candidates.

From around the globe, ambitious candidates seek Zain's expertise, motivation, and guidance to tap into their true potential and achieve their career dreams. If you're looking for advice on any aspect of CIA, CMA, CFA, CPA, FRM, or global accounting standards, Muhammad Zain is your go-to expert.

Don't miss out on the wealth of knowledge available on Zain Academy's YouTube channel. Explore informative videos on CIA, CMA, CFA, CPA, FRM, and more, along with updates on upcoming publications.

Embark on your journey to excellence with Muhammad Zain, where learning is unrestricted and knowledge is for all!

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